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Test Booklet Series

TEST BOOKLET

C

GENERAL STUDIES (P) 2024 – Abhyaas Test – 4361

Time Allowed: Two Hours

Maximum Marks: 200

INSTRUCTIONS

1. IMMEDIATELY AFTER THE COMMENCEMENT OF THE EXAMINATION, YOU SHOULD CHECK THAT THIS BOOKLET DOES **NOT** HAVE ANY UNPRINTED OR TURN OR MISSING PAGES OR ITEMS, ETC. IF SO, GET IT REPLACED BY A COMPLETE TEST BOOKLET.
2. ENCODE CLEARLY THE TEST BOOKLET SERIES **A, B, C** OR **D** AS THE CASE MAY BE IN THE APPROPRIATE PLACE IN THE ANSWER SHEET.
3. You have to enter your Roll Number on the Test Booklet in the Box provided alongside. **Do NOT** write anything else on the Test Booklet.
4. This Test Booklet contains **100** items (Questions). Each item is printed in **Hindi & English**. Each item comprises four responses (answers). You will select the response which you want to mark on the Answer Sheet. In case you feel that there is more than one correct response with you consider the best. In any case, choose **ONLY ONE** response for each item.
5. You have to mark all your responses **ONLY** on the separate Answer Sheet provided. See direction in the answers sheet.
6. All items carry equal marks. Attempt all items. Your total marks will depend only on the number of **correct responses** marked by you in the answer sheet. For **every incorrect** response **1/3rd of the allotted marks** will be deducted.
7. Before you proceed to mark in the Answer sheet the response to various items in the Test booklet, you have to fill in some particulars in the answer sheets as per instruction sent to you with your Admission Certificate.
8. After you have completed filling in all responses on the answer sheet and the examination has concluded, you should hand over to Invigilator only the answer sheet. You are permitted to take away with you the Test Booklet.
9. Sheet for rough work are appended in the Test Booklet at the end.

DO NOT OPEN THIS BOOKLET UNTIL YOU ARE ASKED TO DO SO

1. With reference to the 'general consent' to the Central Bureau of Investigation (CBI), consider the following statements:

1. It refers to the central government's approval for investigations in different states.
2. The Delhi Special Police Establishment (DSPE) Act, 1946 explicitly mentions the provision of general consent to the CBI.
3. The CBI can continue probing old cases even after the general consent is withdrawn.

How many of the statements given above are correct?

- (a) Only one
- (b) Only two
- (c) All three
- (d) None

2. Consider the following:

1. The creation of a separate development board for the underdeveloped regions
2. A state assembly's concurrence to central acts related to cultural practices
3. The creation of local cadres in various classes of employment
4. Restriction on the sale and purchase of land by outsiders

How many of the above mentioned special protections are allocated to some states under Article 371 (A-J) of the constitution?

- (a) Only one
- (b) Only two
- (c) Only three
- (d) All four

3. Consider the following:

1. Cases involving a substantial question of law as to the interpretation of the Constitution
2. A Presidential reference made under Article 143
3. Petition calling in question the election of the President and Vice President

In how many of the cases given above is it mandatory for the matter to be adjudicated by a Constitution bench?

- (a) Only one
- (b) Only two
- (c) All three
- (d) None

4. Consider the following statements regarding the Mediation Act, 2023 which was recently passed:

1. It provides for mandatory mediation before the institution of legal proceedings in all civil and criminal matters.
2. It provides for the establishment of the Mediation Council of India.
3. The act lists down certain disputes such as criminal offences which are not fit for mediation.

How many of the statements given above are correct?

- (a) Only one
- (b) Only two
- (c) All three
- (d) None

5. Consider the following statements:
Statement-I: The Parliament of India cannot repeal or abrogate the Indian constitution itself despite having constituent powers.
Statement-II: The Parliament is bound by the doctrine of basic structure propounded by the Supreme Court.
Which one of the following is correct in respect of the above statements?
- Both Statement-I and Statement-II are correct and Statement-II is the correct explanation for Statement-I
 - Both Statement-I and Statement-II are correct and Statement-II is not the correct explanation for Statement-I
 - Statement-I is correct but Statement-II is incorrect
 - Statement-I is incorrect but Statement-II is correct
6. Consider the following statements:
Statement-I: The Indian Parliament can make laws for the equitable distribution of material resources of the community for the common good even if it hampers the right to equality.
Statement-II: In general, the Directive Principles of State Policies have been deemed superior to the fundamental rights.
Which one of the following is correct in respect of the above statements?
- Both Statement-I and Statement-II are correct and Statement-II is the correct explanation for Statement-I
 - Both Statement-I and Statement-II are correct and Statement-II is not the correct explanation for Statement-II
 - Statement-I is correct but Statement-II is incorrect
 - Statement-I is incorrect but Statement-II is correct
7. Recently, the Supreme Court declared the 'right to be free from the adverse effects of climate change' as a:
- statutory right under the Environment (Protection) Act, 1986.
 - constitutional right under Article 48A of the Indian Constitution
 - fundamental right under Article 14 and Article 21 of the Indian Constitution
 - fundamental right under Article 19 and Article 23 of the Indian Constitution.
8. In the context of Indian Polity, which of the following statements is **not** correct regarding the effects of a national emergency?
- The President can modify the constitutional distribution of revenues between the center and the states.
 - Parliament becomes empowered to make laws on any subject mentioned in the state list.
 - The Centre can give directions to any state only on certain specific matters as listed in the state list.
 - The Parliament may extend the normal tenure of a state legislative assembly.
9. Consider the following statements regarding the Committee on Content Regulation in Government Advertising (CCRG):
- It is a parliamentary committee set up to regulate the content of Central and State government advertisements in all media platforms.
 - It recommends remedial action to the concerned authorities on the complaints received by it.
- Which of the statements given above is/are correct?
- 1 only
 - 2 only
 - Both 1 and 2
 - Neither 1 nor 2

10. With reference to Information Technology Act 2000, consider the following statements:

1. State governments are empowered to issue directives for intercepting information transmitted through any computer resource.
2. It encompasses all intermediaries such as Internet Service Providers (ISPs) involved in the utilization of computer resources and electronic records.
3. Requests for blocking website or application access under the act may either be oral or written.

How many of the statements given above are correct?

- (a) Only one
- (b) Only two
- (c) All three
- (d) None

11. Consider the following map:



Which of the following best describes the shorter route connecting Mumbai to Saint-Petersburg in the above map?

- (a) Belt and Road Initiative corridor route
- (b) Area covering countries which are signatories to the Ashgabat Agreement.
- (c) India-Middle East Economic Corridor
- (d) International North South Transport Corridor

12. Consider the following statements:

Statements-I: Recently, India and the European Union (EU) have launched the 'Trade and Technology Council'.

Statement-II: Through the Trade and Technology Council, India and the EU aim to bypass the stalemate created due to the World Trade Organization Dispute Settlement body.

Which one of the following is correct in respect of the above statements?

- (a) Both Statement-I and Statement-II are correct and Statement-II is the correct explanation for Statement-I
- (b) Both Statement-I and Statement-II are correct and Statement-II is not the correct explanation for Statement-I
- (c) Statement-I is correct but Statement-II is incorrect
- (d) Statement-I is incorrect but Statement-II is correct

13. Consider the following pairs:

Countries often mentioned in news **Reason for being in the news**

1. Serbia : Regional demands for independence
2. Gabon : Occurrence of military coup incidents
3. Sudan : Recent outbreak of Cholera
4. Guyana : Border dispute with Colombia

How many of the pairs given above are correctly matched?

- (a) Only one
- (b) Only two
- (c) Only three
- (d) All four

14. Consider the following statements regarding the Asia Pacific Forum of Human Rights:

1. It is a platform to bring together National Human Rights Institutions (NHRIs) from the Asia Pacific to address human rights challenges in the region.
2. The National Human Rights Commission of India is one of its members.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

15. Which of the following best describes the term 'iterative evolution'?

- (a) It is the process by which an organism evolves in a linear fashion from a common ancestor.
- (b) It involves the repeated evolution of a specific trait or body plan from the same ancestral lineage at different points in time.
- (c) It involves the simultaneous evolution of unrelated species in response to similar environmental pressures.
- (d) It involves the gradual accumulation of genetic changes over time leading to new species.

16. Consider the following statements regarding Captive Elephant (Transfer or Transport) Rules, 2024:

1. Before transferring an elephant outside of a state, its genetic profile must be registered with the Ministry of Environment, Forest and Climate Change (MoEFCC).
2. Before transferring an elephant within the state, a veterinarian must verify its health and the Deputy Conservator of Forests confirm its suitable habitats.
3. An elephant being transferred ought to be accompanied by a mahout and an elephant assistant.

How many of the statements given above are correct?

- (a) Only one
- (b) Only two
- (c) All three
- (d) None

17. Consider the following statements regarding camelids:

1. Dromedary camels have one hump whereas bactrian camels have two humps.
2. India is home to dromedary camels but not bactrian camels.
3. Dromedary camels are the most widespread camels in the world.

How many of the statements given above are correct?

- (a) Only one
- (b) Only two
- (c) All three
- (d) None

18. Consider the following statements:
Statement-I: Some dolphins exhibit cooperative hunting behaviors, such as herding fish into tight groups using bubble nets.
Statement-II: Cooperative hunting allows dolphins to catch prey more efficiently.
Which one of the following is correct in respect of the above statements?
- (a) Both Statement-I and Statement-II are correct and Statement-II is the correct explanation for Statement-I
 - (b) Both Statement-I and Statement-II are correct and Statement-II is not the correct explanation for Statement-I
 - (c) Statement-I is correct but Statement-II is incorrect
 - (d) Statement-I is incorrect but Statement-II is correct
19. How does the concept of 'bottom-up control' differ from 'top-down control' in ecosystem dynamics?
- (a) Bottom-up control refers to the influence of predators on prey populations, while top-down control refers to the influence of resource availability on consumer populations.
 - (b) Bottom-up control occurs when primary producers regulate the abundance of consumers, while top-down control occurs when predators regulate the abundance of prey.
 - (c) Bottom-up control is driven by genetic adaptations within populations, while top-down control is driven by behavioral responses of organisms.
 - (d) Bottom-up control is influenced by external environmental factors, while top-down control is driven by internal ecosystem dynamics.

20. Recently, one of the Europe's largest chemical complexes tried to manufacture a chemical named 'aniline' using sugar as a base material instead of oil to reduce its carbon footprint. In this context, how many of the following contain/cause aniline?
- 1. Smoking tobacco
 - 2. Burning of plastics
 - 3. Synthetic dyes
 - 4. Rubber accelerators
- Select the correct answer using the code given below.
- (a) Only two
 - (b) Only three
 - (c) All four
 - (d) None
21. Consider the following statements regarding the Rugra mushroom which was recently seen in the news:
- 1. It is naturally grown in Jammu and Kashmir.
 - 2. It is a poisonous and bioluminescent mushroom.
- Which of the statements given above is/are correct?
- (a) 1 only
 - (b) 2 only
 - (c) Both 1 and 2
 - (d) Neither 1 nor 2

22. Consider the following statements regarding Collaborative Partnership on Forests (CPF):

1. It was established by the Global Forest Watch in partnership with the Food and Agriculture Organization (FAO).
2. Its mission is to enhance the contribution of all forest types, including trees outside traditional forest areas, in advancing the objectives of the '2030 Agenda for Sustainable Development'.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

23. Consider the following statements regarding the category of 'Vermin' as per the Wildlife Protection Act 1972:

1. The Union Government can notify any wild animal in the Schedule II of the Act as vermin for a period of time in an area/state.
2. An animal is declared as vermin when it poses threat to life and crops.
3. Only licensed shooters by the government can kill an animal which is declared as vermin.

How many of the above statements are correct?

- (a) Only one
- (b) Only two
- (c) All three
- (d) None

24. Consider the following statements with respect to the Technology Development Fund (TDF) scheme:

1. It is a flagship programme of the Ministry of Textiles.
2. It is an initiative to boost the Indian textile industry by enabling ease of doing business, bolstering exports and fuelling employment.
3. The scheme provides grants in aid to Indian industries, including MSMEs, startups, and academic and scientific institutions.

How many of the statements given above are correct?

- (a) Only one
- (b) Only two
- (c) All three
- (d) None

25. How many of the following are the benefits of Pradhan Mantri Vaya Vandana Yojana?

1. Assured pension to subscribed elderly persons aged 60 and above
2. Premature exit without penalty in case of critical or terminal illness of self or spouse
3. Returns on the scheme are exempted from income tax

Select the correct answer using the code given below.

- (a) Only one
- (b) Only two
- (c) All three
- (d) None

26. Consider the following statements:
Statement-I: The Asiatic Society of Bengal was established by Sir William Jones who was an evangelist.

Statement-II: The evangelists argued about the necessity of government intervention to liberate Indians from their religion that was full of superstitions, idolatry, and tyranny of the priest.

Which one of the following is correct in respect of the above statements?

- (a) Both Statement-I and Statement-II are correct and Statement-II is the correct explanation for Statement-I
- (b) Both Statement-I and Statement-II are correct and Statement-II is not the correct explanation for Statement-I
- (c) Statement-I is correct but Statement-II is not correct
- (d) Statement I is not correct but the Statement-II is correct

27. In the context of colonial history of India, *dadani* merchants were those who

- (a) supplied indentured labourers to work in British plantations in colonies outside India.
- (b) carried out British opium trade to China.
- (c) used advances to purchase goods from the market like textiles.
- (d) carried out trade in perfume and spices from South India.

28. Consider the following statements regarding the evolution of separate electorate in India during the colonial period:

- 1. The Nehru report rejected the idea of a separate electorate with reservation of seats for Muslims at the centre and in all provinces.
- 2. Dr BR Ambedkar put a demand for a separate electorate for the depressed class in front of the Simon Commission in the case of universal adult franchise.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

29. Consider the following statements regarding the development of railways in India by the Britishers:

- 1. The East India Company provided guarantee to railways shareholders a 5% return in their capital investment along with a 99- year operating contract.
- 2. The railway lines categorised as 'protective' were those with commercial potential.
- 3. The Railway Board was established during the tenure of the Viceroy Lord Minto for a single point of control and management of railway services.

How many of the statements given above are correct?

- (a) Only one
- (b) Only two
- (c) All three
- (d) None

30. Consider the following statements about Dharma Sabha:

1. It was formed by orthodox members of the Hindu community led by Radhakanta Deb.
2. It filed an appeal in the Privy Council against the Sati ban.

Which of the statements given above is/are *not* correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

31. Consider the following statements with reference to Prepaid Payment Instruments in India:

1. All systems to facilitate Prepaid Payment Instruments have to be authorized/approved by the Reserve Bank of India.
2. All Prepaid Payment Instruments have to be loaded through Indian Rupee only.
3. Prepaid payment instruments can be issued both as cards and paper vouchers.

How many of the statements given above are correct?

- (a) Only one
- (b) Only two
- (c) All three
- (d) None

32. Recently, the Reserve Bank of India (RBI) has set standards for Small Finance Banks (SFBs) desiring to obtain the Universal Banking License. In this context, consider the following statements:

1. The number of SFBs in India has exceeded 20.
2. The Gross Non-Performing Asset (GNPA) of most of the SFBs is below 1%.
3. To get the Universal Banking License, an SFB should have a minimum net worth of Rs 500 crore.

How many of the statements given above are correct?

- (a) Only one
- (b) Only two
- (c) All three
- (d) None

33. Consider the following taxes:

1. Securities Transaction Tax
2. Minimum Alternate Tax
3. Capital Gains Tax

How many of the above taxes are imposed on profits?

- (a) Only one
- (b) Only two
- (c) All three
- (d) None

34. As per NITI Aayog, India is a new frontier of the gig economy. In this context, consider the following:

1. Online platform workers
2. Own-account workers in conventional sectors
3. Casual wage workers
4. All migrant workers

How many of the above are normally categorized as gig workers?

- (a) Only one
- (b) Only two
- (c) Only three
- (d) All four

- 35.** Consider the following statements:
1. The Minimum Support Price (MSP) for this crop is the highest among the oilseeds.
 2. It is grown in India as a Kharif, semi-Rabi, and summer crop.
 3. With India as one of its major exporters, this crop was in the news due to its cultivation in the conflicted zones of Sudan and Ethiopia.

Which of the following oilseeds has been described in the statements given above?

- (a) Mustard
- (b) Groundnut
- (c) Sesamum
- (d) Soybean

- 36.** Consider the following statements with reference to fertilizers in India:

1. India is the world's second-largest consumer of finished fertilizers.
2. India has a policy related to indigenous urea production.
3. The fertilizer subsidy in India is greater than the petroleum subsidy.

How many of the statements given above are correct?

- (a) Only one
- (b) Only two
- (c) All three
- (d) None

- 37.** Consider the following statements:
- Statement-I: As per law, to avoid overlapping interests a subsidiary company cannot hold the shares in its holding company or parent company.

Statement-II: The SEBI Act, 1992, deals with matters related to the regulation of the securities market and investor protection.

Which one of the following is correct in respect of the above statements?

- (a) Both Statement-I and Statement-II are correct and Statement-II is the correct explanation for Statement-I.
- (b) Both Statement-I and Statement-II are correct and Statement-II is not the correct explanation for Statement-I
- (c) Statement-I is correct but Statement-II is incorrect
- (d) Statement-I is incorrect but Statement-II is correct

- 38.** Consider the following statements regarding the export of foodgrains from India:

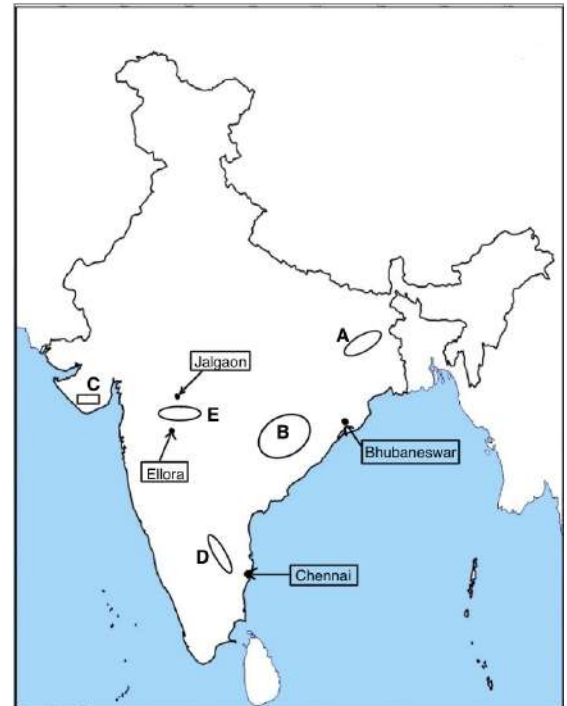
1. The WTO's Agreement on Agriculture places restrictions on the export of wheat and non-basmati rice procured by the government for the Central Pool Stock at MSP.
2. Wheat and rice from Central Pool Stocks cannot be exported to foreign countries even as humanitarian assistance aid.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

39. Consider the following statements with reference to the health sector in India:
1. The National Health Policy 2017 envisages to increase the public expenditure on health to 2.5% of GDP by 2025.
 2. The Ayushman Bharat-Pradhan Mantri Jan Arogya Yojana (AB-PMJAY) is fully funded by the Central Government.
 3. Recently, the maximum age limit for purchasing health insurance policies has been raised to 70 years.
- How many of the statements given above are correct?
- (a) Only one
(b) Only two
(c) All three
(d) None
40. In the context of economics, Luddite Fallacy is a belief most closely related to
- (a) The consolidation of labour codes and the resulting increase in the efficiency of the workforce
(b) The adoption of protectionist measures and saving domestic industries and labour force from outside competition.
(c) The adoption of new technology and possibility of unemployment
(d) Feminization of workforce and the possibility of reduction in efficiency
41. Consider the following statements regarding diamond reserves:
1. China has the world's largest diamond reserves.
 2. In India, the diamond-bearing regions are mainly found in the Aravallis.
 3. Andhra Pradesh is the largest producer of diamonds in the country.
- How many of the above statements are correct?
- (a) Only one
(b) Only two
(c) All three
(d) None

42. Consider the map given below:



The places marked A, B, C, D and E on the map are respectively:

- (a) Kaimur range, Dandakaranya, Gir range, Javadi hills and Ajanta range
(b) Rajmahal hills, Telangana Plateau, Gir range, Palkonda range and Satpura range
(c) Chota Nagpur plateau, Garhjat & Tikarapara hills, Gir range, Shevroy hills and Satmala hills
(d) Rajmahal hills, Dandakaranya, Gir range, Palkonda range and Ajanta range
43. Which of the following countries are part of the Amazon Basin?
1. Colombia
 2. Paraguay
 3. Guyana
 4. Chile
- Select the correct answer using the code given below.
- (a) 2 and 4 only
(b) 1 and 3 only
(c) 1, 3 and 4 only
(d) 1, 2 and 3 only

44. Consider the following pairs:

<i>Tribes</i>	<i>State/UT</i>
1. Gadda Brahmin	: Jammu and Kashmir
2. Konda Savaras	: Andhra Pradesh
3. Mankidia	: Madhya Pradesh
4. Paddari	: Odisha

How many pairs given above are correctly matched?

- (a) Only one
- (b) Only two
- (c) Only three
- (d) All four

45. Consider the following statements regarding the phenomenon of twin cyclones:

- 1. These are formed due to equatorial Rossby waves.
- 2. They occur when a cyclone splits into two smaller cyclones due to atmospheric conditions.
- 3. The ocean surface temperature should be between 20°C to 25°C for their formation.

How many of the statements given above are correct?

- (a) Only one
- (b) Only two
- (c) All three
- (d) None

46. Which of the following is **not** correct regarding maize crop in India?

- (a) Bulk of the maize production in India is used as poultry feed.
- (b) In India, production of maize is mainly carried out during Kharif season.
- (c) Karnataka is the largest producer of maize in India.
- (d) India is the largest producer of maize in the world.

47. In the context of India's energy resources, consider the following pairs:

<i>Energy Project</i>	<i>Description</i>
1. Khavda	: world's largest renewable energy park
2. Muppandal	: largest onshore wind farm in India
3. Sasan	: India's largest integrated power generation plant

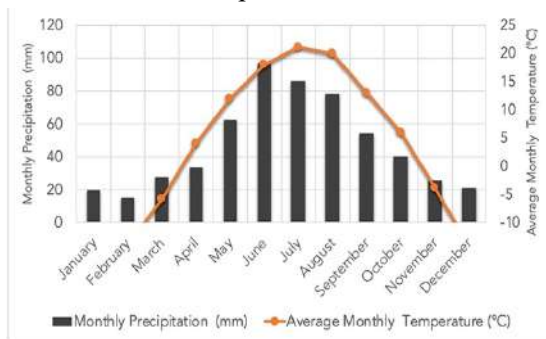
How many of the above pairs are correctly matched?

- (a) Only one
- (b) Only two
- (c) All three
- (d) None

48. The term 'Wilson cycle', often discussed in the context of geology, refers to the:

- (a) cyclical process of erosion and deposition along river banks.
- (b) alternating phases of glaciation and deglaciation during ice ages.
- (c) continuous formation and destruction of oceanic crust through seafloor spreading and subduction.
- (d) periodic variations in Earth's axial tilt, leading to changes in climate over geological time scales.

49. The graph given below shows the mean average monthly temperatures (in °C) and mean monthly rainfall (in mm) of a place in the Northern Hemisphere:



The above-given graph is indicative of which one of the following climatic zones of the world?

- (a) Temperate marine
(b) Warm Temperate Western Margin
(c) Temperate grassland
(d) Mid-latitude deserts
50. Consider the following pairs:
- | Lakes | Formed By |
|------------------|-----------------------------|
| 1. Kettle lakes | : Volcanic activity |
| 2. Ribbon lakes | : Meteor impact |
| 3. Tarn lakes | : Glaciation |
| 4. Barrier lakes | : Landslides and avalanches |
- How many of the pairs given above are correctly matched?
- (a) Only one
(b) Only two
(c) Only three
(d) All four
51. In the context of ancient Indian history, the Kalabhra revolt was
- (a) to overthrow the Nanda dynasty because of the humble origin of the dynasty's founder and excessive taxation.
(b) against the oppressive rule of Mahipala II.
(c) a powerful peasant protest directed against the landed brahmanas.
(d) against forced labour or vishti during the Gupta period.

52. Consider the following rulers:

1. Magas of Cyrene
2. Ptolemy II of Egypt
3. Alexander III of Macedonia

How many of the above Hellenistic rulers were contemporaries of King Ashoka?

- (a) Only one
(b) Only two
(c) All three
(d) None

53. In the context of the Sangam Age, the term 'tippayadal' refers to a

- (a) ritual dance performed during festivals and royal ceremonies.
(b) practice where a widow voluntarily ends her life on her husband's funeral pyre.
(c) form of ancient Tamil martial art used for self-defence and warfare.
(d) customary tribute paid by subordinate kings to their sovereign.

54. With reference to agricultural practices during the Delhi Sultanate, which of the following crops were predominantly grown?

1. Maize
2. Red chillies
3. Tobacco
4. Sugarcane
5. Cotton

Select the correct answer using the code given below.

- (a) 1, 2, 4 and 5 only
(b) 4 and 5 only
(c) 1, 2 and 3 only
(d) 1, 2, 3, 4 and 5

55. With reference to the Indian medieval history, which of the following statement best describes the concept of *Khalisa* land?

- (a) It refers to the rent-free land held by the nobilities in lieu of service.
- (b) It refers to those land whose revenues were reserved for the imperial treasury.
- (c) It refers to rent-free land provided to intellectuals and artists as gifts.
- (d) It refers to the land owned by the community and outside the state regulations.

56. In the context of medieval India, which of the following rulers issued the *Mahzar*, a new state policy and conferred upon himself “the attributes of infallibility”?

- (a) Muhammad bin Tughlaq
- (b) Akbar
- (c) Aurangzeb
- (d) Balban

57. Consider the following treatise:

- 1. Natya Shastra
- 2. Brihaddeshi
- 3. Svaramelakalanidhi
- 4. Abhinaya Darpan

How many of the above treatises are associated with the evolution of music in India?

- (a) Only one
- (b) Only two
- (c) Only three
- (d) All four

58. Arrange the following Jyotirlinga temples from east to west.

- 1. Mahakaleshwar
- 2. Nageshwar
- 3. Mallikarjuna
- 4. Trimbakeshwar

Select the correct answer using the code given below.

- (a) 3-4-2-1
- (b) 4-3-2-1
- (c) 1-2-3-4
- (d) 3-1-4-2

59. Consider the following pairs regarding Buddhist canonical texts:

<i>Text</i>	<i>Description</i>
1. Sutta Pitaka	: Rules and Regulations of daily life.
2. Vinaya Pitaka	: Dialogues and discourses on morality.
3. Abhidhamma Pitaka	: Deals with philosophy and metaphysics.

How many of the pairs given above are correctly matched?

- (a) Only one
- (b) Only two
- (c) All three
- (d) None

60. Which of the statements given below are **not** correct regarding the architecture at Ellora caves?

- (a) It houses the monolithic Kailasanatha Temple having well painted porch walls.
- (b) Ardhanarishvara and Maheshamurti are the famous sculptures found here.
- (c) Paintings on Jain temples found here belong to the Digambara sect of Jainism.
- (d) Jain temples here were constructed during the reign of Amoghvarsha.

61. With reference to the nanofibers, consider the following statements:

1. They are much smaller than traditional fibers and have many unique properties, such as high strength and low weight.
2. They can be used for removing volatile organic compounds from the atmosphere.
3. Their high surface area and porosity make them ideal for improving the performance of energy storage devices.

How many of the statements given above are correct?

- (a) Only one
- (b) Only two
- (c) All three
- (d) None

62. With reference to the TSAT-1A, consider the following statements:

1. It is India's first indigenously privately-built sub-metre resolution surveillance satellite.
2. It was built by Tata Advanced Systems Limited in collaboration with the Indian Space Research Organization.
3. It was launched using the PSLV C-58 rocket.

How many of the statements given above are correct?

- (a) Only one
- (b) Only two
- (c) All three
- (d) None

63. The terms 'Airawat', 'Pratyush', and 'Mihir' are often seen in the news. What are they?

- (a) Mini satellites
- (b) Cloud seeding techniques
- (c) Supercomputers
- (d) Missile defence systems

64. Consider the following statements:

Statement-I: The new Kulasekarapattinam launch site will boost ISRO's attempts to enhance payload capacities for Small satellite launch vehicles.

Statement-II: The geographical location of Kulasekarapattinam allows a direct southward and smaller trajectory without having to overfly Sri Lanka

Which one of the following is correct in respect of the above statements?

- (a) Both Statement-I and Statement-II are correct and Statement-II is the correct explanation for Statement-I
- (b) Both Statement-I and Statement-II are correct and Statement-II is not the correct explanation for Statement-I
- (c) Statement-I is correct but Statement-II is incorrect
- (d) Statement-I is incorrect but Statement-II is correct

65. 'White Rabbit Technology' recently seen in the news, in the context of

- (a) enhancing the nutritional value of genetically modified crops
- (b) advanced wireless communication systems
- (c) high-precision network time synchronization in distributed systems
- (d) development of biofertilizers for sustainable agriculture

66. Consider the following statements about the Agni Prime missile:

1. The missile is capable of carrying both conventional and nuclear warheads, providing flexibility in operational use.
2. Agni Prime is a three-stage missile that uses solid propulsion technology.
3. The Agni Prime missile is heavier than all the earlier Agni series of missiles.

How many of the statements given above are correct?

- (a) Only one
- (b) Only two
- (c) All three
- (d) None

67. Arrange the following parts of the human digestive system (including vestigial parts) from top to bottom:

1. Vermiform Appendix
2. Liver
3. Stomach
4. Small Intestine

Select the correct answer using the code given below.

- (a) 1-2-3-4
- (b) 2-3-4-1
- (c) 3-2-4-1
- (d) 3-4-1-2

68. Consider the following statements:

1. Electrons move in circular orbits around the nucleus of an atom.
2. Unlike the earlier beliefs, electrons are not particles but waves.
3. Wormholes have not yet been discovered in the universe.

How many of the above statements are supported/ have been proven by quantum mechanics and modern physics?

- (a) Only one
- (b) Only two
- (c) All three
- (d) None

69. Consider the following statements about the lifecycle of stars:

1. The red giant phase marks the youngest phase of a star's life.
2. Our sun and stars of similar size and mass will die to form neutron stars.
3. The Chandrasekhar limit is the maximum mass that a stable white dwarf star can attain before it collapses into a neutron star or a black hole.

How many of the statements given above are correct?

- (a) Only one
- (b) Only two
- (c) All three
- (d) None

70. Consider the following statements about the C-Dome Défense System:

1. The C-Dome is an advanced naval version of the Iron Dome air defense system.
2. The system has been developed by Israel.
3. The C-Dome system is equipped with its own radar technology to detect any incoming threats.

How many of the statements given above are correct?

- (a) Only one
- (b) Only two
- (c) All three
- (d) None

71. Consider the following statements with reference to Jal Jeevan Mission (JJM):

1. Rashtriya Jal Jeevan Kosh which is being set up under Jal Jeevan Mission will receive funds from Water Cess for construction of water infrastructure.
2. All water conservation activities under the Natural Resource Management (NRM) component of MGNREGA can be converged with JJM.
3. It mandates that a minimum of 50% of members of village water and sanitation Committees shall be women.

How many of the statements given above are correct?

- (a) Only one
- (b) Only two
- (c) All three
- (d) None

72. Arrange the following Acts relating to women, in the chronological order of their enactment, starting from the earliest.

1. The Commission of Sati (Prevention) Act
2. The Dowry Prohibition Act
3. Protection of Women from Domestic Violence Act
4. The Immoral Traffic (Prevention) Act

Select the correct answer using the code given below.

- (a) 4-2-1-3
- (b) 2-4-1-3
- (c) 4-2-3-1
- (d) 1-2-3-4

73. Consider the following statements regarding the curative jurisdiction of the Supreme Court of India:

1. Curative jurisdiction is a judicial innovation to correct 'grave miscarriage of justice' after a final judgment is passed.
2. The concept of curative petition was first introduced in the Rupa Ashok Hurra vs Ashok Hurra case.
3. It is a part of the review jurisdiction of the Supreme Court.

How many of the statements given above are correct?

- (a) Only one
- (b) Only two
- (c) All three
- (d) None

74. The decadal census of India 2021 has been delayed for some time now. In this context, consider the following:

1. Coverage of beneficiaries under the Public Distribution System (PDS)
2. Women's representation in the Indian parliament
3. Data on migrant population and migration patterns
4. Devolution of funds on the basis of Finance Commission recommendations
5. Reservations for SCs, STs as well as Other Backward Classes.

How many of the above will be impacted by the delay in Census 2021?

- (a) Only two
- (b) Only three
- (c) Only four
- (d) All five

75. Consider the following statements:
- Statement-I: Since the enactment of the 74th Constitutional Amendment Act, elections to the urban local bodies have been invariably held at regular intervals in India.
- Statement-II: Article 243U stipulates that the duration of urban local governments is five years and that an election to constitute an urban local government should be completed “before the expiry of its duration”.
- Which one of the following is correct in respect of the above statements?
- (a) Both Statement-I and Statement-II are correct and Statement-II is the correct explanation for Statement-I
 - (b) Both Statement-I and Statement-II are correct and Statement-II is not the correct explanation for Statement-I
 - (c) Statement-I is correct but Statement-II is incorrect
 - (d) Statement-I is incorrect but Statement-II is correct
76. How many of the following strengthen the idea of cooperative federalism in India?
1. NITI Aayog
 2. Inter-State Council
 3. GST Council
- Select the correct answer using the code given below.
- (a) Only one
 - (b) Only two
 - (c) All three
 - (d) None

77. Which of the following is **not** enshrined as a Fundamental Duty in the Indian Constitution?
- (a) To promote harmony amongst all the people of India transcending linguistic diversities.
 - (b) To have compassion for living creatures.
 - (c) To strive for excellence in collective activities.
 - (d) To promote international cooperation and provide humanitarian aid.
78. Consider the following statements:
1. Mandal Commission recommendations were implemented during the tenure of Prime Minister V.P. Singh.
 2. P.V. Narsimha Rao was the first Prime Minister to head a coalition government.
 3. Manmohan Singh was the first member of Rajya Sabha to be appointed as a Prime Minister.
- How many of the statements given above are correct?
- (a) Only one
 - (b) Only two
 - (c) All three
 - (d) None
79. The Lobito rail corridors being revived through the support of the USA is in:
- (a) Africa
 - (b) South America
 - (c) North Europe
 - (d) East Europe
80. In the context of India, the State Energy Efficiency Index has been developed by which organization in partnership with the Alliance for an Energy Efficient Economy (AEEE)?
- (a) Bureau of Energy Efficiency (BEE)
 - (b) NITI Aayog
 - (c) The Energy and Resources Institute (TERI)
 - (d) Indian Institute of Science (IISc)

81. Consider the following species with reference to microalgae:

1. Microalgae are found only in marine water.
2. Seagrass is the most common type of microalgae found in water bodies as a result of eutrophication.
3. All microalgae are harmful for human consumption due to high levels of toxins present in them.

How many of the statements given above are correct?

- (a) Only one
- (b) Only two
- (c) All three
- (d) None

82. Global Summit on Extreme Heat which was held recently is a joint initiative of

- (a) International Federation of Red Cross and Red Crescent Societies and World Meteorological Organization
- (b) United States Agency for International Development and International Federation of Red Cross and Red Crescent Societies
- (c) World Health Organization and Intergovernmental Panel on Climate Change
- (d) United Nations Environment Programme and World Meteorological Organization

83. Recently, the forest officials installed GPS transmitters on the Northern River Terrapin in the Indian Sundarbans. In this context, consider the following statements with reference to Northern River Terrapin:

1. It is a species of riverine turtle endemic to India and Bangladesh.
2. It is omnivorous in nature.
3. It is the smallest turtle species found in Asia.

How many of the statements given above are correct?

- (a) Only one
- (b) Only two
- (c) All three
- (d) None

84. In the context of ecology, the term 'brumation' refers to a

- (a) form of migration in birds
- (b) type of mating behavior in mammals
- (c) period of dormancy in reptiles
- (d) process of plant growth during winter

85. Recently, which of the following Indian states has become the first state to set up a Blended Finance Facility in partnership with the World Bank?

- (a) Himachal Pradesh
- (b) Goa
- (c) Tamil Nadu
- (d) Karnataka

- 86.** Consider the following statements:
Statement-I: During the first half of 1860s, raw cotton exports from India steadily declined.
Statement-II: The American Civil War broke out in 1861.
Which one of the following is correct in respect of the above statements?
- Both Statement-I and Statement-II are correct and Statement-II is the correct explanation for Statement-I
 - Both Statement-I and Statement-II are correct and Statement-II is not the correct explanation for Statement-I
 - Statement-I is correct but Statement-II is incorrect
 - Statement-I is incorrect but Statement-II is correct
- 87.** Consider the following statements:
Statement-I: The Pitt's India Act provided a parliamentary prohibition on imperial expansion but Lord Wellesley reinitiated the policy of imperial expansion in India through 'subsidiary alliance'.
Statement-II: Napoleonic invasion of Egypt in 1798 was used as a pretext by Wellesley to keep India safe from the French.
Which one of the following is correct in respect of the above statements?
- Both statement-I and statement-II are correct and statement-II is the correct explanation for statement-I
 - Both statement-I and statement-II are correct and statement-II is not the correct explanation for statement-I
 - Statement-I is correct but statement-II is not correct
 - Statement-I is not correct but the statement-II is correct
- 88.** In the context of Indian history, the Risley Papers revolved around:
- the philosophy and motivations behind the revolutionary movement during 1900s.
 - the transfer plan for the partition of Bengal and Central Province.
 - the emergence of socialist activities in India in first half of twentieth century.
 - the corruption charges levied on British officers in India.
- 89.** Consider the following statements regarding the system of Indian indentured labour during colonial India:
- Indentured labour is a form of contract labour in which labourers enter into an official agreement with their employer to work for an indefinite time.
 - Indian Indentured labourers were employed only in British colonies outside India.
 - Suriname was the first British colony to receive indentured labourers.
- How many of the statements given above are correct?
- Only one
 - Only two
 - All three
 - None
- 90.** Recently, there was a demand to include the Kui language in the eighth schedule of the Indian Constitution. Which of the following statements given below regarding the Kui language is correct?
- It belongs to the Tibeto-Burman Family of languages and is spoken by the Mizo people.
 - It belongs to the Indo-Aryan Family of languages and is spoken by the Katkari Tribes of Maharashtra.
 - It belongs to the Dravidian Family of languages and is spoken by the Kandh Tribes of Odisha.
 - It belongs to the Austro-Asiatic Family of languages and is spoken by the Khasi tribe of Meghalaya.

91. Which of the following statements is correct regarding the Khalimpur Copper Plate Inscription?

- (a) It was a pre-Mauryan inscription that talked about the famine relief efforts.
- (b) It belonged to the Pala dynasty that talked about how King Gopala was elected by the people to replace an unfit ruler.
- (c) It was a Jain inscription that depicted the Western Ganga Dynasty King Ereganga Nitimarga I attained death through the practice of Sallekhana.
- (d) It was the oldest known Kannada language inscription showing the earliest evidence of the usage of Kannada as an administrative language.

92. Consider the following statements regarding the city of Fatehpur Sikri:

- 1. It was built by Akbar.
- 2. The tomb of Sheikh Salim Chisti is the only building in the city that uses white marble.
- 3. Most buildings here follow the trabeate style of architecture.

How many of the statements given above are correct?

- (a) Only one
- (b) Only two
- (c) All three
- (d) None

93. In the context of India's economy, CAROTAR, 2020 deals with

- (a) verifying the rules of origin under the free trade agreements (FTAs).
- (b) regulation and inspection of generic pharma industries.
- (c) domestic content requirement in manufacturing of solar panels.
- (d) tax treaties signed between countries.

94. Recently, the International Sugar Organization (ISO) was seen in the news. In this context, consider the following statements:

- 1. India assumed the chair of ISO for the year 2024.
- 2. ISO is an intergovernmental body devoted to improving the world's sugar market conditions.
- 3. India accounts for 25 percent of the total global sugar consumption.

How many of the statements given above are correct?

- (a) Only one
- (b) Only two
- (c) All three
- (d) None

95. India is a founding member of which of the following institutions of the World Bank Group?

- 1. International Bank for Reconstruction and Development (IBRD)
- 2. International Development Association (IDA)
- 3. International Finance Corporation (IFC)
- 4. Multilateral Investment Guarantee Agency (MIGA)
- 5. International Centre for Settlement of Investment Disputes (ICSID)

Select the correct answer using the code given below.

- (a) 1, 2 and 3 only
- (b) 2 and 4 only
- (c) 1, 3 and 4 only
- (d) 1, 2, 3, 4 and 5

96. The term 'Greenium' is prevalent in the energy market. In this context, which of the following correctly defines the logic behind 'Greenium'?
- Green bond investors may willingly accept a lower yield.
 - Extra taxes should be imposed by governments on companies creating environmental pollution.
 - A new currency can be created backed by environmentally-consistent assets.
 - Alcohol and hydrogen-based fuels can be promoted.
97. Consider the following instruments:
- External Commercial Borrowings (ECBs)
 - Depository Receipts (DRs)
 - Participatory Notes (P-notes)
- Using how many of the instruments given above can foreign investors invest in the Indian market?
- Only one
 - Only two
 - All three
 - None
98. This instrument is used for temporary mismatches in the cash flow of the Government of India. It was introduced in 2010 after the consultation of the Government of India with the Reserve Bank of India. Which of the following instruments has been described in the passage given above?
- Ways and Means Advance
 - Treasury-bills
 - Special Zero Coupon Recapitalisation Bonds
 - Cash Management Bills
99. The term Neerakshi, recently in the news, was seen in the context of:
- Breakthroughs in ocean water purification technology.
 - Developments in renewable energy sources.
 - Innovations in biodegradable materials.
 - Autonomous underwater vehicle designed to detect mines.
100. Consider the following statements:
- Statement I: Dr. Homi J Bhabha, father of India's nuclear programme, devised a three-stage nuclear power programme in 1950s.
- Statement II: India possesses one of the largest shares of global thorium reserves, while it holds only about 2-3% of the world's uranium reserves.
- Which one of the following is correct in respect of the above statements?
- Both Statement-I and Statement-II are correct and Statement-II is the correct explanation for Statement-I
 - Both Statement-I and Statement-II are correct and Statement-II is not the correct explanation for Statement-I
 - Statement-I is correct Statement-II is incorrect
 - Statement-I incorrect but Statement-II is correct



ANSWERS & EXPLANATION

GENERAL STUDIES (P) 2023 – ABHYAAS TEST – 4361

Q 1.B

- The Central Bureau of Investigation (CBI) is a federal investigative agency in India. It is governed by the Delhi Special Police Establishment Act (DPSEA). The DSPE Act establishes the Central Bureau of Investigation (CBI) as a specialized agency for investigating certain offenses.
- **Delhi Special Police Establishment (DSPE) Act, 1946: Section 6 of the Act provides for the granting of 'general consent' by a state to the CBI to investigate any offense specified in the Schedule of the Act. Hence, statement 2 is correct.**
- **General consent is a form of legal authority granted by a state government to the Central Bureau of Investigation (CBI) and other agencies covered under the DSPE Act, 1946. This consent allows these agencies to exercise their powers and functions within the respective state's jurisdiction.**
 - **Key Points:**
 - ✓ **Approval by State Government:** General consent is granted by the state government concerned through a notification. **Hence, statement 1 is not correct.**
 - ✓ **Scope of Authority:** The consent authorizes the CBI and other agencies to investigate and prosecute offenses under various central laws within the state's boundaries.
 - ✓ **Withdrawal of Consent:** **The state government can withdraw its general consent at any time by issuing a notification.** In such cases, the CBI and other agencies would lose their authority to operate within that state.
 - **Withdrawal of General Consent**
 - ✓ **Individual states have the power to grant or withdraw general consent** for the Central Bureau of Investigation (CBI) to investigate cases within their territory.
 - ✓ Withdrawal of general consent means that the CBI can no longer initiate new investigations in that state without obtaining specific approval.
 - ✓ **Even after the withdrawal of general consent, the CBI can continue to investigate old cases that were initiated before the withdrawal. Hence, statement 3 is correct.**

Q 2.D

- There are 12 other States that enjoy special powers under Article 371 of the Constitution. These include Arunachal Pradesh, Gujarat, Maharashtra, Manipur, Nagaland and Sikkim.
- Special provisions provided to other States are listed in Articles 371 (A-J) in Part XXI of the Constitution, which deals with “temporary, transitional and special powers for certain States.” While Articles 370 and 371 have been a part of the Constitution since 1950, Articles 371 (A-J) were incorporated in subsequent years through amendments.
- **Article 371 deals with the States of Maharashtra and Gujarat. As per the provision, the Governor of Maharashtra has a special responsibility to establish separate development boards for Vidarbha, Marathwada, and the rest of the State.** The Governor of Gujarat has a similar responsibility towards Saurashtra, Kutch and the rest of Gujarat. These responsibilities cover equitable allocation of funds for development expenditure, providing adequate facilities for technical education and vocational training, and adequate opportunities for employment in services under the control of the State Government.
- Article 371A was made part of the Constitution in 1962 after a deal was struck between the Centre and the Naga People's Convention creating the State of Nagaland. **Under the provision, no Act of Parliament about the religious or social practices of the Nagas, their customary law and procedure, including civil and criminal justice matters, and ownership or transfer of land and resources will apply to Nagaland, unless the State's Legislative Assembly passes a resolution to do so.** It further gives the

Governor a 'special responsibility' regarding law and order in the State. Further, non-residents can't buy land in Nagaland.

- Articles 371D and E include special provisions for Andhra Pradesh. The President can pass an order to provide equitable opportunities and facilities to people belonging to different parts of Andhra Pradesh in public employment and education. **It empowers the President to specify any part of the State as a 'local area' for the creation of local cadres in various classes of employment and allotment of civil posts. It provides for the constitution of an Administrative Tribunal for Andhra Pradesh. Only the Supreme Court has any power of superintendence over this tribunal, as per the provision. Article 371E, meanwhile, deals with the establishment of a university in the State.**
- Similarly, Article 371J accords special status to the Hyderabad-Karnataka Region (Kalyana Karnataka), and provides for the establishment of a separate development board for the area. It provides for equal allocation of funds and equal opportunities for locals in employment and education. As per the provision, a proportion of seats must be reserved for individuals from the region in educational institutions and vocational training institutes.
- Article 371F takes into account the unique status of Sikkim after its merger with India in 1975 and grants protection to existing laws. It states that the State Legislative Assembly shall consist of at least 30 members. The Governor has a special responsibility for the maintenance of peace and equitable arrangements for ensuring the social and economic advancement of different sections. **Only descendants of Sikkim subjects (those who lived in the State before its merger with India) whose names were mentioned in the 1961 register have the right to own land and get State government jobs in Sikkim. They were also exempted from paying income tax.**
- Hence, option (d) is the correct answer.

Q 3.C

- The Supreme Court is at the apex of the Indian judicial system. As the guardian of the Constitution, it is the primary duty of the SC to uphold the fundamental rights of citizens and protect their liberties. Whenever a matter of law arises that requires a provision or provision of the Constitution to be interpreted, or there is a "significant legal question", it is required to be decided by a Bench involving a minimum of five judges of the Supreme Court. Such a Bench is called a Constitution Bench.
- **Constitution Bench can be formed in the following cases:**
 - Every case involving a substantial question of law as to the interpretation of the Constitution under Article 145(3) or any Reference made under Article 143 of the Constitution shall be heard by a Bench consisting of not less than five Judges.
 - Every petition calling in question the election of the President and Vice-President under Article 71 of the Constitution read with Part III of the Presidential and Vice-Presidential Elections Act, 1952, shall be posted before a Bench of five Judges.
 - The Chief Justice may, from time to time, constitute a Bench consisting of five or more Judges for the purpose of hearing any other cause, appeal or matter.
 - Hence, option (c) is the correct answer.

Q 4.B

- Historically, mediation has been commonplace in the Indian context, with the Gram Panchayats (Village Council) serving as the mediator at the community level to help resolve disputes of all kinds. The concept of mediation is not new—it has always been available as an alternative dispute resolution (ADR) mechanism for parties in a dispute. Mediation also stands out as a mechanism that can be tailored to the needs of the disputants, offering a highly flexible and adaptable process for dispute resolution. Therefore, the recent Mediation Act, 2023, is seen as progressive by the industry and the legal fraternity.
- Features of the Mediation Act:
 - The Mediation Act recognizes the mediation process where irrespective of any mediation agreement, the parties voluntarily undertake for settlement of disputes prior to filing of any case of civil or commercial nature before the court.
 - **It provides for voluntary mediation before the institution of legal proceedings in civil and criminal matters. Hence statement 1 is not correct.**
 - **The Central Government shall, by notification, establish for the purposes of this Act, a Council to be known as the Mediation Council of India to perform the duties and discharge the functions under this Act. Hence statement 2 is correct.**

- Mediation proceedings must be typically conducted within the jurisdiction of the competent court or tribunal responsible for deciding the disputed subject matter. However, parties can agree to hold mediation at another location by mutual consent. Additionally, parties have the option to conduct mediation online, provided they formally agree to it in writing.
- The mediation process must be concluded within 120 days from the mediator's first appearance, with an option to extend it by up to 60 days if all parties agree. Similar changes are proposed in the Commercial Courts Act 2015, to align pre-litigation mediation timelines.
- **The act contains disputes not fit for mediation that includes disputes relating to claims against minors or persons of unsound mind, involving criminal proceedings, affecting the rights of third parties. Hence statement 3 is correct.**
- The act will apply to mediations conducted in India involving only domestic parties, at least one foreign party, and relating to a commercial dispute.

Q 5.A

- The parliament of India uses its constituent powers to amend the constitution under Article 368, but the amendment powers were limited by the Kesavanand Bharati judgment which propounded the basic structure doctrine.
- Upholding the significance of the doctrine, the Supreme Court in the Minerva Mills case held that since the Constitution had conferred a limited amending power on the Parliament, the Parliament cannot under the exercise of that limited power enlarge that very power into an absolute power. **Indeed, a limited amending power is one of the basic features of the Constitution and, therefore, the limitations on that power cannot be destroyed.**
- **In other words, Parliament cannot, under Article 368, expand its amending power so as to acquire for itself the right to repeal or abrogate the Constitution or to destroy its basic features.** The donee of a limited power cannot by the exercise of that power convert the limited power into an unlimited one.
- Hence, Statements I and II are correct and Statement II is the correct explanation of Statement I

Q 6.C

- Article 39(b) states that the state shall direct policy towards securing that the ownership and control of the material resources of the community are so distributed as best to serve the common good. This involves equitable redistribution of material resources of the community.
- **The 25th amendment to the constitution inserted new Article 31C which states that no law which seeks to implement the socialistic Directive Principles specified in Article 39 (b) and (c) shall be void on the ground of contravention of the Fundamental Rights conferred by Article 14 (equality before law and equal protection of laws), Article 19 (protection of six rights in respect of speech, assembly, movement, etc).**
- **Thus, only these two articles under fundamental rights are made subservient to pursue the policies under Article 39(b) and 39(c). In general, the fundamental rights are not subservient to the Directive principles. Hence, statement I is correct but statement II is not correct.**

Q 7.C

- **The Supreme Court of India has held that the people have a right to be free from the adverse effects of climate change within the ambit of fundamental rights, according to a judgment in April, 2024.**
- **Articles 14 and 21 of the Constitution of India guarantee the fundamental rights to equality and life respectively. The right to a clean environment has earlier been recognized as a fundamental right within the ambit of the right to life by the Court in a plethora of decisions.**
- The Apex Court passed its judgment in a case titled **M K Ranjitsinh & Ors versus Union of India & Ors** regarding the protection of two critically endangered bird species on the IUCN Red List – the great Indian bustard (GIB) and the lesser florican. Both are scheduled species listed under Part III of Schedule I of the Wild Life (Protection) Act, 1972.
- Other Provisions of the Constitution: **Article 48A** of the Constitution provides that the State shall endeavour to protect and improve the environment and to safeguard the forests and wildlife of the country. **Clause (g) of Article 51A** stipulates that it shall be the duty of every citizen of India to protect and improve the natural environment including forests, lakes, rivers and wild life, and to have compassion for living creatures. Although these are not justiciable provisions of the Constitution, they are indications that the Constitution recognises the importance of the natural world.

- In **MC Mehta vs Kamal Nath, 2000**, the Court held that **Articles 48A and 51A(g) must be interpreted in light of Article 21. Any disturbance of the basic environment elements, namely air, water and soil, which are necessary for “life”, would be hazardous to “life” within the meaning of Article 21 of the Constitution.**
- **The right to equality may also be violated** in ways that are more difficult to remedy. For example, a person living in say, the Lakshadweep Islands, will be in a disadvantageous position compared to person living in say, Madhya Pradesh when sea levels rise and oceanic problems ensue. Similarly, forest dwellers or tribal and indigenous communities are at a high risk of losing not only their homes but also their culture, which is inextricably intertwined with the places they live in and the resources of that place.
- This acknowledgement of human rights in the context of climate change is underscored in the preamble of the Paris Agreement, which recognizes the interconnection between climate change and various human rights, including the right to health, indigenous rights, gender equality, and the right to development.
- **Hence, option (c) is the correct answer.**

Q 8.C

- While a proclamation of a National Emergency is in force, the normal fabric of the Centre-state relations undergoes a basic change. This can be studied under three heads, namely, executive, legislative and Financial.
 - **During a national emergency, the executive power of the Centre extends to directing any state regarding the manner in which its executive power is to be exercised.** In normal times, the Centre can give executive directions to a state only on certain specified matters. However, during a national emergency, the Centre becomes entitled to give executive directions to a state on ‘any’ matter. Thus, the state governments are brought under the complete control of the Centre, though they are not suspended. **Hence, option (c) is the correct answer.**
- **During a national emergency, the Parliament becomes empowered to make laws on any subject mentioned in the State List.** Although the legislative power of a state legislature is not suspended, it becomes subject to the overriding power of the Parliament. Thus, the normal distribution of the legislative powers between the Centre and states is suspended, though the state Legislatures are not suspended. In brief, the Constitution becomes unitary rather than federal.
- **While a proclamation of national emergency is in operation, the President can modify the constitutional distribution of revenues between the centre and the states.** This means that the president can either reduce or cancel the transfer of finances from Centre to the states. Such modification continues till the end of the financial year in which the Emergency ceases to operate. Also, every such order of the President has to be laid before both the Houses of Parliament.
- While a proclamation of National Emergency is in operation, the life of the Lok Sabha may be extended beyond its normal term (five years) by a law of Parliament for one year at a time (for any length of time). However, this extension cannot continue beyond a period of six months after the emergency has ceased to operate. **Similarly, the Parliament may extend the normal tenure of a state legislative assembly (five years) by one year each time (for any length of time) during a national emergency, subject to a maximum period of six months after the Emergency has ceased to operate.**

Q 9.B

- **A three-member body, the CCRGA was formed by the Union Ministry of Information and Broadcasting in April, 2016, on the directions of the Supreme Court in its judgment in the Common Cause vs. Union of India case in May, 2015.** The body is set up to regulate the content of Central and State government advertisements in all media platforms. The SC had also mandated states to constitute their own respective bodies. While some States have set up committees to regulate public advertising content, some have given consent to the CCRGA to monitor their advertisements. **Hence, statement 1 is not correct.**
- The SC, in its order had also issued a set of guidelines for public-funded advertising by governments. Some of the guidelines mentioned include that government advertising should maintain political neutrality and avoid glorification of political personalities or projecting a positive impression of the party in power or a negative impression of parties critical of the government. They also should not be used at patronising media houses.
- **The CCRGA examines complaints received by it on the basis of suo-motu cognizance regarding violation of content regulation guidelines for Government Advertisements and recommends remedial action to the concerned authorities. Hence, statement 2 is correct.**

Q 10.B

- **Information Technology Act, 2000** provide a legal framework for electronic transactions, digital signatures, and other aspects of information technology (IT) to facilitate e-commerce and e-governance in India.
- ITA 2000 has undergone several amendments, including the Information Technology (Amendment) Act, 2008, to address evolving technological advancements and cyber threats.
- **Section 69: Direction for Interception, Monitoring or Decryption of Information**
 - This provision empowers both the Central and State governments (Hence, statement 1 is correct.) with the authority to issue directives for the following purposes:
 - ✓ **Interception:** To intercept electronic communications passing through or being processed in a computer system.
 - ✓ **Monitoring:** To monitor or view electronic communications passing through or being processed in a computer system.
 - ✓ **Decryption:** To decrypt encrypted electronic communications stored or transmitted through a computer system.
- In the context of cyberspace, "intermediaries" refer to entities that facilitate the transmission, storage, or retrieval of electronic records. These include platforms such as internet service providers (ISPs), search engines, social media websites, and e-commerce portals.
- **The ITA explicitly covers all "intermediaries" within its scope. Hence, statement 2 is correct.**
- **Section 69A empowers the Central Government of India to order the blocking of public access to certain information on the internet for reasons of national security, sovereignty, or public order.**
- **Procedure for Blocking:**
 - The request for blocking must be made in writing and not orally (Hence, statement 3 is not correct.) and state the specific reasons for the request.
 - The government must specify the period for which the information should be blocked.
 - The government may direct any agency of the government or any intermediary, such as an internet service provider, to block access to the specified information.

Q 11.D

- **The shorter route connecting Mumbai to Saint-Petersburg is the route for the International North-South Transport Corridor (INSTC). Proposed in 2000, the International North-South Transport Corridor (INSTC) was initially meant to transport goods from India to Russia via Iran as an alternative to the conventional Suez Canal route. Hence, option (d) is the correct answer.**
- Apart from India, Russia and Iran, 10 more countries — Azerbaijan, Armenia, Kazakhstan, Kyrgyzstan, Tajikistan, Turkey, Ukraine, Belarus, Oman, and Syria — signed onto the project later but implementation waned. In July 2022, the first major commercial consignment through the INSTC was sent from Russia to India. The corridor is again in focus due to the initiative of India, Russia and Iran.

Q 12.C

- The first Ministerial meeting of India- European Union Trade and Technology Council (TTC) took place in Brussels, Belgium on 16th May 2023. Union Minister for Commerce & Industry, Textiles, Consumer Affairs, Food & Public Distribution, Shri Piyush Goyal was one of the co-chairs along with External Affairs Minister (EAM) and Union Minister of State for Electronics and Information Technology (MeitY). The EU side was co-chaired by Executive Vice Presidents Mr. Dombrovskis and Ms. Vestager.
- **The formation of the TTC was announced by the Prime Minister, Shri Narendra Modi and H.E. Ms. Ursula von der Leyen, President of the European Commission in New Delhi in April, 2022 with the objective of creating a High-level coordination platform to tackle strategic challenges at the nexus of trade, trusted technology and security. Hence, statement I is correct.**
- **The EU and India have set up the TTC as a coordination platform to address key trade, trusted technology and security challenges. The TTC is a key forum to deepen the strategic partnership on trade and technology between the two partners. Geostrategic challenges have reinforced the EU and India's common interest in ensuring security, prosperity and sustainable development based on shared values. Hence, statement II is not correct.**
- It is a forum to prioritise their partnership through cooperation in digital governance, green and clean energy technologies, and resilient value chains, trade and investment. The Council is a response to increasing great-power competition and a desire on both sides to bolster their respective strategic

autonomies, with the potential to have a significant impact on the EU's ambitions in the political, economic and technology spheres.

- The bilateral forum includes three working groups on cooperation in digital governance and connectivity, green and clean energy technologies, and resilient value chains, trade and investment.

Q 13.C

- **Recently, Gabon was in news due to a military coup. A military coup thrust the Central African nation of Gabon into turmoil recently**, unseating the president – whose family had held power for more than half a century – just minutes after he was named the winner of a contested election. Ousted President Ali Bongo Ondimba, also known as Ali Bongo, has faced accusations of election fraud and corruption since he began ruling the oil-rich but poverty-stricken nation nearly 14 years ago. Following the coup, residents in the country's capital were seen celebrating and embracing soldiers on the street. **Hence, pair 2 is correctly matched.**
- **Guyana and Venezuela are currently engaged in a border dispute.** The current dispute is over the Essequibo region, a massive area of 61,600 square miles with a population of 125,000 people, the large majority of whom speak English. The region is administered by Guyana but is now being claimed by Venezuela. The Essequibo is notable for two reasons: it is massive, making up roughly two-thirds of the entire land area of Guyana, even larger than the size of England; and it is rich in natural resources, primarily petroleum. **Hence, pair 4 is not correctly matched.**
- **European Union recently announced new funding of €800,000 to Sudan, to contribute to the response to the cholera epidemic that has been claiming lives in the country since September 2023.** World Health organization had reported the outbreak of Cholera in the African country recently. **Hence, pair 2 is correct matched.**
- **Kosovo is a small landlocked region in the southwest of Serbia sharing borders with Albania, Montenegro, North Macedonia. Many serbs consider Kosovo to be the birthplace of the nation. Serbs and Albanians are ethnicities who have been living in this region for centuries. Serbs are Eastern Orthodox Christians, while the Albanians in Kosovo are majority Muslims. Other ethnic groups, such as the Bosnians and the Turks, are minority populations. Serbs are in the majority in Serbia while Albanians are in the majority in the Kosovo region. Kosovo's majority ethnic Albanians view Kosovo as belonging to them, and accuse Serbia of occupation and repression. Kosovo had declared independence in 2008 but is not recognized by Serbia as an independent nation. Hence, pair 1 is correctly matched.**

Q 14.C

- Recently, a 2-day Conference of National Human Rights Institutions of Asia Pacific known as the Asia Pacific Forum (APF) of Human Rights concluded by adopting the 'Delhi Declaration'.
- **The APF has created a strong and united platform that brings together National Human Rights Institutions (NHRIs) from all corners of the Asia Pacific to address some of the most serious human rights challenges in our region.** It works closely with the Office of the High Commissioner for Human Rights (OHCHR) and the United Nations Development Programme (UNDP) to promote the establishment and strengthening of national human rights institutions in the Asia Pacific. **Hence, statement 1 is correct.**
- It collaborates with two of the UN's most significant agencies working to advance the human rights of women and girls - UN Women and the United Nations Population Fund - in order to provide tailored support to our members.
- It organizes APF annual and biennial conferences and also provides advice and support to APF members seeking accreditation or re-accreditation with the Global Alliance of National Human Rights Institutions.
- **The APF was established in 1996 with five members. Today, the membership includes 26 National Human Rights Institutions from all corners of the Asia Pacific including the National Human Rights Commission of India. Hence, statement 2 is correct.**

Q 15.B

- **Recent context:** The **flightless Aldabra rail** went extinct over 130,000 years ago, but the species then reappeared via iterative evolution.
 - Aldabra rail is a subspecies of the white-throated rail and is the only living flightless bird in the Indian Ocean.
- **New research has found that it had once gone extinct, but rose from the dead thanks to a rare process called “iterative evolution”.**

- **Iterative evolution means the repeated evolution of similar or parallel structures from the same ancestor but at different times.**
- A 2019 study published in the Zoological Journal of the Linnean Society examined the fossil record of rails in Aldabra and found evidence of a flightless rail on the atoll from before it was submerged beneath the waves 136,000 years ago.
- Study found that on two occasions, separated by tens of thousands of years, a rail species was able to colonise an island called Aldabra and subsequently became flightless on both occasions. The last surviving colony is still found on the island. In losing its ability to fly once again, the Aldabra rail has essentially evolved twice, rising from the dead through a process called iterative evolution where a species goes extinct, but then another comes along and evolves the same traits to become near-identical to the one that was lost.
- This is the first time that iterative evolution has been seen in rails and is one of the most significant in bird records.
- **Hence option (b) is the correct answer.**

Q 16.C

- The Centre has notified a set of rules called the Captive Elephant (Transfer or Transport) Rules, 2024 that liberalise the conditions under which elephants may be transferred within or between states.
- The circumstances under which captive elephants can be transferred are when an owner is no longer in a position to maintain the elephant, the elephant will likely have a better upkeep than in the present circumstances; or when a state's Chief Wildlife Warden "deems it fit and proper" in the circumstances of the case for better upkeep of the elephant.
- **Before a transfer within the state, an elephant's health has to be ratified by a veterinarian and the Deputy Conservator of Forests must establish that the animal's current habitat and prospective habitat are suitable. Hence statement 2 is correct.**
 - The Chief Wildlife Warden on receipt of such documents may choose to reject or approve the transfer.
- **If the transfer involves moving the elephant outside of a state, similar conditions apply. Before a transfer is effected, the "genetic profile" of the elephant has to be registered with the Ministry of Environment, Forest and Climate Change. Hence statement 1 is correct.**
- **The elephant being transferred ought to be accompanied by a mahout and an elephant assistant. Hence statement 3 is correct.**
- Health certificate from a veterinary practitioner to the effect that the elephant is fit for transport and is not showing any sign of musth or infectious or contagious disease, is to be obtained.
- The transport shall be carried out after the mandatory quarantine period as advised by the veterinary practitioner is over, in case of contagious disease.
- The elephant shall be properly fed and given water before loading.
- Necessary arrangements shall be made to provide food and water to the elephant en route.
- Tranquilizers/sedatives shall be used to control nervous or temperamental elephants upon prescription by the veterinary practitioner.

Q 17.B

- The United Nations declared 2024 the International Year of Camelids (IYC 2024). The Year will highlight how camelids are key to the livelihoods of millions of households in hostile environments across over 90 countries, particularly Indigenous Peoples and local communities.
- From alpacas to Bactrian camels, dromedaries, guanacos, llamas, and vicuñas, camelids contribute to food security, nutrition and economic growth as well as holding a strong cultural and social significance for communities across the world.
- **There are two species of camel: the Bactrian camel (*Camelus bactrianus*) and the dromedary camel (*Camelus dromedarius*). About 90% of the world's camels are dromedary camels, also known as the Arabian camel. Hence statement 3 is correct.**
- All existing dromedary camels are domesticated. Meanwhile, there are two types of Bactrian camels: wild and domesticated.
- The Bactrian camel is 10 to 11.5 feet long, 5.2 to 5.9 feet tall at shoulder height, and weighs 990 to 1,100 pounds. Meanwhile, the dromedary camel is 7.2 to 11.2 feet long, 5.9 to 6.6 feet tall at the shoulder height, and weighs 880 to 1,320 pounds.

- **One of the camel's most notable physical traits is its hump or humps. While the dromedary camel has one hump, the Bactrian camel has two humps. They store fat in these humps, which later can be used as an energy source. Hence statement 1 is correct.**
- They have other features that allow them to survive in the desert, such as double rows of extra-long eyelashes to help keep sand out of the eyes. They are also able to close their nostrils to keep out sand.
- Bactrian camels are native to the Gobi Desert in China and the Bactrian steppes of Mongolia. Domesticated dromedary camels are found throughout desert areas in North Africa and the Middle East. A feral population of dromedary camels lives in Australia.
 - **A small population of bactrian camels exists in the Nubra valley of Ladakh. Hence statement 2 is not correct.**
- Camels are social animals that live in groups, called herds. The herd consists of a dominant adult male, females, and their young

Q 18.A

- Dolphins are highly social marine mammals known for their cooperative behaviors, including hunting. Cooperative hunting involves dolphins working together in coordinated efforts to increase their hunting success.
 - **One notable strategy employed by some dolphin species, such as bottlenose dolphins, involves using bubble nets to catch prey. In this behavior, dolphins blow a ring of bubbles underwater to create a circular barrier or net around a school of fish. Hence statement-I is correct.**
 - This bubble net effectively corrals the fish into a concentrated area, making it easier for the dolphins to capture them.
- The reasoning behind cooperative hunting in dolphins is that it enhances their ability to capture prey. Dolphins are highly adaptable predators, and cooperative hunting provides several advantages.
 - **By working together, dolphins can corral and concentrate prey, making it easier for individual dolphins to capture fish within the confined area of the bubble net. This strategy allows them to conserve energy while maximizing hunting success. Hence statement-II is correct and it is correct explanation of statement-II.**
- Moreover, cooperative hunting among dolphins often involves sophisticated coordination and communication among group members. Dolphins can coordinate their actions to encircle and trap fish effectively, displaying complex social behaviors and communication skills.
- **Hence option (a) is the correct answer.**

Q 19.B

- Bottom-up control and top-down control are two fundamental concepts in ecology that describe different mechanisms by which ecosystems are regulated.
- **Bottom-up control:**
 - **Bottom-up control refers to the influence of resource availability and productivity at the lower trophic levels of an ecosystem on the abundance and dynamics of higher trophic levels.**
 - **In bottom-up control, changes in the availability of resources, such as nutrients, light, and primary productivity (e.g., plants), directly affect the abundance and distribution of consumers (e.g., herbivores) and subsequently impact higher trophic levels (e.g., carnivores).**
 - For example, in a terrestrial ecosystem, an increase in nutrient availability or primary productivity can lead to an increase in plant biomass. This abundance of plants can support a larger population of herbivores, which in turn may support a larger population of predators that feed on those herbivores.
 - **Therefore, bottom-up control emphasizes the importance of resource availability and productivity in shaping ecosystem structure and dynamics.**
- **Top-down control:**
 - **Top-down control refers to the influence of higher trophic levels, particularly predators, on the abundance and dynamics of lower trophic levels in an ecosystem.**
 - **In top-down control, predators regulate the abundance of prey species, indirectly influencing the populations of lower trophic levels by controlling the number of individuals they consume.**
 - For example, in a marine ecosystem, the abundance of zooplankton (prey) may be regulated by the population size of their predators, such as fish. When fish populations increase due to favorable conditions or reduced predation pressure, they may consume more zooplankton, leading to a decrease in zooplankton abundance.

- As a result of this predation pressure, the abundance of primary producers (e.g., phytoplankton) may increase, as they experience less grazing pressure from herbivorous zooplankton.
- **Therefore, top-down control emphasizes the importance of predator-prey interactions and trophic dynamics in shaping ecosystem structure and function.**
- **Hence option (b) is the correct answer.**

Q 20.C

- **Recent Context:** In climate push, German chemical maker swaps oil for sugar to manufacture "aniline", a chemical used in making foams — used widely in mattresses and armchairs, as well as building insulation.
- **Aniline is a colourless to brown, oily liquid which darkens on exposure to air and light.**
 - It has a characteristic amine odour and burning taste.
 - Hygroscopic. Moderately soluble in water.
 - Aniline is miscible with alcohol, benzene, chloroform, carbon tetrachloride, acetone, and most organic solvents.
 - It is combustible. The substance decomposes on heating at temperatures above 190°C, or on burning producing toxic and corrosive fumes (ammonia, nitrogen oxides, and carbon monoxide) and flammable vapours.
- Aniline can be formed from the breakdown of certain pollutants found in outdoor air, from **the burning of plastics, or from burning tobacco**. Airborne exposure to aniline may occur from breathing contaminated air, from smoking tobacco or proximity to someone who is smoking, or from being near industrial sources that use large quantities of aniline. **Hence options 1 and 2 are correct.**
 - Occupational exposure to aniline could occur in industries that use aniline to make other chemicals.
 - Small amounts of aniline may be found in some foods, such as corn, grains, rhubarb, **apples, beans, and rapeseed cake** (animal feed). Aniline has also been found as a volatile component of **black tea**. Aniline has been detected in drinking water and has also been found in surface water.
- **Aniline is used in rubber accelerators and anti-oxidants, synthetic dyes** and intermediates, photographic chemicals, as isocyanates for urethane foams, in pharmaceuticals, explosives, petroleum refining; and in production of diphenylamine, phenolics, herbicides and fungicides. Aniline, an organic base used to make dyes, drugs, explosives, plastics, and photographic and rubber chemicals. **Hence options 3 and 4 are correct.**
 - **Aniline forms the basis for the modern synthetic dye industry.** Aniline dyes are known for their wide range of bright colours that do not fade (unlike many natural dyes).
- **Short-term exposure** to the substance irritates the eyes, and may cause effects on the blood: methaemoglobinaemia, resulting in cyanosis, brain damage and kidney failure. It is potentially lethal to humans.
 - **Long-term or repeated exposure** may have effects on the liver, kidneys, blood and spleen. Aniline (benzenamine) is a probable carcinogen.
 - **It does not remain in the body due to its breakdown and removal.** Aniline is a possible reproductive hazard.

Q 21.D

- **Rugra, a variety of edible mushroom grows naturally in some places of Jharkhand. So this mushroom is indigenous to Jharkhand. Rugra is washed well before cooking to remove the coating of soil.**
 - **Thus it is not poisonous and does not have any bioluminescent properties. Hence both statements 1 and 2 are not correct.**
- Rugra is not only tasty but also constitutes higher protein, vitamins and minerals than other mushroom with almost zero carbohydrates.
- Unlike other species of Mushroom which grow above the ground, it is found below the ground, specifically under Sal tree. The dense Sal forest of Bundu, Tamar, Pithoria, Chaibasa, Lohardaga and some other areas in and around the state capital Ranchi exclusively provides ambient moist and humid condition to grow this mushroom.
- A very interesting fact about Rugra is that it is not cultivated in any region commercially. It grows naturally well on the onset of the first monsoon around Sal tree.
- Rugra does not have seeds but has powdery basidiospores. When it falls on moisture in the ground, it forms fibre (mycelium). Then, when the mycelium is formed, it forms the rugra.

Q 22.B

- The Collaborative Partnership on Forests (CPF), a partnership of 16 global organisations chaired by the United Nations Food and Agriculture Organization (FAO), has launched the Joint Call to Action for Forests towards 2030.
 - The Call to Action for Forests Towards 2030 has four focal areas: Implementation and action; data, science and innovation; finance for forests; and communication and awareness-raising.
- **The Collaborative Partnership on Forests (CPF) is an innovative voluntary interagency partnership on forests that was established in April 2001 in response to an invitation issued in the resolution 2000/35 by the Economic and Social Council of the UN (ECOSOC) that established the International Arrangement on Forests and the UN Forum on Forests (UNFF). Hence statement 1 is not correct.**
 - The Partnership is currently comprised of fifteen international organizations, institutions and secretariats that have substantial programmes on forests.
- **The mission of the CPF is to help enhance the contribution of all types of forests and trees outside forests to the 2030 Agenda for Sustainable Development** and other internationally agreed development goals, promote the sustainable management of all types of forests and to strengthen long-term political commitment to that end. **Hence statement 2 is correct.**
- The core functions of the CPF are to:
 - support the work of UNFF and its member countries;
 - provide scientific and technical advice to the Forum and governing bodies of other CPF members, at their request
 - enhance coherence, cooperation as well as policy and programme coordination at all levels, including through joint programming and the submission of coordinated proposals to members' governing bodies, consistent with their mandates
 - **promote the implementation of the UN Forest Instrument and the United Nations Strategic Plan for Forests as well as the contribution of forests and trees to the 2030 Agenda for Sustainable Development and other major forest-related agreements.**

Q 23.B

- Recently, Kerala Legislative Assembly unanimously passed a resolution urging the Union Government to amend relevant sections of the Wildlife Protection Act of 1972 to address the escalating human-animal conflict in the state.
- Section 11 of the 1972 Act regulates hunting of wild animals. As per clause (1)(A) of the section, the Chief Wildlife Warden (CWLW) of a state may — if satisfied that a wild animal specified in Schedule I (mammals) has become dangerous to human life or disabled or diseased beyond recovery — permit hunting or killing of such animal. The section gives powers to the CWLW to order killing of such a wild animal, if it cannot be tranquillised or relocated after capturing.
- Kerala also wants the Centre to declare wild boar as vermin, **as per section 62 of the Wildlife Protection Act. According to this section, the Union Government can notify any wild animal in Schedule II of the Act (which protects it from hunting), as vermin for a period of time in an area/state. Hence statement 1 is correct.**
- **An animal is declared as vermin when it poses threat to life and crops.** Once declared as vermin, wild boar would lose its protection from hunting, thus enabling the state and citizens to cull the wild boar population to protect against the menace the species poses to life and livelihood. **Once declared as vermin anyone can kill the animal and there is no need for licensed shooters by the government to kill them.**
 - **Hence statement 2 is correct and statement 3 is not correct.**
- Wild boars, in particular, are notorious for devastating farmland. After the Centre previously rejected demands for declaring the species as vermin, Kerala, in 2022, empowered local self-governing bodies to deploy licenced shooters to kill wild boars that raid crops and foray into human settlements.
 - Once killed, forest officials were to be informed about the act, and the carcass of the boar was to be burned on the spot. However, this step did not help address the wild boar menace for want of licenced shooters in villages, as well as the long-winding process of involving the forest department after every killing.

Q 24.A

- The Technology Development Fund (TDF) scheme is a flagship programme of the Ministry of Defence executed by DRDO under the 'Make in India' initiative. Hence, statement 1 is not correct.
- The main objectives of the scheme are enumerated below:
 - To provide Grants in Aid to Indian industries, including MSMEs and startups, as well as academic and scientific institutions for the development of Defence and dual-use technologies that are currently not available in the Indian Defence Industry. Hence, statement 3 is correct.
 - To engage with the private industries especially MSMEs and Start-ups to bring in the culture of Design & Development of Military Technology and support them with Grant in Aid.
 - To focus on Research, Design & Development of Niche technologies which are being developed for the first time in the country.
 - To create a bridge between the Armed Forces, research organizations, academia and qualifying/certifying agencies with private sector entities.
 - To support the futuristic technologies having a Proof of Concept and converting them into prototypes.
- **Funding Support:**
 - The funding will be through the provision of grants to the Industry.
 - The project cost of up to INR 10 Cr will be considered for funding, subject to a maximum of 90% of the total project cost.
 - Industry may work in collaboration with academia or research institutions.
 - The work involvement of academia cannot exceed 40% of the total project cost.
 - The funding will be linked to mutually agreed milestones.
 - Funds will be released either in advance against a bank guarantee of the same amount as collateral, or reimbursement based on the completion of milestones.
 - Subsequent instalments will be released on the successful completion of milestones.
- **Project Duration:** The maximum development period will be two years.
- The Ministry of Textiles introduced the Technology Upgradation Fund Scheme (TUFS) in 1999 to boost the Indian Textile Industry by enabling ease of doing business, bolstering exports & fuelling employment. Hence, statement 2 is not correct.

Q 25.A

- Pradhan Mantri Vaya Vandana Yojana (PMVVY) is a Pension Scheme announced by the Government of India exclusively for the senior citizens aged 60 years and above which was available from 4th May, 2017 to 31st March, 2020. The scheme is now extended up to 31st March, 2023.
- **Objective:**
 - To provide social security during old age and to protect elderly persons aged 60 and above against a future fall in their interest income due to uncertain market conditions.
- **Benefits of the scheme:**
 - On payment of an initial lump sum amount ranging from a minimum purchase price of Rs.150000/- for a minimum pension of Rs.1000/- per month to a maximum purchase price of Rs.750000/- for a maximum pension of Rs.5000/- per month.
 - PMVVY is a social security scheme for senior citizens intended to give an **assured minimum pension** to them based on an assured return on the purchase price / subscription amount. **Hence option 1 is correct.**
 - ✓ To allow **initially an assured rate of return** of 7.40 % per annum for the year 2020-21 per annum and thereafter to be reset every year. Annual reset of assured rate of interest with effect from April 1st of financial year in line with revised rate of returns of Senior Citizens Saving Scheme (SCSS) upto a ceiling of 7.75% with fresh appraisal of the scheme on breach of this threshold at any point.
 - Pension is payable at the end of each period, during the policy term of 10 years, as per the frequency of monthly/ quarterly/ half-yearly/ yearly as chosen by the pensioner at the time of purchase.
 - **The scheme is exempted from Goods and Services Tax (GST).**
 - ✓ Usually GST is levied at the time of buying an insurance plan. This includes 18% GST on term insurance, 4.5% GST in the first year and 2.25% in the second year on traditional endowment insurance. Similarly, all general insurance schemes such as health insurance, motor insurance etc. have GST at 18%. **However, as a relief to senior citizens, no GST is levied on the exempted insurance schemes which include Pradhan Mantri Vaya Vandana Yojana Scheme.**

- On survival of the pensioner to the end of the policy term of 10 years, the Purchase price along with the final pension instalment shall be payable. Loan up to 75% of Purchase Price shall be allowed after 3 policy years (to meet the liquidity needs).
- Loan interest shall be recovered from the pension instalments and the loan to be recovered from claim proceeds.
- **The scheme also allows for premature exit for the treatment of any critical/ terminal illness of self or spouse. On such premature exit, 98% of the Purchase Price shall be refunded. So, a 2 percent penalty on principal amount is there. Hence statement 2 is not correct.**
- On the death of the pensioner during the policy term of 10 years, the Purchase Price shall be paid to the beneficiary.
- The ceiling of maximum pension is for a family as a whole, the family will comprise of pensioner, his/her spouse and dependents. Life Insurance Corporation (LIC) of India is the implementing agency for this scheme.
- Investors under the scheme are **not eligible to claim deductions under section 80C of the Income-tax Act**. Returns from this plan will be taxed at the time of receipt in accordance with current tax laws and the applicable tax rate. **Hence option 3 is not correct.**

Q 26.D

- The Asiatic Society of Bengal was founded during the Company rule in India to enhance and further the cause of "**Oriental research**" (in this case, research into India and the surrounding regions). **It was founded by William Jones on 15 January 1784 (during the tenure of Governor-General Warren Hastings) in a meeting presided over by Justice Robert Chambers in Calcutta, the then-capital of the Presidency of Fort William.**
 - **William Jones was an orientalist rather than an evangelist. Hence Statement-I is not correct.**
 - **The Asiatic Society aimed to provide a platform for European scholars to study, discuss, and document various aspects of Asian societies**, including history, literature, science, and religion. Initially, the objective of the society is to understand Indian cultural past in a better manner so that the rule of the East India Company can be smoothened and facilitated.
- The **Orientalist approach** to governing **focused on disrupting the Indian way of life as little as possible to minimize reasons for revolt and discontent among the indigenous population.**
 - Orientalism as an "**official policy**" was adopted between 1774- 1785 under Governor General **Warren Hastings**. Some historians view this policy as a direct response to the exploitive and distant nature between government/traders and Indians; where its administrators fit seamlessly into the fabric of daily indigenous life by changing the English approach rather than forcing foreign policies on the colonized subject. The Administrators worked through existing institutions rather than enforcing new ones.
 - Orientalism's expression in translation was seen through the upsurge in popularity of texts from India's past.
- On the other hand, **the evangelists were actively involved in spreading a religious faith**. In the context of British India, **many evangelists believed it was their duty to convert Indians to Christianity**. They argued about the **necessity of government intervention to liberate Indians from their religion that were full of superstitions, adultery, and tyranny of the priest**. **Hence Statement II is correct.**

Q 27.C

- The term **Dadani** comes from the Persian word '**Dadan**' meaning "**advance**". The **East India Company** provided advances (**Dadan**) to local merchants known as **Dadani merchants**. These **Dadani merchants used advances to purchase goods from the market like textiles.**
 - They might even **subcontract the procurement** to smaller players like **dalal or paikars**.
 - The **Dadani merchants were expected to deliver the goods within a stipulated time frame** and according to the specifications in the earlier agreement.
 - For their services, **Dadani merchants received a fixed commission.**
 - In 1753, the **company abolished the Dadani system because of the concerns like a failure in delivery of goods within the time framework or the disappearance of Dadani merchants with advanced fund**. This system was replaced by a new system where the company employed its **own workers called gomustahs** (Indian agent of East India Company) who directly procured goods.
- **Hence option (c) is the correct answer.**

Q 28.B

- In 1906, a group of Muslim elites led by Agha Khan met Lord Minto and demanded separate electorate for the Muslims. Their demands were met by the **Morley-Minto Reform of 1909**, which for the first time provided separate electorates for Muslims to the Central Council elections. After that, separate electorate has become an important instrument in the hand of British government to appease a small section of the Muslim elites.
- A sub-committee was appointed under the chairmanship of Motilal Nehru in All-Party Conference of 1928. Nehru committee recommended for various range of issues like dominion status, linguistic province, nineteenth fundamental rights, complete dissociation of State from religion. Apart from these, one prominent recommendation was rejection of separate electorate and instead of it, a demand for joint electorate with reservation of seats for Muslim at the centre and in province where they were in minority in proportion to the Muslim population there with respect to contest additional seats. This recommendation was in opposition to Muslim League's Delhi Proposals where it demanded representations to Muslims in Punjab and Bengal (where they were in majority) in proportion to their population. **Hence statement 1 is not correct.**
- In front of Simon Commission, Dr. B. R. Ambedkar in the case/issue of Universal Adult Franchise demanded the political protection of the depressed class and representation on the same line as the Mohammedan minority. So he put forward two conditional demands, either separate electorate for depressed classes or reservation of seats for the depressed classes. **Hence statement 2 is correct.**
 - He wanted to reserve seats for the depressed class if the Universal Adult Franchise was granted and in case of Universal Adult Franchise was not granted, he asked for a separate electorate for the depressed class.
- In the **Second Round Table conference**, there was a debate over separate electorate to the depressed class. However, after it, in 1932, the **Communal Award was announced by British Prime Minister Ramsey Macdonald**. As per this, separate electorates were provided for Muslim, Europeans, Sikh, Indian Christian, Anglo-Indians, depressed class and even to Marathas for some seats in Bombay.

Q 29.A

- The railway network in the first half of the 19th century played an important role in England which inspired British East India Company to commission the development of a railway network in India as well. However, the company could not afford to finance such large project and at the same time British capitalists were not that much interested in investing in India. The Court of Directors of the British India Company decided to come up with a system of guarantee. **As per this system, the East India Company would guarantee the railway shareholders a 5% return on their capital investment, free use of land and a 99-year operating contract. Hence statement 1 is correct.**
- In the 1880s, the government came out with a new model. As per this the companies were to be guaranteed a 4% return on their capital but for a limited period of 5 years. Also, the government retained the option of purchasing the line after 30 years. The extent to which the state should be actively involved in the construction and maintenance of individual lines turned on whether the lines were categorized as 'protective' or 'productive'. Lines with commercial potential were categorized as productive where private investors were much interested. Lines which lacked commercial potential but were important to the company's overall infrastructure were categorized as protective. **Hence statement 2 is not correct.**
- Till the end of 19th century, the management of various Indian railway undertakings had become very complex. In some instances, the company manager was responsible to the board member of company, to the British government in London, and to the Government of India. **The Viceroy Lord Curzon established a Railway Board as a single point of control in 1905. The board comprises a Commissioner of Railway, a Secretary to the Government of India, a Chief Inspector of Railway, and a team of Government Inspectors. Hence statement 3 is not correct.**
- The Railway Board was formed in 1901 on the advice of Sir Thomas Robertson's committee. In 1905, Lord Curzon formalized its power. The Board was under the Department of Commerce and Industry, and had government railway official who served as its chairman, a railway manager from England and an agent of one of the company railways as the other two members.
- After 1910, the British Government adopted a plan to purchase railway lines from private companies and by 1920, the government owned 73% of the total railway mileage in India. This plan was further accelerated and by 1940, all the trunk lines were owned and managed by the British Government.

Q 30.D

- Dharma Sabha was formed in 1830 in Calcutta by orthodox members of the Hindu community led by Radhakanta Deb. Hence statement 1 is correct.
 - The organization was established mainly to counter the ongoing social reform movements led by protagonists such as Raja Ram Mohun Roy and Henry Derozio.
 - More specifically, the impetus of forming the organization came from a new law enacted by the colonial British rule which banned the barbaric practice of burning widows alive (sati) in the country; the focus of the new association was to repeal the law which was seen as an intrusion by the British into the religious affairs of the indigenous people by some sections of the Hindu community.
 - The Dharma Sabha thus, filed an appeal in the Privy Council against the ban on Sati by Lord William Bentinck as, according to them, it went against the assurance given by George III of non-interference in Hindu religious affairs; however, their appeal was rejected and the ban on Sati was upheld in 1832. Hence statement 2 is correct.
 - It published a newspaper called Samachar Chandrika.
 - It also campaigned against the Hindu Widow Remarriage Act, 1856 and submitted a petition against the proposal with nearly four times more signatures than the one submitted for it by Ishwar Chandra Vidyasagar. However, Lord Dalhousie personally finalized the bill despite the opposition and it being considered a flagrant breach of Hindu customs as prevalent then, and it was passed by Lord Canning.

Q 31.B

- Prepaid payment instruments (PPIs) are payment instruments that facilitate purchase of goods and services against the value stored on such instruments. The value stored on such instruments represents the value paid for by the holder, by cash, by debit to a bank account, or by credit card; **example phonepe wallet, Google Wallet.**
- PPIs can be issued as cards, wallets, and any such form/instrument that can be used to access the PPI and to use the amount therein. PPIs in the form of paper vouchers cannot be issued. **Hence statement 3 is not correct.**
- Banks and non-bank entities issue PPIs in the country after obtaining necessary approval/authorisation from RBI under the Payment and Settlement Systems Act, 2007 (PSS Act). The RBI circular clearly states that No entity can set up and operate payment systems for PPIs without prior approval / authorisation of RBI. **Hence statement 1 is correct.**
- PPIs can be loaded/reloaded by cash (not permitted in one type of Small PPI), debit to a bank account, credit and debit cards, PPIs (as permitted from time to time) and other payment instruments issued by entities regulated in India and in Indian Rupees (INR) only. **Hence statement 2 is correct.**
- Small PPIs can be of two types:
 - PPIs upto ₹10,000/- (with cash loading facility). These PPIs shall be converted into full-KYC PPIs within 24 months.
 - PPIs upto ₹10,000/- (with no cash loading facility).

Q 32.D

- The universal banks include the Public Sector Banks or the PSBs and private sector banks which can provide all banking services and products. On the other hand, under differentiated banking licence, the Small Finance Banks and Payments Banks can provide only selected products (and in prescribed geographies). To have access to greater capital and increase profitability, the Small Finance Banks (SFBs) had requested the Reserve Bank of India to drop the 'small finance' tag, but the Reserve Bank of India (RBI) has turned down the request.
- At present, there are 12 Small Finance Banks. There has been a merger of AU Small Finance Bank and Fincare Small Finance Bank effective from 1st April 2024. So, now, there are 11 SFBs in India. **Hence statement 1 is not correct.**
- **Small Finance Banks:**
 - Small Finance Banks (SFB) registered as public limited companies under the Companies Act, 2013 intend to provide financial services to the unserved and unbanked region of the country.
 - They primarily undertake basic banking activities of acceptance of deposits (both demand and time deposits) and lending (with limits) to un-served and underserved sections including small business units, small and marginal farmers, micro and small industries, and unorganized sector entities, but without any restriction in the area of operations, unlike Regional Rural Banks or Local Area Banks.

- They are subject to all prudential norms and regulations of RBI as applicable to existing commercial banks like maintenance of Cash Reserve Ratio (CRR) and Statutory Liquidity Ratio (SLR).
- They can issue both debit cards and credit cards.
- And the minimum paid-up capital required for small finance banks is Rs 200 crores.
- Of the 12 small finance banks, the AU small finance bank has the lowest NPA ratio, but its Gross NPA is 1.67%. All the other small finance banks have a higher GNPA. **Hence statement 2 is not correct.**
- The eligibility criteria for an SFB to transition into a Universal bank will now be as follows:
 - scheduled status with a satisfactory track record of performance for a minimum period of five years;
 - shares of the bank should have been listed on a recognized stock exchange;
 - having a minimum net worth of ₹1,000 crore as at the end of the previous quarter (audited); **Hence statement 3 is not correct.**
 - meeting the prescribed CRAR requirements for SFBs;
 - having a net profit in the last two financial years; and
 - having GNPA and NNPA of less than or equal to 3 percent and 1 percent respectively in the last two financial years.

Q 33.B

- **Minimum Alternate Tax (MAT):** Corporations take advantage of various provisions of the Income Tax Act by claiming exemptions, deductions, depreciation, etc. to reduce their tax liability. One such provision available to companies for deducting their corporate tax burden is their Minimum Alternate Tax (MAT). **MAT is calculated as 15% of the book profit of the tax assessee.** Under existing rules, book profit is calculated as per Section 115JB of the Income Tax Act, 1961. **Hence option 2 is correct.**
- **Securities Transaction Tax (STT):** It is a kind of financial transaction tax that is similar to tax collected at source (TCS). STT is a direct tax levied on every purchase and sale of securities that are listed on the recognized stock exchanges in India. STT is governed by the Securities Transaction Tax Act (STT Act), and the STT Act has specifically listed down various taxable securities transactions i.e., transactions on which STT is leviable. STT is paid regardless of the profit or loss of the asset that is involved any transaction. **Hence option 1 is not correct.**
- **Capital Gains Tax:** Capital gains tax is a tax imposed on the profits realized from the sale of assets such as stocks, bonds, real estate, and other investments. It is the tax applied to the difference between an asset's purchase price (or "cost basis") and its selling price. **Hence option 3 is correct.**

Q 34.C

- **A gig worker in an economy refers to a person who performs work or participates in a work arrangement and earns from such activities outside of the traditional employer-employee relationship.**
- The rapidly burgeoning gig workforce is ushering in a new economic revolution globally. India – with its demographic dividend of half-a-billion labour force and the world's youngest population, rapid urbanization, widespread adoption of smartphones, and associated technology – is the new frontier of this revolution.
- **As per a report published by the NITI Aayog in 2022, gig workers – those engaged in livelihoods outside the traditional employer-employee arrangement – can be broadly classified into platform and non-platform-based workers.**
 - **Platform workers are those** whose work is based on online software apps or digital platforms.
 - **Non-platform gig workers are generally casual wage workers and own-account workers in the conventional sectors,** working part-time or full-time. Emigrant workers or migrant workers are those workers who leave their residence to work elsewhere. Emigrant workers can work in organized sectors on high salaries. **Hence only options 1, 2, and 3 are correct.**
- The emergence of tech-enabled platforms, demand flexible work arrangements, and focus on skills has led to the rise of the gig economy.
- The gig economy is prevalent mostly among blue-collar jobs in India. However, in recent times, the demand for gig workers in white-collar jobs such as project-specific consultants, salespeople, web designers, content writers, and software developers is also emerging.
- As per a NITI Aayog report, in 2020-21, 77 lakh (7.7 million) workers were engaged in the gig economy, and the gig workforce is expected to expand to 2.35 crore (23.5 million) workers by 2029-30.
- **Hence option (c) is the correct answer.**

Q 35.C

- **Sesame (Sesamum indicum L.) is an oilseed crop of Indian origin (also grown in other parts of Asia and Africa).** It is chiefly confined to tropical and sub-tropical regions and has vintage value for high-quality edible oil and seed for direct confectionery use. It is the fourth largest oilseed crop cultivated in India (1.72 million ha) after soybean, rapeseed-mustard, and groundnut. The importance of sesame is recognized globally and therefore holds a prime position in international trade. **Sudan and India are the major exporters of Sesame (sesamum)**
- **Sesamum can be taken as kharif, semi-rabi, and summer crop.** Kharif crop is sown in June-July at the onset of monsoon. Summer crops are sown in March and rabi crop is sown in November.
- The Minimum Support Price for Sesamum is currently Rs 8365/quintal which is the highest for any crop.
- The sesame trade has been a mainstay of local livelihoods in Ethiopia and Sudan. Amid civil war and territorial rivalry on both sides of the border, it now plays a central role in a conflict economy that perpetuates violence and political instability. Competition to control sesame revenues has reshaped local agricultural markets, and provided a strategic motivation for conflict participants and members of political and economic elites. If left unaddressed, these shifts threaten to prolong and intensify conflict and worsen the inequalities faced by people living in the border regions.
- **Hence option (c) is the correct answer.**

Q 36.C

- As per a government report in 2020, India consumed about 500 Lakh Metric Tonnes (LMT) of fertilizer per year in the last 10 years. The total consumption of all fertilizer products at 639.2 LMT during 2022-23 showed a decline of 0.03% over 2021-22.
- As per the Department of Fertilisers, India is the second-largest consumer and third-largest producer of finished fertilizer in the world. India is a net importer of fertilizer, both raw materials and finished fertilizers. **Hence statement 1 is correct.**
- Fertilizer, water, and seeds are vital inputs for higher agricultural production. Increasing use of fertilizers has contributed significantly in enhancing food grain production and bringing self-sufficiency in food grain production in the country. India is the second-largest consumer and third-largest producer of finished fertilizers in the world. India is a net importer of fertilizers, both finished products as well as raw materials.
- India launched a New Urea Policy (NUP) in 2015. **Hence statement 2 is correct.**
- The objectives of the policy are
 - To maximize indigenous urea production.
 - To promote energy efficiency in the urea units.
 - To rationalize the subsidy burden on the Government of India.
- The Union Budget for fiscal year 2024-25 (FY25) allocated ₹1.64 trillion for fertilizer subsidy, lower than the revised estimate of ₹1.88 trillion allotted for FY24. In terms of actuals also (2021-22), the fertilizer subsidy is much greater than the petroleum subsidy. **Hence statement 3 is correct.**

Major subsidies by the government

(In ₹ Crores)				
Subsidy Scheme Name	2021-2022 Actuals	2022-2023 Budget Estimates	2022-2023 Revised Estimates	2023-2024 Budget Estimates
Food	288968.54	206831.09	287194.05	197350.00
1. Food Subsidy	288968.54	206831.09	287194.05	197350.00
Fertiliser	153758.10	105222.32	225220.16	175099.92
1. Nutrient based Subsidy	52769.97	42000.00	71122.23	44000.00
2. Urea Subsidy	100988.13	63222.32	154097.93	131099.92
Petroleum	3422.60	5812.50	9170.50	2257.09
1. LPG Subsidy	3421.07	5812.50	9170.50	2257.09
2. Direct Benefit Transfer-Kerosene	1.53	...	...	...

Q 37.B

- A parent company is a company that owns more than half of the stock of an Indian subsidiary and can therefore control its operations. A parent company is a holding company that owns a significant number of voting shares of another company. **As per the provisions of Section 19 of the Companies Act, 2013, a subsidiary company is prohibited from holding any shares in its parent company, by itself or its**

nominee. In other words, the Indian subsidiary has no shareholding rights in its holding company. A provision of the Act also prohibits a holding company from allotting any shares in its Indian subsidiary. **Hence statement-I is correct.**

- The SEBI Act which came into being in 1992 provides for the establishment of a Board to protect the interests of the investors securities and to promote the development of, and to regulate, the securities market. **Thus statement-II is correct, but it does not explain statement-I (which is backed by the Companies Act, 2023).**
- **Hence option (b) is the correct answer.**

Q 38.A

- Import/Export Section of the Department of Food and Public Distribution deals with the issue/supply of wheat and non-basmati rice from Central Pool Stocks available with FCI, under the WTO norms which inter-alia restricts the export of foodgrains from Public Stocks procured for the domestic consumption through welfare schemes.
- **The wheat and non-basmati rice procured for the Central Pool Stock at MSP (Minimum Support Price) and available with FCI, constitute Public Stock-holding, and** therefore certain restrictions have been imposed on exports to foreign countries as per the provisions of the Agreement on Agriculture (AoA) under the WTO framework. **Hence statement 1 is correct.**
- **Wheat and Rice are issued from Central Pool Stocks to foreign countries as Humanitarian Assistance / Aid.** The humanitarian assistance/aid, fully in grant form, to the deserving countries, is only allowed through the Ministry of External Affairs. **Hence statement 2 is not correct.**

Q 39.A

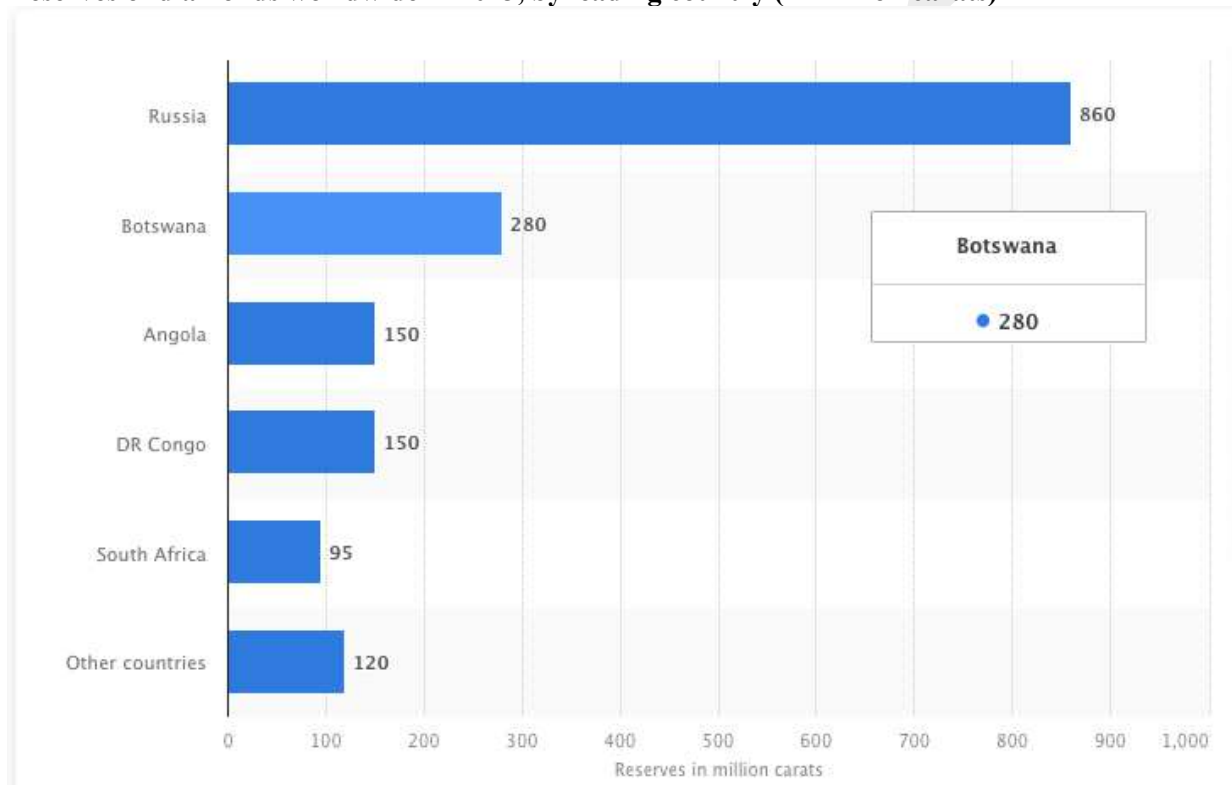
- The due importance given to the health sector in India is supported by the fact that between Fiscal Year 2014(FY14) and FY19, the share of government health expenditure in total health expenditure has increased from 28.6 percent to 40.6 percent.
- This is aligned with the National Health Policy, 2017 which envisages “the attainment of the highest possible level of health and well-being for all at all ages, through a preventive and promotive healthcare orientation in all developmental policies, and universal access to good quality healthcare services without anyone having to face financial hardship as a consequence.
- Accordingly, the Policy recommended an increase in the Government’s health expenditure **from the existing 1.2 percent to 2.5 percent of GDP by 2025. Hence statement 1 is correct.**
- The policy aimed at achieving universal health coverage and delivering quality healthcare services to all at affordable cost. Accordingly, Ayushman Bharat Pradhan Mantri Jan Arogya Yojana (AB-PMJAY) was launched on 23rd September 2018. AB-PMJAY is completely funded by the Government and costs are shared between Central and State Governments in the ratio as per the extant directives issued by the Ministry of Finance. **Hence statement 2 is not correct.**
- Recently, to widen the market and foster adequate protection from healthcare expenses, the insurance regulator **IRDAI (Insurance Regulatory and Development Authority of India)** has removed the age limit of 65 years for individuals buying health insurance policies. Thus, there is no upper age limit now. **Hence statement 3 is not correct.**

Q 40.C

- **The term Luddite Fallacy has often been seen in the news in the context of the emergence of Artificial Intelligence and the possibility of job loss.**
- The Luddite fallacy is the mistaken belief that new technology leads to higher overall unemployment in the economy. New technology may cause disruption and some workers to lose their job, but the improved technology will also create jobs in other sectors of the economy – balancing out any jobs lost. **Hence option (c) is the correct answer.**
- **Luddite is also a term used (usually pejoratively) to describe people who oppose the introduction of new technology.** The idea that new technology leads to job losses has persisted, despite the fact that economists are almost universally united in stating that new technology will not increase the long-term unemployment rate.

Q 41.D

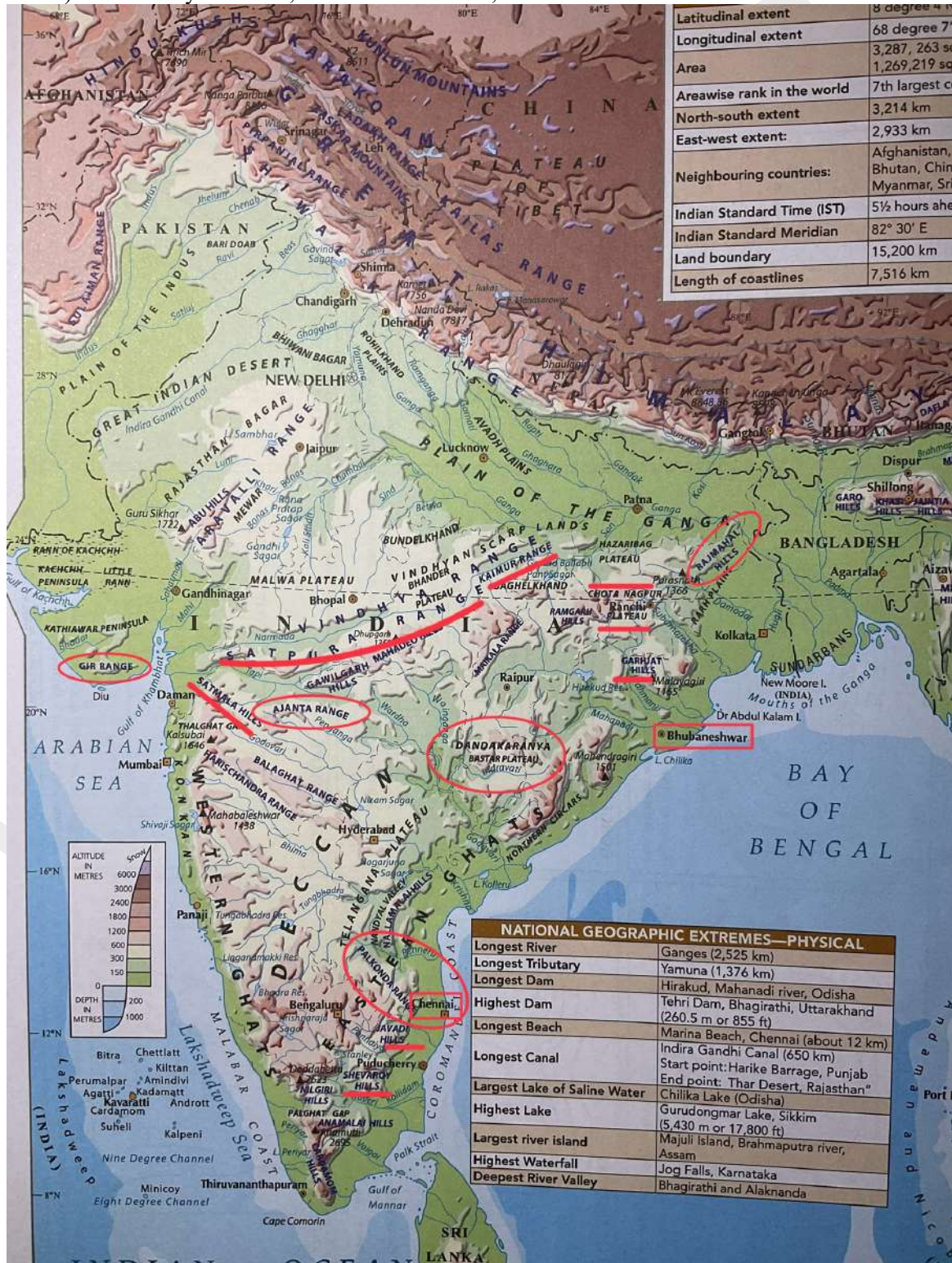
- As of 2023, Russia's natural industrial diamond reserves were estimated at approximately 860 million carats, representing about 52% of the global capacity. Thus, Russia had the largest industrial diamond reserves worldwide. Hence statement 1 is not correct.
 - Following Russia, Botswana held the world's second-largest diamond reserves, totaling 280 million carats as of 2023.
 - Based on production volume, Russia and the Democratic Republic of Congo are the world's largest diamond producers.
 - Russia mined 18 million carats of industrial diamonds in 2023 and the Democratic Republic of Congo produced eight million carats during that year. Demand for rough diamonds is increasing worldwide, and is expected to reach some 292 million carats by 2050.
- Reserves of diamonds worldwide in 2023, by leading country (in million carats)



- Almost half of the world's mined diamonds are used for industrial purposes as rough diamonds. Industrial diamonds can be used in cutting, drilling, grinding, and polishing.
- Diamond occurrences are reported since prehistoric times in the country. Presently, diamond fields of India are grouped into four regions:
 - South Indian tract of Andhra Pradesh, comprising parts of Anantapur, Kadapa, Guntur, Krishna, Mahabubnagar and Kurnool districts;
 - Central Indian tract of Madhya Pradesh, comprising Panna belt;
 - Behradin-Kodawali area in Raipur district and Tokapal, Dugapal, etc. areas in Bastar district of Chhattisgarh; and
 - Eastern Indian tract mostly of Odisha, lying between Mahanadi and Godavari valleys.
- Vindhayan system has diamond-bearing regions from which Panna and Golconda diamonds have been mined. They are devoid of any kind of metalliferous minerals but provide large quantities of durable stones, ornamental stones, limestone, pure glass-making sand, etc. Hence statement 2 is not correct.
- According to the Indian Bureau of Mines (IBM) inventory report, India's diamond reserves are concentrated in three states: Madhya Pradesh, Andhra Pradesh, and Chhattisgarh with Madhya Pradesh accounting for more than 90% of the reserves. National Mineral Development Corporation Ltd. (NMDC), a CPSE under the Ministry of Steel has its diamond mine at Panna in Madhya Pradesh, which is the only state in India that accounts for 90% of the total diamond resource for our country. Currently, there are two reporting mines, both under the Public Sector located in the Panna district of Madhya Pradesh. Hence statement 3 is not correct.

Q 42.D

- **Option (a) can be eliminated** as Chennai is located north-east of Javadi hills also a northern chain of the Vindhyas continues eastwards as Bhandar Plateau and Kaimur Range, which runs north of the Son River.
- **Option (b) can be eliminated** as the Satpura range is a range of hills that runs between the Narmada and Tapi rivers i.e. north of Jalgaon and Tapi river.
- **Option (c) can be eliminated as:**
 - The Chota Nagpur Plateau is a plateau in eastern India, which covers much of Jharkhand state as well as adjacent parts of Chhattisgarh, Odisha, West Bengal and Bihar.
 - Garhjat hills are located to the north-west of Bhubaneswar.
 - The Servarayan hills, with the anglicised name Shevaroy Hills, are a towering mountain range (1620 m) near the city of Salem, in Tamil Nadu state, southern India. It is located south of Chennai.



- Hence option (d) is the correct answer.

Q 43.B

- Amazon basin, the drainage basin of the Amazon River covers about 34 percent of the land of South America (about 6,100,000 square km) and is located in the center and eastern portions of the continent. The Amazon basin is the largest drainage basin in the world, more than twice as large as that of the Congo River in Africa. Approximately one-fifth of all freshwater runoff on Earth passes through this watershed.
- The Amazon basin includes roughly 60 per cent of the world's rainforest and harbours 10 per cent of the planet's known forms of life—including more than 40,000 plant species, 1,300 bird species, and several million species of insects and other invertebrates.
- **The Amazon basin covers an immense area that extends from the Andes Mountains in Peru to the northeastern coast of Brazil, where the river empties into the Atlantic Ocean. It includes parts of the countries of Bolivia, Brazil, Colombia, Ecuador, Guyana, Peru, Suriname, and Venezuela.**



- Hence option (b) is the correct answer.

Q 44.B

- The Parliament of India has recently has passed the Constitutional (Jammu and Kashmir) Scheduled Tribes Order (Amendment) Bill, 2024 to include Pahari Ethnic Group, Paddari Tribes, Koli and Gadda Brahmin communities in the list of STs of UT of Jammu and Kashmir. Hence pair 1 is correctly matched.
 - In 1991, four communities namely **Gujjar, Bakerwals, Gaddis and Sippis** were provided with **ST status in J&K**. These four communities are entitled to 10% reservation in government jobs and admission in the educational institutes. In 2019 a 10% quota in Lok Sabha and Assembly seats of J&K empowered the politically too. However, the recent expansion in ST list has triggered protest in the valley by Gujjar-Bakerwals.
 - **Gadda Brahmins and Kolis are very small communities**. Also, Gadda Brahmins are a branch of the Gaddis while Kolis are a sub-caste of Sippis – both these communities are already in the ST list.
 - The **Pahari are Hindus, Muslims, and Sikhs and include the people of Kashmiri origin who settled in the districts of Rajouri and Poonch**.
 - **Paddari tribes are live in the remote Paddar area of the hilly Kishtwar district**. As per 2011 census, their population was 21548 comprises 83.6% Hindus, 9.5% Buddhists and 6.8% Muslims. Hence pair 4 is not correctly matched.
- Recently the parliament has passed the Constitutional (Scheduled Tribes) Order (Amendment) Bill, 2024 to modify the list of STs in Andhra Pradesh. The Bill added the three Particularly Vulnerable

Tribal Groups (PVTGs) Bondo Porja, Khond Porja and Konda Savaras to the list of STs in Andhra Pradesh. Hence pair 2 is correctly matched.

- The other bill the Constitutional (Scheduled castes and Scheduled Tribes) Order (Amendment) Bill, 2024 aimed at modify in the ST list of Odisha. This bill added four PVTGs namely Pauri Bhuyan and Paudi Bhuyan (included as synonyms of Bhuyan Tribes), Chuktia Bhunjia (recognised as Bhunjia Tribes), Bondo community (sub-tribe of Bondo Poraja) and Mankidia community (synonym of Mankirdia tribes) to the list of STs in Odisha. Hence pair 3 is not correctly matched.
- These above two bills only added synonyms, phonetic variations and sub-tribes of the existing STs in Andhra Pradesh and Odisha. The number of PVTGs in India remains unchanged at 75.
- Further two new tribes were added in STs list of Odisha – Muka Dora and Konda Reddy. There was a removal of Tamadia and Tamudia communities from the SC list of Odisha and added them in ST list of Odisha.

Q 45.A

- Recently satellite images have shown a pair of cyclones in the Indian Ocean region, one in the northern hemisphere and one in the southern hemisphere. Named cyclone 'Tej' in the Arabian Sea and cyclone 'Hamoon' in the Bay of Bengal.
- Twin cyclones are not rare. The interplay of the wind and the monsoon system combined with the Earth system produces these synchronous cyclones. **Twin tropical cyclones are caused by what are called equatorial Rossby waves. Hence, statement 1 is correct.**
- Rossby waves are huge waves in the ocean with wavelengths of around 4,000–5,000 kilometres. This system has a vortex in the northern hemisphere and another in the southern hemisphere, and each of these is a mirror image of the other. The vortex in the north spins counterclockwise and has a positive spin, while the one in the southern hemisphere spins in the clockwise direction and therefore has a negative spin. Both have positive values of the vorticity which is a measure of the rotation. Very often twin cyclones are formed from these Rossby waves. Rossby waves are named for famous meteorologist Carl-Gustaf Rossby who was the first to explain that these waves arose due to the rotation of the Earth. When the vorticity is positive in both the Northern and Southern hemispheres, as is the case with Rossby waves, the air in the boundary layer, which is moist, is lifted slightly. That is enough to set off a feedback process. **Hence, statement 2 is not correct.**
- When the air is lifted slightly, the water vapour condenses to make clouds. As it condenses, it lets out the latent heat of evaporation. The atmosphere warms, this parcel of air rises, and positive feedback is set off by this process. The warmer parcel of air can rise further because it is lighter than the surrounding air, and it can form deeper clouds. Meanwhile, moisture comes in from both sides. This leads to the formation of a cyclone if certain conditions are present. **"The ocean's surface temperature has to be 27 degrees or warmer; the wind shear in the atmosphere must not be too high,"** For example, if you have westerly winds at the lower level and easterly winds at the upper level, if the difference between them is too high, cyclones will not form. But if the difference is modest, cyclones will still form. **"There will be a big, tall vortex with all sorts of clouds inside. Once they are stronger, they will spin faster and faster and organize themselves into the big storms,"** **Hence, statement 3 is not correct.**

Q 46.D

- Maize, the queen of cereals, is becoming a vital source of raw material not only to increase ethanol production but also to feed growing poultry, animal feed, starch and other industries in India.
- **The United States of America (USA) is the largest producer of maize contributes nearly 35 %** of the total production in the world and maize is the driver of the US economy. The USA has the highest productivity ($> 9.6 \text{ t ha}^{-1}$) which is double the global average (4.92 t ha^{-1}). Whereas, the average productivity in India is 2.43 t ha^{-1} . **Hence option (d) is the correct answer.**
- In India, maize is principally grown in two seasons, rainy (kharif) and winter (rabi). **Kharif maize represents around 83% of the maize area in India**, while rabi maize corresponds to 17% maize area. Over 70% of the kharif maize area is grown under rainfed conditions with a prevalence of many biotic and abiotic stresses. The stress-prone ecology contributes towards lower productivity of kharif maize (2706 kg/ha) as compared to rabi maize (4436 kg/ha), which is predominantly grown under an assured ecosystem.
- Major wheat-growing states in India are Karnataka, Madhya Pradesh, Bihar, Tamil Nadu, Telangana, Maharashtra and Andhra Pradesh. **Karnataka is the largest producer of maize in India, contributing 15.53% to the country's total maize output during 2021-22.**

- **Maize has emerged as an important crop in the non-traditional regions i.e. peninsular India as the states like Andhra Pradesh** which ranks 5th in area (0.79 m ha) has recorded the highest production (4.14 m t) and productivity (5.26 t ha⁻¹) in the country although the productivity in some of the districts of Andhra Pradesh is more or equal to the USA.
- **Bulk of the maize production in India, approximately 47%, is used as poultry feed.** Of the rest of the produce, 13% is used as livestock feed and food purpose each, 12% for industrial purposes, 14% in starch industry, 7% as processed food, and 6% for export and other purposes.

Q 47.C

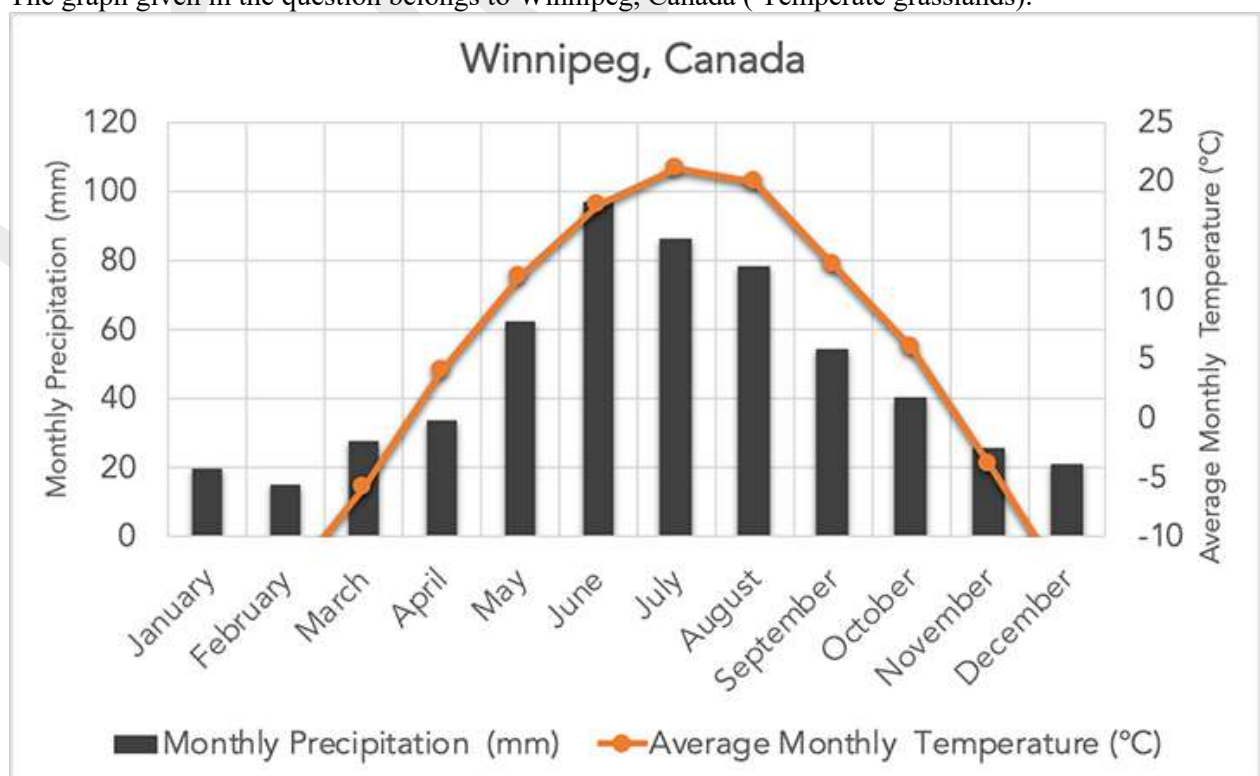
- **Pair 1 is correctly matched:** The Khavda Renewable Energy Park, also known as the Gujarat Hybrid Renewable Energy Park or Khavda Solar Park, is a renewable energy park under construction in the Kutch district of Gujarat, India. It's located near Vighakot village, close to the Pakistan border. Adani Green Energy Limited is developing 30 GW renewable energy plant on barren land at Khavda in Kutch of Gujarat over area of 538 sq km after which the project would be the planet's largest power plant.
- **Pair 2 is correctly matched:** Muppandal in Tamil Nadu is home to India's largest operational onshore wind farm in terms of installed capacity. This project is located in Kanyakumari district, Tamil Nadu. The project was developed by Tamil Nadu Energy Development Agency. Its installed capacity is 1,500 MW.
- **Pair 3 is correctly matched:** Sasan Ultra Mega Power Plant or Sasan UMPP is one of the four Ultra Mega Power Projects awarded by the Ministry of Power, Government of India. It is located in Sasan village near Waidhan in Singrauli district of Madhya Pradesh. **Sasan UMPP is India's largest integrated power generation and coal mine project with 3,960 MW power plant.**

Q 48.C

- **The Wilson cycle describes the cyclical process of the formation and breakup of supercontinents, driven by the movement of tectonic plates.** It involves several stages, including the opening of ocean basins through seafloor spreading, the convergence of plates leading to subduction and the closure of ocean basins, and ultimately the collision of continents to form supercontinents. This process repeats over geological time scales, shaping the Earth's surface and influencing the distribution of landmasses and oceans. **Hence option (c) is the correct answer.**

Q 49.C

- The graph given in the question belongs to Winnipeg, Canada (Temperate grasslands).



- Bordering the deserts, away from the Mediterranean regions and in the interiors of continents are the **temperate grasslands**. Though they lie in the Westerly wind belt, they are so remote from maritime influence that the grasslands are practically treeless.
- **Climate:**
 - **Temperature:**
 - ✓ Their location in the heart of continents means that they have little maritime influence. Their climate is thus continental with extremes of temperature. Summers are very warm, over 20°C in Winnipeg for July, as illustrated in the graph.
 - ✓ **Depending on latitude, the annual range in temperate grasslands can be between -20°C to 30°C.**
 - ✓ Winters are very cold in the continental steppes of Eurasia because of the enormous distances from the nearest sea. The winter months are well below freezing and in Winnipeg, the January reading is -20°C, below freezing-point.
 - ✓ The annual range of temperature is great, a direct result of continentality. Winters are so cold that parts of the Eurasian Steppes are snow-covered for several months. The snow melts with the return of spring and by mid-summer, temperatures soar to over 20°C. It is really hot for its latitude.
 - **Precipitation:**
 - ✓ In its continental position, the annual precipitation of the Steppe Climate can be expected to be light. Winnipeg receives just more than 500 mm of rainfall per year with a distinct summer maximum from convectional sources, when the continental interiors are intensely heated. **The heaviest rain comes in the middle of the year with 90 mm (approx) each in June and July.** Most of the winter months have about 20 mm of precipitation, brought by the occasional depressions of the Westerlies and coming in the form of snow.
 - ✓ **Hence option (c) is the correct answer.**
- **Temperate marine (The Cool Temperate Western Margin (British Type) Climate:** The cool temperate western margins are under the permanent influence of the Westerlies all round the year. They are also regions of much cyclonic activity typical of Britain and are thus said to experience the British type of climate.
 - The mean annual temperatures are usually between 5 °C (in winters) and 16°C (in summers).
 - **Monthly temperatures of over 18 °C even in mid-summer are rare.**
 - Winters are abnormally mild and no stations actually record mean January temperatures below freezing-point in north-western Europe.
 - This is attributable to the warming effect of the warm North Atlantic Drift and the prevalence of South Westerlies.
 - It has adequate rainfall throughout the year with a tendency towards a slight winter or autumn maximum (September and October) from cyclonic sources. Since the rain-bearing winds come from the west, the western margins have the heaviest rainfall.
- **Warm Temperate Western Margin (Mediterranean) Climate:** They are entirely confined to the western portion of continental masses, between 30° and 45° north and south of the equator.
 - Clear skies and high temperatures; hot, dry summers and cool, wet winters.
 - Temperature of warmest month greater than or equal to 10° C.
 - **Temperature of the coldest month is less than 18° C but greater than -3° C**
 - Climate is not extreme because of cooling from water bodies.
 - **Mean annual precipitation ranges from 35 – 90 cm.**
 - **The Mediterranean lands receive most of their precipitation in winter when the Westerlies shift equatorwards.**
 - **The rain comes in heavy showers and only on a few days with bright sunny periods between them.**
- **Mid-Latitude Deserts:**
 - The mid-latitude desert is an excessively dry region with an arid climate located between 30°N and 60°N and S of the equator. The placement of these deserts is found in the interior parts of the continents. The precipitation and humidity levels in these regions are extremely low. There is no or rare vegetation in mid-latitude deserts.
 - A stark seasonal contrast in temperature can be observed in the mid-latitude deserts. In the hot summer of July, the temperature increases up to 20 °C and decreases to -27 °C in the harsh winter of January.

- The summer temperature in mid-latitude deserts is not very hot due to its location in the temperate zone, which is cooler than the tropical zone.
- **Deserts, whether hot or mid-latitude have an annual precipitation of less than 25 cm.** Rain normally occurs as violent thunderstorms of the convectional type. It 'bursts' suddenly and pours continuously for a few hours over small areas.

Q 50.B

- Lakes are amongst the most varied features of the earth's surface. They occupy the hollows of the land surface in which water accumulates. They vary tremendously in size, shape, depth, and mode of formation. Lakes are formed by the earth's movement, glaciation, volcanic activity erosion, and human and biological activities.
- - **These are depressions in the outwash plain left by the melting of masses of stagnant ice.** They are irregular because of the uneven morainic surface and are never of any great size or depth. **Examples-** The meres of Shropshire in England, and the kettle-lakes of Orkney in Scotland. **Hence, pair 1 is not correctly matched.**
- **Ribbon Lakes-** The glacier on its downward journey, fed by ice from several corries like tributaries that join a river begins to wear away the sides and the floor of the valley down which it moves. It scratches and grinds the bedrock removing any rock debris and surface soil. While it tends to straighten any protruding spurs on its course. The interlocking spurs are thus blunted to form truncated spurs and the floor of the valley is deepened. A valley which has been glaciated takes a characteristic U-shape, with a wide, flat floor and very steep sides. **After the disappearance of the ice, the deep sections of these long narrow glacial troughs may be filled with water forming ribbon lakes. They are sometimes referred to as trough lakes or finger legs. Examples-** The meres of Shropshire in England, and the kettle-lakes of Orkney in Scotland. **Hence, pair 2 is not correctly matched.**
- **Cirque lakes or tarns-** A glacier on its way down the valley leaves behind circular hollows in the heads of the valleys up in the mountains. Such hollows are the arm-chair-shaped cirques or corries. Their over-deepened floors may be filled with water to become cirque or tarn lakes. **Examples-** Red Tarn in the English Lake District. **Hence, pair 3 is correctly matched.**
- **Barrier Lakes-** Lakes formed due to landslides, screes, and avalanche processes are known as **barrier lakes.** Landslides or screes may block valleys so that rivers are dammed. Such lakes are short-lived because the loose fragments that pile across the valleys will soon give way under the pressure of water. When they suddenly give way the dammed water rushes down, causing floods. **Examples-** Lake Gormire in Yorkshire, blocked by landslides; Ffynnon French on Snowdon blocked by screes. **Hence, pair 4 is correctly matched.**

Q 51.C

- The Kalabhra Revolt:
 - Although the period between AD 300 and 750 was extremely important for state formation and agrarian expansion in the peninsula, very little is known about what happened at the tip of the peninsula after the eclipse of the Cholas, the Cheras, and the Pandyas. The only important event is a revolt led by the Kalabhras in the sixth century.
 - **The Kalabhras seem to have been a tribal people who captured power, particularly at the cost of the Cholas, and ruled for seventy-five years.** Their rule also affected the Pallavas as well as their neighboring contemporaries. The Kalabhras are called evil rulers, who overthrew innumerable kings and established their hold on the Tamil land.
 - **The Kalabhra revolt was a powerful peasant protest directed against the landed brahmanas. The repeated teaching that those who attack land grants are condemned to hell for sixty thousand years failed to change their minds.** They put an end to the Brahmadeya rights granted to the brahmanas in numerous villages. It appears that the Kalabhras were of Buddhist persuasion as they patronized Buddhist monasteries. **Hence option (c) is the correct answer.**
 - **The Kalabhras' revolt was so widespread that it could be quelled only through the joint efforts of the Pandyas, the Pallavas, and the Chalukyas of Badami.** By the last quarter of the sixth century, according to tradition, the Kalabhras had imprisoned the Chola, the Pandya, and the Chera kings, which underlines how formidable their revolt was. The confederacy of the kings formed against the Kalabhras, who had revoked the land grants made to the Brahmanas, shows that the revolt was directed against the existing social and political order in south India.
 - It, therefore, appears that some land grants had been made between AD 300 and 500 to the Brahmanas by the kings of the deep south. The Sangam texts tell us that villages were granted to the warriors by

the chief for their acts of bravery. Land grants seem to have stimulated agrarian expansion under the Pallavas in south Andhra and north Tamil Nadu from the end of the third century onwards, but they seem to have adversely affected the peasants.

Q 52.B

- Specifically, his Major Rock Edict No. 13 refers to five Hellenistic rulers by name. These kings were likely contemporaries of Ashoka and where Ashoka demonstrated his commitment to spreading the teachings of the Dharma by sending emissaries to the great kings of the West:
 - Antiyoga (Antiochus I/II of the Seleucid Empire or Syria),
 - **Tulamaya (Ptolemy II Philadelphus of Egypt), Hence option 2 is correct.**
 - Antekina (Antigonos II Gonatas of Macedon),
 - **Maka (Magas of Cyrene), Hence option 1 is correct.**
 - Alikyashudala (Alexander II of Epirus)
- **Alexander III of Macedonia or Alexander the Great was the king of Macedonia (336–323 BCE),** who overthrew the Persian empire, carried Macedonian arms to India, and laid the foundations for the Hellenistic world of territorial kingdoms. **Hence option 3 is not correct.**
 - In 326 BC, Alexander invaded India, after crossing the river Indus he advanced towards Taxila. He then challenged King Porus, ruler of the kingdom between the rivers Jhelum and Chenab.
 - Alexander continued eastward up to the Beas River, but his army refused to go further due to the formidable power of Magadha, war-weariness, disease, and homesickness. Thus, Alexander was forced to retreat, ending his dream of an eastern empire. During his return, he died of typhoid in Babylon.
 - His conquest opened the door for the Maurya Empire to grow there. Chandragupta Maurya watched Alexander's army at work and learned some information that enabled him to defeat the Nandas.

Q 53.B

- The Sangam age refers to that period in the early history of south India when large numbers of poems in Tamil were composed by a number of authors. The term Sangam refers to an assembly or “meeting together” of Tamil poets.
- Women in the Sangam period appear to have been educated. This is testified by many poems contributed by women poets to the Sangam literature. **Women are also described as engaged in various economic activities such as paddy plantation, cattle rearing, basket-making, spinning, etc.**
- **However, the cruel practice of Sati was also prevalent in Tamil society, and it was known as tippayadal.** But it was not obligatory as there are references to widows present in society. However, their position was miserable as they were prohibited to decorate themselves or participate in any form of amusement.
- **Hence option (b) is the correct answer.**

Q 54.B

- **One of the most remarkable features of the agriculture of the time was the large number of crops grown by the peasants of the Delhi Sultanate.** This has perhaps no parallel in other parts of the world except perhaps in South China.
- Ibn Battutah, who traveled all over India, has left us a detailed account of the food grains and various other crops, fruits, and flowers produced in the country. Most of them are familiar to us, **with rice and sugarcane being produced in the east and south**, and wheat, oilseeds, etc., in the north.
- **Cotton was grown widely, as well as barley, sesame, and other inferior crops.** Ibn Battuta was struck by the multiplicity of crops grown and described in sufficient detail the various crops grown in the two cropping seasons. **He also suggests that in the region around Delhi, double cropping was also practiced, that is, on the same soil both the khalif and the rabi crops were raised.**
- Rice was sown three times a year. Some of the crops were the basis of village industries, such as oil processing, making jaggery, indigo spinning, and weaving, among others.
- **Maize, red chilies, and tobacco, which were introduced during the 17th century or later, are, of course, missing.**
- **Hence option (b) is the correct answer.**

Q 55.B

- The concept of Khalisa land in Indian medieval history referred to land that was directly owned and managed by the state or the ruler. Income from Khalisa villages went directly to the royal exchequer. Hence, option (b) is the correct answer.
- **Inam land:** land granted as a reward or exemption from revenue to individual religious institutions and officials.
- **Jagir:** land grant on assignment given by the Mughal emperor to a Noble or a military officer in exchange for the services rendered. The Mughal governing class was usually paid either in the form of territorial revenue assignments or in cash from the treasury. The territorial assignments were called Jagirs. The holders were termed as Jagirdars.

Q 56.B

- **The Mahzar, and The Beginning of a New State Policy:**
 - *The Mahzar or attested statement signed by seven leading ulama*, including Shaikh Abdun Nabi and Abdullah Sultanpuri, and including Shaikh Mubarak, father of Abul Fazl, **issued in 1579**, has led to a good deal of controversy. Vincent Smith was of the opinion that, copying the examples of the Pope, **Akbar had conferred upon himself "the attributes of infallibility."** Some others thought that it was meant to free Akbar from the allegiance of the Ottoman Khalifa, and the Shia rulers of Iran. Some historians in Pakistan have dubbed it as "a dishonest document," because it permitted Akbar to interpret laws whereas he was hardly literate. Hence, a careful study of the document is necessary. **Hence option (b) is the correct answer.**
 - In the first place, the document declared that Akbar was "the Sultan of Islam, the asylum of mankind, the commander of the faithful, the shadow of God over worlds." These are the attributes of the khalifa of the age. There was no opposition in declaring Akbar as the khalifa of the age because the Timurids had never accepted any outside authority as khalifa. Hence, the question of countering Ottoman or Iranian claims hardly arises because such claims of allegiance over India were never made (except in the south in the case of the Shah of Iran).
 - Second, it was argued, citing Quran and some Hadis a few of them being spurious, that as a just and wise ruler Akbar not only had the right to claim the allegiance of everyone, but that his position was higher than a mutahid (interpreter of holy laws) in the eyes of God; third, should "a religious question arise in future, and the opinions of the mujtahids be at variance" the Emperor could adopt any one of them "for the welfare of mankind and proper functioning of the administrative affairs of the world."
 - Lastly, it was argued that Akbar himself could issue any decree which did not go against the nas i.e. explicit decree of Quran, and the hadis and is "calculated to benefit humanity at large." Any opposition to such a decree passed by His Majesty "shall involve divine displeasures in this world and the next."

Q 57.C

- The **Natyashastra** is a Sanskrit work on drama, performing arts, **theater, dance, music and various other topics**. The word natyashastra also refers to a global category of literature encompassing this ancient Indian tradition of dramatic performance. The work is authored by Bharata Muni. Other treatises on music after Bharata, such as the **Brihaddesi of Matanga, Sangeeta Ratnakara of Sharangadeva, Sangeet Sudhakara of Haripala, Svaramelakalanidhi of Ramamatya**, etc. **Hence option 1 is correct.**
- The **Brihaddeshi** authored by **Matanga Muni**, probably belonging to the sixth century CE, is a landmark in **Sangeetshashtra**. It is the **first extant text to describe Rāga, to introduce sā ri gā mā notation**, to usher in a fresh approach towards śruti, svara, grāma, mūrchanā, etc., and to introduce prabandha, the compositional form independent on drama, to establish the concept of deśi and its counterpart mārṅa and to mention ethnic groups like śabara, pulinda and nāga in the context of bhāṣās (varieties) of grāma-rāgas. **Hence option 2 is correct.**
- **Svaramelakalanidhi** is a **musicological treatise** written by Ramamatya, a minister under the Vijayanagar King Achutayaraya around 1550 AD. The King commissioned Ramamatya to write a treatise reconciling the contradictions existing at that time. It has 5 chapters - Preface, Swaraprakarana, Veenaprakarana, Melaprakarana and Ragaprakarana. **Hence option 3 is correct.**
- **Nandikesvara's Abhinaya Darpana** attributed to the 4th to 6th C. AD is a **standard text on dance Nrtya which is followed by many schools of classical Indian dance. It is not related to the evolution of music. Hence option 4 is not correct.**

Q 58.D

- **Jyotirlingas are a collection of 12 revered Shiva shrines in India.** They are considered to be the most sacred abode of Lord Shiva.
- **Mahakaleshwar**, which translates to the great lord of time. It is believed to be a Swayambhu shrine, meaning self-manifested and unlike other Shivalinga that are established through rituals. Mahakaleshwar is also revered as one of the Shakti Peethas sacred sites associated with the presence of Sati Devi, the consort of Lord Shiva. **It is located in Ujjain, Madhya Pradesh.**
- **Nageshwar Jyotirlinga is located near Dwarka in Gujarat.** It is believed that the demon Daruka once oppressed a devotee of Lord Shiva named Supriya. In response to Supriya's prayer for help, Lord Shiva appeared as Nageshwar Jyotirlinga and defeated the demon.
- **Mallikarjun Jyotirlinga is situated in the town of Srisailem, Andhra Pradesh.** It is located on the banks of the **Krishna River in Nallamala hills of Andhra Pradesh.** It is constructed in the Dravidian style of architecture. It is also associated with the role of Lord Shiva in resolving the dispute between Brahma and Vishnu.
- **Trimbakeshwar Jyotirlinga is located near the city of Nashik, Maharashtra. It is situated in Brahmagiri Hills, near the source of the Godavari River.** According to mythology, the sage Gautama performed penance here to please Lord Shiva, who was pleased with his devotion and released the sacred Ganga River from his matted locks. Thus, the Godavari River, also known as the Gautami Ganga.



- Hence option (d) is the correct answer.

Q 59.A

- The earliest Buddhist works were written in Pali, which was spoken in Magadha and South Bihar. The Buddhist works can be divided into the canonical and the non-canonical.
- The canonical literature is best represented by the **“Tripitakas”**, that is, three baskets - Vinaya Pitaka, Sutta Pitaka, and Abhidhamma Pitaka.
- **Vinaya Pitaka** deals with the rules and regulations of daily life. **Hence pair 1 is not correctly matched.**
- **Sutta Pitaka** contains dialogues and discourses on morality and deals with Dharma. **Hence pair 2 is not correctly matched.**

- **Abhidhamma Pitaka** deals with philosophy and metaphysics. It includes discourses on various subjects such as ethics, psychology, theories of knowledge, and metaphysical problems. **Hence pair 3 is correctly matched.**
- The non-canonical literature is best represented by the Jatakas. Jatakas are the most interesting stories about the previous births of the Buddha. It was believed that before he was finally born as Gautama, the Buddha practicing Dharma passed through more than 550 births, in many cases even in the form of animals. Each birth story is called a Jataka. The Jatakas throw invaluable light on the social and economic conditions ranging from the sixth century BC to the second century BC.

Q 60.B

- The rock-cut shrines at Ellora and Elephanta located in Maharashtra belong to the period of Rashtrakutas who made splendid contributions to Indian art.
- The Ellora cave complex contains the features of Buddhist, Hindu, and Jain monuments and artwork. **Amoghavarsha I** espoused Jainism and there are **five Jain cave temples at Ellora** ascribed to his period. **Hence option (d) is correct.**
- Jaina group of caves (caves 30 – 34) is exquisitely carved with fine, delicate sculptures, and includes fine paintings dedicated to the **Digambara sect. Hence option (c) is correct.**
- The most striking structure at Ellora is the creation of the **Monolithic Kailasanath Temple with well painted porch.** The temple was hewn out of a single rock during the time of Krishna I in the 8th century. It is similar to the Lokeshvara temple at Pattadakal, in Karnataka, built by Chalukya king Vikramaditya II to commemorate his victory over the Pallavas. **Hence option (a) is correct.** The sculptured panels of Dasavataara Bhairava, Ravana shaking Mount Kailasa, dancing Shiva and Vishnu, and Lakshmi listening to music are the best specimens of sculpture inside the temple.
- The **main shrine at Elephanta** is artistically superior to the shrines at Ellora. The sculptures such as **Nataraja and Sadashiva** excel even that of the Ellora sculptures in beauty and craftsmanship. **Ardhanarishvara and Maheshamurti** are the other **famous sculptures at Elephanta caves.** The latter, a three-faced bust of Lord Shiva, is 25 feet tall and considered one of the finest pieces of sculpture in India. **Hence, option (b) is not correct.**

Q 61.C

- **Nanofibers are ultrafine fibers with diameters typically ranging from a few nanometers (nm) to a few hundred nanometers.** To put this in perspective, a nanometer is one billionth of a meter, and the width of a human hair is about 80,000 nm. Super small!
- **Nanofibers possess unique properties derived primarily from their nanoscale dimensions.** Nanofibers are precisely engineered fibers that **are much smaller than traditional fibers and have many unique properties**, such as high strength, low weight, a high specific surface area, exceptional durability, high aspect ratio, and high surface area-to-volume ratio. They are very versatile and can be used to fabricate a variety of 3-dimensional structures with unique properties. Nanofibers have a significantly larger surface area per unit mass than bulk materials, making them suitable for applications where surface interactions are important. They also have a high length-to-diameter ratio, resulting in exceptional mechanical properties, including flexibility and strength. **Hence statement 1 is correct.**
- **Applications of nanofibres:** Nanofibers have found applications across various industries and fields, including biomedical and healthcare, environmental remediation, textiles and apparel, electronics, and energy.
 - **Medical Applications:** Nanofibers have made significant strides in the field of medicine. Researchers have developed nanofiber-based wound dressings that can accelerate wound healing by providing a conducive environment for tissue regeneration.
 - **Environmental Remediation:** Nanofibers are being used to address environmental challenges, such as water purification and air filtration. Electrospun nanofibers are useful for removing volatile organic compounds (VOC) from the atmosphere. **Hence statement 2 is correct.**
 - **Nano-Electronics:** Nanofibers hold great promise in the field of electronics. They can be used to create flexible and stretchable electronic components, including sensors and conductive textiles.
 - **Tissue Engineering:** Nanofibers are playing a crucial role in tissue engineering and regenerative medicine. Researchers are working on developing nanofiber scaffolds that mimic the extracellular matrix, promoting the growth of functional tissues and organs.

- **Energy Storage:** Nanofibers are being investigated for energy storage applications, particularly in the development of advanced batteries and supercapacitors. Their high surface area and porosity make them ideal for improving the performance of energy storage devices. **Hence statement 3 is correct.**
- **Space Exploration:** Nanofibers could find applications in space exploration due to their lightweight and strong characteristics. They could be used in the development of advanced materials for spacecraft and space suits, enhancing the durability and safety of space missions.
- **Sportswear textile:** Sportswear textile with a nanofiber membrane inside is based on modern nanofiber technology where the core of the membrane consists of fibers with a diameter 1000× thinner than human hair. **Nanofiber apparel and shoe membranes consist of polyurethane so its production is not harmful to nature.** Membranes to sportswear made from nanofiber are recyclable.

Q 62.A

- Recently, SpaceX's Falcon 9 rocket launched TSAT-1A satellite from the Kennedy Space Center, Florida, in the US on Sunday (April 7), Tata Advanced Systems Limited (TASL) said in a statement. TASL & Satellogic launched TSAT-1A, India's first private sub-metre resolution earth observation satellite. It is equipped with both multispectral and hyperspectral imaging capabilities. This technology allows TSAT-1A to collect data across a wide range of wavelengths within the electromagnetic spectrum, providing a deeper and more nuanced understanding of land, water, and various natural resources. **Hence statement 1 is correct and statement 3 is not correct.**
- TSAT-1A has been designed to deliver high-resolution imagery with remarkable precision, providing a resolution of less than one metre per pixel. The satellite is expected to cater primarily to the needs of the Indian government and armed forces, offering military-grade imagery for various applications.
- Tata Advanced Systems Ltd (TASL), a subsidiary of Tata Sons, has successfully launched India's first military-grade geospatial satellite built entirely by the private sector. TASL has partnered with Satellogic Inc., a leading provider of sub-metre resolution earth observation satellites, for the development of TSAT-1A. Satellogic's expertise in this field has been instrumental in the successful design and manufacture of the satellite. **Hence statement 2 is not correct.**

Q 63.C

- **A supercomputer is a high-performance, high-powered computer capable of processing massive quantities of data and performing complex calculations at lightning speed. It's put to use when a lot of processing power is needed.**
- **Pratyush** is one of the fastest supercomputers in India, developed by the Indian Institute of Tropical Meteorology (IITM) Pune in 2018. The primary function of this supercomputer is to predict monsoons and their severity accurately. The machine is currently being used by two government bodies which include IITM Pune and the National Center for Medium-Range Weather Forecasting (NCMRWF), Noida.
- **Mihir** is an Indian-built supercomputer also developed by the Indian Institute of Tropical Meteorology in the year 2018. Mihir has a maximum capacity of 2.5 petaflops. Pratyush and Mihir are used together to perform weather forecasting tasks.
- **The AI supercomputer 'AIRAWAT' is installed at C-DAC, Pune. AIRAWAT is a product of the National Artificial Intelligence Program, an initiative by the Indian Government.** The supercomputer has been named in the 61st edition of the Top 500 Global Supercomputing List released recently. Airawat PSAI, stands as India's largest and fastest AI supercomputing system, with a remarkable speed of 13,170 teraflops (Rpeak). AIRAWAT's manufacturer is Netweb Technologies.
- **Hence option (c) is the correct answer.**

Q 64.A

- **The Prime Minister laid the foundation stone of the second rocket launchport of the Indian Space Research Organisation (ISRO) at Kulasekarapattinam on February 28. Situated at a geographically advantageous location in coastal Tamil Nadu's Thoothukudi district, the facility costs Rs 986 crore. It will be extensively and exclusively used for commercial, on-demand, and small satellite launches in the future.**
- **With this, India will soon be among the elite league of nations that own and operate two or more spaceports - facilities from where rockets can be assembled and launched into space. So far, only the US, Russia, China, and Japan are known to operate two or more spaceports.** Having two or more spaceports offers flexibility and the ability to perform a greater number of launches, thereby increasing commercial activity.

- Sriharikota is best suited for launching heavier rockets into equatorial orbits (orbits where the satellite circles on a path above the earth's equator). Rockets launched from anywhere on the east coast (provided it is closer to the equator), are aided by the velocity of the earth's West-East spin and get an additional boost.
- However, when launching rockets towards polar orbits (orbits where the satellite circles above the earth's poles), the rockets that fly south from Sriharikota face the risk of flying over the Sri Lankan landmass. There are numerous risks of rockets flying over inhabited regions, given that rockets detach their used-up stages mid-flight. Rocketry is also a high-risk domain where mid-flight mishaps occur. Therefore, **all launches towards the south pole being performed from Sriharikota are done in a manner that the rocket takes a detour and avoids overflying Sri Lanka. Instead of flying directly and in a fuel-efficient manner, the rocket is made to take a diversion and spend more fuel.**
- **This affects the performance and payload-carrying capacity of rockets. This effect is not felt much in the larger rockets, but it affects smaller rockets, such as India's Small Satellite Launch Vehicle (SSLV) and the ones being built by Indian startups. Given the location of Kulasekarapattinam, just by the tip of the Indian peninsula, rockets can fly straight from the upcoming spaceport to the south pole, without having to overfly Sri Lanka. This way, the rockets fly in a fuel-efficient manner and can deliver maximum payloads to Polar orbits.**
- **Hence Statement-I and Statement-II are correct and Statement-II is the correct explanation for Statement-I**

Q 65.C

- **White Rabbit (WR) is a technology developed at CERN, in collaboration with institutes and companies, to synchronise devices in the accelerators down to sub-nanoseconds and solve the challenge of establishing a common notion of time across a network.** Indeed, at a scale of billionths of a second, the time light takes to travel through a fiber-optic cable and the time the electronics take to process the signal are no longer negligible. To avoid potential delays, the co-inventors of White Rabbit designed a new ethernet switch.
- First used in 2012, the application of this fully open-source technology has quickly expanded outside the field of particle physics. In 2020, it was included in the worldwide industry standard known as Precision Time Protocol (PTP), governed by the Institute of Electrical and Electronics Engineers (IEEE).
- White Rabbit is used in the finance sector as well as in many research infrastructures, and it is currently being evaluated for application in the future quantum internet. The technology could also play a key role in the future landscape of global time dissemination technologies, which currently rely heavily on satellites. The infrastructure for the dissemination of time is essential to the economy and underpins the most critical national infrastructure. Governments and industries across the globe are therefore striving to find alternatives to distribute a reference time, such as the one WR could offer via optical fiber, with telecom and power grid companies starting to test WR in their networks.
- **Hence option (c) is the correct answer.**

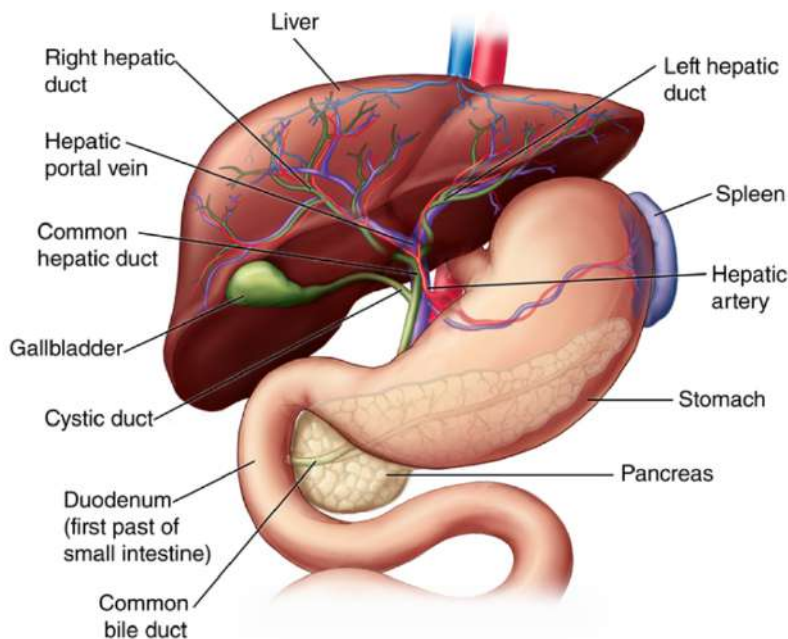
Q 66.A

- **Agni Prime is designed to carry both conventional and nuclear warheads.** This versatility allows for flexible use in various strategic scenarios, accommodating different mission requirements and enhancing the missile's utility in India's strategic arsenal. **Hence statement 1 is correct.**
- **Agni-P is a two-stage canisterised solid propellant ballistic missile with a dual redundant navigation and guidance system.** It has a range of 1,000-2,000km and was tested for the first time in June 2021. Since the missile is canisterised, it can be transported by road and rail and stored for longer periods, significantly reducing the time required for preparation and launch. According to reports, the missile uses a cold launch mechanism and can be fired in salvo mode. **Hence statement 2 is not correct.**
- **The 'Agni Prime' missile is lighter than all the earlier Agni series of missiles.** It weighs at least 50 per cent less than the Agni 3 missile and has new guidance and propulsion systems. It is designed to be more agile and advanced in terms of technology. It is a canisterized version with specific improvements aimed at mobility and flexibility rather than increased weight. **Hence statement 3 is not correct.**

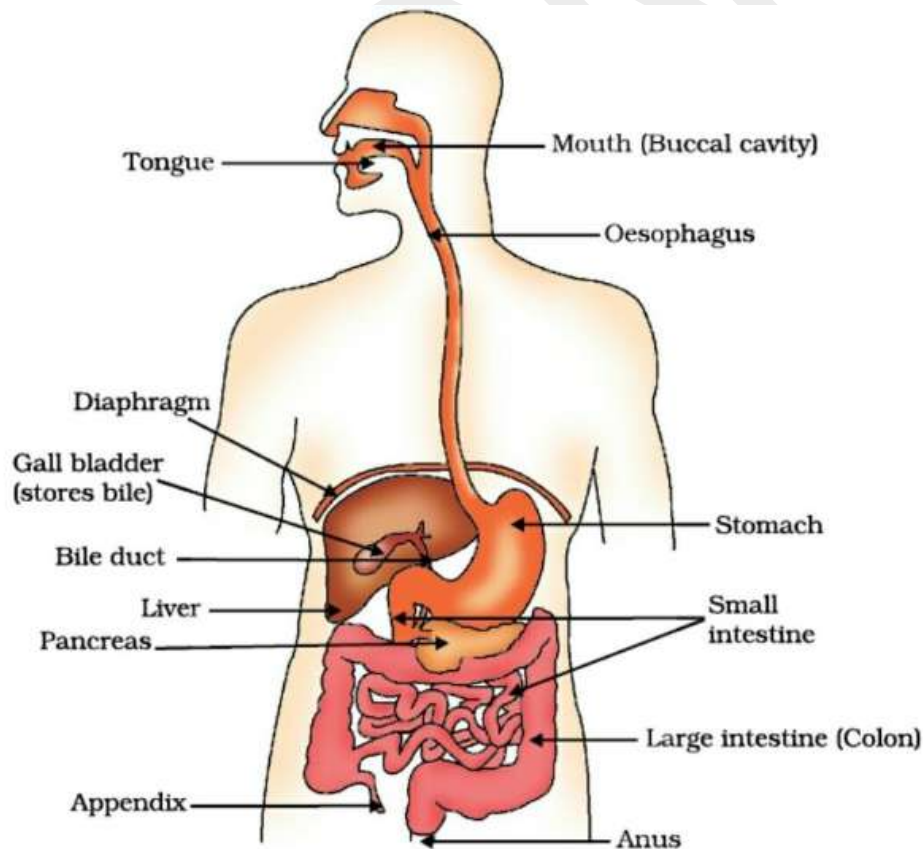
Q 67.B

- The human body uses the process of digestion to break down food into a form that can be absorbed and used for fuel. The organs of the digestive system are the mouth, esophagus, stomach, pancreas, liver, gallbladder, small intestine, large intestine and anus.

- The liver is located in the upper right-hand portion of the abdominal cavity, beneath the diaphragm, and on top of the stomach, right kidney, and intestines. Shaped like a cone, the liver is a dark reddish-brown organ that weighs about 3 pounds. Thus among the four parts given the liver lies at the top.



- The stomach is a J-shaped sac connecting the esophagus above and the small intestine below. The first part of the small intestine is known as the duodenum. Thus the stomach lies above the small intestine.



- The Vermiform Appendix lies at the bottom of the digestive system, attached to the large intestine. It is a cylindrical muscular structure attached to the large intestine, originating near the ileocecal valve from the cecal fundus or the posteromedial border of the cecum, approximately 2 cm from the junction of the cecum with the ileum.
- Hence option (b) is the correct answer.

Q 68.A

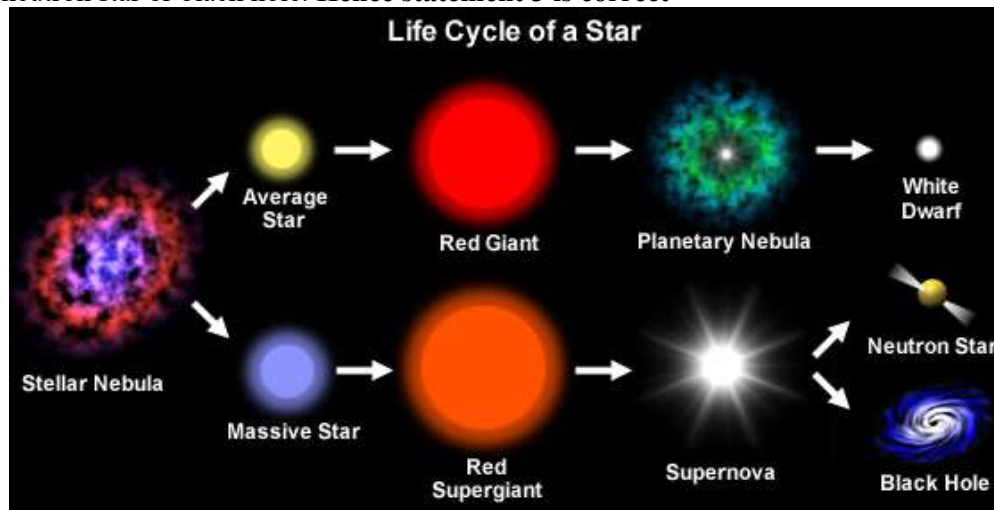
- In the decade from 1895 to 1905, three very important discoveries were made: J.J. Thomson proved the existence of the electron, Becquerel identified natural radioactivity, and Rontgen discovered X-rays. In addition, two important theoretical developments were put in place: Max Planck's hypothesis that radiant energy in its interaction with matter behaves like particles with discrete quanta of energy, and Einstein's Special Theory of Relativity. These events defined the beginning of modern physics.
- The birth of Quantum Mechanical Theory, followed by the Theory of Relativity, has totally changed the way physicists (and many other professionals) regard the world.
- The orbit picture (Bohr's model) gives the right answers for many physical properties, including the energy needed for the electron to jump from one energy (orbit) level to another one. This switch in energy (orbit) levels occurs when the electron gains or loses energy through the absorption or emission of a photon (which is a light particle with a certain amount of energy, also called a quantum of energy). In the more accurate quantum picture, the motion of the electron is described by probability functions and **there is no fixed orbit. Thus an electron, as per the orbital theory (quantum mechanics) in a stable atomic state does not move in the sense of a solid little ball zipping around in circles like how the planets orbit the sun, since the electron is spread out in a wave. Furthermore, an electron in a stable atomic state does not move in the sense of waving through space. The orbital electron does move in the sense of vibrating in time. Hence statement 1 is not correct.**
- When electrons pass through a double slit and strike a screen behind the slits, an interference pattern of bright and dark bands is formed on the screen. This proves that electrons act like waves, at least while they are propagating (traveling) through the slits and to the screen. The double-slit experiment shows the wave aspect of the electron, while the photoelectric effect shows the electron's particle aspect. The wave-particle duality developed from the research related to modern physics. As per the duality, an object may behave like a wave or like a particle, but never both simultaneously. **Hence statement 2 is not correct.**
- A wormhole is a hypothetical structure connecting disparate points in space-time, and is based on a special solution of the Einstein field equations. Einstein's theory of general relativity mathematically predicts the existence of wormholes, but none have been discovered to date. **Hence statement 3 is correct.**

Q 69.A

- **Most of the stars in the universe are main sequence stars — those that convert hydrogen into helium in their cores via nuclear fusion.**
- **Nebulae:** Stars begin their lives as clouds of dust and gas called nebulae. These nebulae contain hydrogen, helium, and trace amounts of other elements. Gravitational forces cause regions within the nebula to condense and form protostars.
- **Protostar:** As the protostar contracts due to gravity, it heats up. Eventually, the core becomes hot enough for nuclear fusion reactions to occur, primarily converting hydrogen into helium. At this point, the protostar becomes a main-sequence star.
- **Main Sequence:** The main-sequence phase is the longest stage in a star's life cycle. During this phase, the star is stable, with the inward pull of gravity balanced by the outward pressure generated by nuclear fusion in its core. Our Sun, for example, is currently in the main sequence phase.
- **Red Giant or Supergiant:** A red giant is a dying star in the final stages of stellar evolution. In about five billion years, our own sun will turn into a red giant, expand and engulf the inner planets — including Earth. Once a main-sequence star (like our sun) exhausts its core hydrogen fuel, it begins to evolve. For stars like our Sun, this evolution involves swelling into a red giant. For more massive stars, the process leads to becoming a supergiant. During this phase, the star's outer layers expand, while its core contracts. **Hence, statement 1 is not correct.**
- **Helium Fusion:** In the core of a red giant or supergiant, helium fusion reactions may occur, depending on the star's mass. These reactions convert helium into heavier elements such as carbon and oxygen.
- **Planetary Nebula or Supernova:** In the late stages of a star's life, what happens next depends largely on its mass. Low to medium-mass stars like our Sun, will eventually shed their outer layers, forming a planetary nebula, leaving behind a hot core called a white dwarf. In contrast, high-mass stars undergo a dramatic explosion known as a supernova. This explosion disperses heavy elements into space and may result in the formation of neutron stars or black holes.
- **White Dwarf, Neutron Star, or Black Hole:** After a low to medium-mass star (like our Sun) sheds its outer layers, the remaining core becomes a white dwarf, a dense object roughly the size of Earth but with the mass of a star. In the case of a high-mass star, if the core is massive enough, it may collapse into either

a neutron star or a black hole, depending on its mass. The sun will shrink down to become a dying ember of a star, known as a **white dwarf (and not a neutron star)**, only a little larger than Earth. **Hence, statement 2 is not correct.**

- **The Chandrasekhar limit** is the maximum mass that a stable white dwarf star can attain before it collapses into a neutron star or a black hole. The Chandrasekhar limit is approximately 1.4 times the mass of the Sun. If a white dwarf accumulates mass beyond this limit, it cannot support itself against gravitational collapse, leading to a cataclysmic event such as a Type Ia supernova or the formation of a neutron star or black hole. **Hence statement 3 is correct**



Q 70.B

- C-DOME, developed by Rafael Advanced Defense Systems, is essentially the naval version of the Iron Dome system. This advanced point defence setup is specifically designed with maritime vessels in mind. Its main objective is to provide effective protection against a broad range of threats such as rockets, artillery, mortars, and other airborne risks that sea-based vessels might encounter. **Hence statement 1 is correct.**
- The C-Dome is developed by Rafael Advanced Defense Systems, an Israeli defence contractor. The development and deployment are within the purview of Israeli technological and defence capabilities, with no mention of external collaborative efforts in the primary development stages from the provided sources. **Hence statement 2 is correct.**
- Unlike the Iron Dome, which has its own dedicated radar, the C-Dome is integrated into the ship's radar to detect incoming targets. C-Dome ensures "full-circular vessel protection and high kill probability against a full spectrum of modern threats—maritime and coastal." C-Dome uses the combat-proven Iron Dome interceptor that has already achieved more than 1,500 successful interceptions. Interceptors are maintenance-free and are stored in a sea-proof canister. **Hence statement 3 is not correct.**

Q 71.B

- Jal Jeevan Mission, is envisioned to provide safe and adequate drinking water through individual household tap connections by 2024 to all households in rural India.
- The programme will also implement source sustainability measures as mandatory elements, such as recharge and reuse through gray water management, water conservation, rain water harvesting.
- As part of Indian ethos to help in providing drinking water; various individuals, corporate/ industrial houses, charitable institutions, etc. contribute/ donate regularly.
- Further, as per Companies Act, 2013, every company having net worth beyond the prescribed thresholds during any financial year shall spend at least 2% of its average net profits towards Corporate Social Responsibility (CSR) activities.
- In order to facilitate all such donations/ contributions, Rashtriya Jal Jeevan Kosh is being set up under JJM which will serve as a receptacle for charitable contributions/ donations and CSR fund to achieve goals of JJM. **Hence statement 1 is not correct.**
- In view of its ambitious target to provide tap water connections to all rural households by 2024, the government will now focus on source sustainability through river rejuvenation and water harvesting works under the MGNREGA scheme. Under the JJM various central scheme are converged with JJM.

- All water conservation activities under Natural Resource Management (NRM) component of MGNREGA can be converged with JJM. Hence statement 2 is correct.
- Women empowerment is one of the components of Jal Jeevan mission. It entails Involvement of women in planning, decision-making, implementation, monitoring and O&M
- The operational guidelines of JJM mandate that a minimum of 50% of members of Village Water and Sanitation Committees shall be women. Hence statement 3 is correct.

Q 72.A

- **The Commission of Sati (Prevention) Act, 1987 was enacted in 1988.** The Act provides for the prevention of the commission of Sati and its glorification. Under this Act, attempt to commit Sati, abatement of commission of Sati and glorification of Sati is a punishable offence.
- Recognising the need to address the social evil of dowry, **the Dowry Prohibition Act was enacted in 1961.** The Act defines dowry and penalises the giving, taking or abetting the giving and taking of dowry. It also lays down a built-in implementation mechanism in the form of Dowry Prohibition Officers to ensure effective enforcement of the law.
- With regard to violence occurring within the private space of home, the key legislation is the **Protection of Women from Domestic Violence Act (PWDVA), 2005.** The objective of the law is to prevent violence and provide immediate and emergency relief in case of such situations irrespective of the status of woman's relationship with the respondent. The Act recognises women's right to live free from violence of any kind occurring within the family and for matters connected therewith or incidental thereto.
- **The Immoral Traffic (Prevention) Act, 1956 (ITPA)** is the central legislation dealing with the issue of prostitution in the country. The Act defines prostitution as commercial sexual exploitation and penalizes those who facilitate and abet commercial sexual exploitation, including clients and those who live off the earnings of prostitutes. It also provides for welfare measures towards rehabilitation of victims in the form of protective homes to be set up and managed by state governments.
- Hence the correct order is 4-2-1-3.

Q 73.B

- The curative jurisdiction and its applicability was recently in debate in the Delhi Metro Rail Corporation Ltd. ("DMRC") vs Delhi Airport Metro Express Pvt Ltd ("DAMEPL").
- The Supreme Court of India is a rather extraordinary court. In addition to its being the highest court of appeal, a federal court, as well as a court of advisory jurisdiction, it also possesses the power to appoint and transfer judges of the High Courts and the Court itself.
- **In 2002, the Court took on a new power called the "Curative Jurisdiction". It is a power to correct its judgments after they have become final. This is distinct from the power of review under Indian law, which enables all courts to rectify errors which are apparent from their records. In 2002, in Rupa Hurra v Ashok Hurra, the SC allowed curative writs as a last resort to correct judgments that are "oppressive to judicial conscience and would cause perpetuation of irremediable injustice." It is allowed to negate grave miscarriage of justice. Hence, statements 1 and 2 are correct.**
- **The Court has a constitutional role to declare the law. The law must, and often does, progress with the growth in human knowledge and with societal change. The judgments of courts must reflect and sometimes trigger the changes in law. It is for this reason that courts modify their views. Examples of changes in the Court's views include the right of privacy, the decriminalization of homosexuality and so on. Curative Jurisdiction is different. This is not merely the Court changing its view on a position of law but is a reversal of the Court's own view in a specific case, above and beyond even the power of review. Hence, statement 3 is not correct.**
- A curative petition may be filed after a review plea against the final conviction is dismissed. It is meant to ensure there is no miscarriage of justice and to prevent abuse of process. A curative petition is usually decided by judges in chamber, unless a specific request for an open-court hearing is allowed. The Hurra judgment held that technical difficulties and apprehensions over the reopening of cases had to give way to a final forum for removing errors in a judgment where the administration of justice may be affected.
- The court ruled that a curative petition can be entertained if the petitioner establishes there was a violation of the principles of natural justice, and that he was not heard by the court before passing an order. It will also be admitted where a judge failed to disclose facts that raise the apprehension of bias.
- The SC has held that curative petitions must be rare rather than regular, and be entertained with circumspection. A curative petition must be accompanied by certification by a senior advocate, pointing out substantial grounds for entertaining it. It must be first circulated to a bench of the three senior-most

judges, and the judges who passed the concerned judgment, if available. Only when a majority of the judges conclude that the matter needs hearing should it be listed — as far as possible, before the same Bench.

Q 74.C

- Recently, the Additional Registrar General of India communicated to States in January 2024, that the date of freezing of administrative boundaries has been extended till June 30. The Census can only begin three months after the boundaries have been frozen. With this recent notification, the Census 2021 may be postponed to October 2024.
- India's first proper or synchronous Census, one which begins on the same day or year across regions of the country, was carried out in **1881 by the colonial administration and has since happened every 10 years, except the one that was supposed to be carried out in 2021.**
- The Census is still conducted under the **Census Act of 1948**, which predates the Constitution. Notably, the Act does not bind the government to conduct the Census on a particular date or to release its data in a notified period. The Centre's intent to conduct the 2021 Census was notified in the Gazette of India on March 28, 2019. The freeze on administrative boundaries was to be effective from January 1, 2020 to March 31, 2021, before the COVID-19 pandemic.
- The decennial census is carried out by lakhs of enumerators empanelled and trained by the government in two phases.
- **The first phase is the housing Census**, where data on housing conditions, household amenities and assets possessed by households are collected and **the second phase is where data on population, education, religion, economic activity, Scheduled Castes and Tribes, language, literacy, migration, and fertility are collected.**
- The freezing of boundary limits of administrative units such as districts, sub-districts, tehsils, and police stations, happens between two consecutive censuses as State administrations often create new districts or merge, or reorganise the existing units.
- **The Census data is crucial for various administrative functions, welfare schemes, and other surveys.**
 - **The Finance Commission allocates funds to States on the basis of Census figures and any delay could put them at a disadvantage.**
 - Besides, outdated Census information (available from the last Census in 2011) often becomes unreliable and affects those who do and do not receive the benefits of welfare schemes. **As per the National Food Security Act, 2013, 75% of the rural population and 50% of the urban population — totalling 67% of the country's population — are entitled to receive subsidised food grains from the government under the targeted public distribution system (PDS).** According to the 2011 Census, India's population was about 121 crore, and PDS beneficiaries were approximately 80 crore. However, economists Jean Dreze and Reetika Khera have pointed out that population growth over the last decade means that if the 67% ratio is applied to 2020's projected population of 137 crore, PDS coverage should have increased to around 92 crore people.
- Besides, the **Census is crucial to determine the population of migrants and migration patterns.** The start of the pandemic saw a sea of migrants on the country's roads, and the only data available with the government was from 2011, which could not answer queries on the numbers, causes and patterns of migration.
- According to the **106th Constitutional Amendment Act, 2023, also known as the Nari Shakti Vandan Adhiniyam, the one-third reservation of seats for women in the Lok Sabha and State Assemblies shall come into effect after an exercise of delimitation is undertaken** based on the relevant figures of the first Census recorded after the Act has commenced. Delays in the census 2021 can impact its speedy implementation.
- **Every Census in independent India from 1951 to 2011 has published data on Scheduled Castes and Scheduled Tribes, but not on other castes. Before that, every Census until 1931 had data on caste. For 2021 too, the Government of India has decided as a matter of policy not to enumerate caste-wise population other than SCs and STs in Census.**
- Hence, option (c) is the correct answer.

Q 75.D

- One of the major issues with municipal elections is that they are not being held on time. This is in violation of the Constitution. As per Janaagraha's Annual Survey of India's City-Systems 2023 study, over 1,400 municipalities did not have elected councils in place, as of September 2021.
- **The CAG's audit reports of 17 States on the implementation of the 74th Constitution Amendment Act (74th CAA) observes that over 1,500 municipalities did not have elected councils in place during the audit period of 2015-2021 across States. Among larger cities, the Greater Chennai Corporation had elections in 2022 after a gap of nearly six years, and the Municipal Corporation of Delhi had elections after a delay of seven months, while the municipal corporations of Mumbai and Bengaluru are awaiting elections for over one and a half years and three years after the expiry of the term of their previous elected councils. Hence, statement I is not correct.**
- Where elections to urban local governments were held, in certain cases, councils were not constituted, and elections of mayors, deputy mayors and standing committees were delayed. In Karnataka, there was a delay of 12-24 months in the formation of elected councils after the declaration of election results in most of the 11 city corporations. As per reports, in 214 urban local governments (out of 304) in Karnataka, there was a 26-month delay in the formation of councils and election of chairpersons and standing committees for the first 2.5-year term, after election results were announced in September 2018. Further, after the expiry of the first term of 2.5 years in May 2023, the same urban local governments did not hold elections for chairpersons and standing committees for more than eight months as of 2023.
- **Article 243U of the 74th CAA stipulates that the duration of urban local governments is five years and that an election to constitute an urban local government should be completed "before the expiry of its duration". Further, in case of dissolution of the elected council by the State, the election should be held before the expiration of a period of six months from the date of its dissolution. Despite the Supreme Court stating in Suresh Mahajan v. State of Madhya Pradesh (2022) that this constitutional mandate is inviolable, State governments do not hold timely elections for urban local governments. Hence, statement II is correct.**

Q 76.C

- Cooperative federalism is a model of intergovernmental relations that recognizes the overlapping functions of the national and state governments. This model can be contrasted with the model of dual federalism, which maintains that the national and state governments have distinct and separate government functions.
- In general, cooperative federalism asserts that governmental power is not concentrated at any governmental level or in any agency. Instead, the national and state governments share power. For instance, bureaucratic agencies at the national and state level normally carry out governmental programs jointly. Because the governments' responsibilities are split between many levels of government, citizens and organized interests have many access points to influence public policy.
- **Interstate Council is a mechanism that was constituted "to support Centre-State and Inter-State coordination and cooperation in India". The Inter-State Council was established under Article 263 of the Constitution, which states that the President may constitute such a body if a need is felt for it. The Council is basically meant to serve as a forum for discussions among various governments. The main functions of the Council are inquiring into and advising on disputes between states, investigating and discussing subjects in which two states or states and the Union have a common interest, and making recommendations for the better coordination of policy and action. Thus, it works on the idea of cooperative federalism.**
- **NITI Aayog, formed in 2015 has replaced the erstwhile Planning Commission. It has been constituted to actualise the important goal of cooperative federalism and to enable good governance in India.** On the premise that strong states make a strong nation, NITI Aayog acts as the quintessential platform for the Government of India by bringing States together as 'Team India' to work towards the national development agenda.
- **The One Hundred and First Amendment to the Constitution, which introduced the Goods and Service Tax (GST) regime, is a watershed moment in the evolution of cooperative Federalism as enshrined in the Constitution of India.** Before the implementation of GST, the Indian taxation system was a medley of Central, State and Local area levies. While structuring a completely new indirect tax system, first challenge was to preserve the federal nature of the Indian polity and thus, 101st Amendment Act was introduced. Article 246A confers concurrent jurisdiction of both the Union and State to levy tax on supply of goods and services, thus, preserving the federal nature of the Constitution. Hence, option (c) is the correct answer

Q 77.D

- Fundamental Duties are our responsibilities towards our country. Under Sardar Swaran Singh's recommendation, the fundamental duties are included in the constitution of India.
- The Fundamental Duties are dealt with in Article 51A under Part-IV A of the Indian Constitution
- 51A. Fundamental duties. It shall be the duty of every citizen of India—
 - (a) to abide by the Constitution and respect its ideals and institutions, the National Flag and the National Anthem;
 - (b) to cherish and follow the noble ideals which inspired our national struggle for freedom;
 - (c) to uphold and protect the sovereignty, unity and integrity of India;
 - (d) to defend the country and render national service when called upon to do so;
 - (e) **to promote harmony** and the spirit of common brotherhood amongst all the people of India **transcending religious, linguistic and regional or sectional diversities**; to renounce practices derogatory to the dignity of women;(f) to value and preserve the rich heritage of our composite culture;
 - (g) to protect and improve the natural environment including forests, lakes, rivers and wild life, and **to have compassion for living creatures**;
 - (h) to develop the scientific temper, humanism and the spirit of inquiry and reform;
 - (i) to safeguard public property and to abjure violence;
 - (j) **to strive towards excellence** in all spheres of individual and **collective activity** so that the nation constantly rises to higher levels of endeavour and achievement;
 - (k) who is a parent or guardian to provide opportunities for education to his child or, as the case may be, ward between the age of six and fourteen years.
- **Hence option (d) is the correct answer.**

Q 78.A

- **Chandra Shekhar became the first Prime Minister of India (November 10, 1990 to June 21, 1991) to head a coalition government at the centre. He was instrumental in preventing India from defaulting on sovereign repayments. Hence, statement 2 is not correct.**
- **Vishwanath Pratap Singh served as Prime Minister of India from 1989 to 1990. The Mandal Commission Report was implemented and the Scheduled Caste and Scheduled Tribe Act of 1989 was passed during his tenure. Hence, statement 1 is correct.**
- Manmohan Singh served as Prime Minister from 2004 till 2014. He was a member of Rajya sabha during his term of office. The National Rural Employment Guarantee Act (NREGA) and the Right to Information Act were passed during his tenure as Prime Minister. **But he was not the first PM to who was a member of Rajya Sabha.**
- **Indira Gandhi, H.D. Deve Gowda and Inder Kumar Gujral were also the prime ministers who hailed from Rajya Sabha. Hence, statement 3 is not correct.**

Q 79.A

- **Sitting on troves of unexplored critical mineral resources such as cobalt, copper, and lithium, the Democratic Republic of Congo (DRC), Tanzania, and Zambia are emerging as theatres of great power competition in Africa. A new infrastructure undertaking which was conceived under the United States' (US) Partnership for Global Infrastructure and Investment (PGII) initiative—the Lobito Corridor, connecting these three countries, is rapidly emerging as a counter to China's Belt and Road Initiative (BRI) in these countries. Hence, option (a) is the correct answer.**
- Located in Central Africa, the Corridor proposes to build a railway line connecting the critical minerals mines in Zambia and DRC with the Lobito port in Angola. The project also proposes investments in green energy development, sustainable mining, energy storage to social infrastructure development and public health initiatives. The wide array of economic sectors covered under the PGII's Lobito Corridor is aimed at presenting a counterweight to Chinese presence in the region.
- The aim is to link Copperbelt mines directly with the Atlantic Ocean, reducing both the cost and the carbon foot-print of the current trucking corridor to South African ports.

Q 80.A

- The State Energy Efficiency Index (SEEI) has been developed by Bureau of Energy Efficiency (BEE) in collaboration with Alliance for an Energy Efficient Economy (AEEE) to track EE initiatives in the states and UTs. It provides insights about sub-national energy efficiency policies, programmes, and investments.
 - SEEI 2023 is the fifth edition of the index, following the successful execution of State Energy Efficiency Preparedness Index 2018, SEEI 2019, SEEI 2020, and SEEI 2021-22.

Objectives



- SEEI 2023 assesses the EE progress of the 36 states and UTs for FY 2022-23 using 65 qualitative, quantitative, and outcome-based indicators distributed across 7 demand sectors, namely: Buildings, Industries, Municipal Services, Transport, Agriculture, DISCOMs, and Cross-Sector. The maximum score for the index is 100.
- The Government of India set up Bureau of Energy Efficiency (BEE). on 1st March 2002 under the provisions of the Energy Conservation Act, 2001. The mission of the Bureau of Energy Efficiency is to assist in developing policies and strategies with a thrust on self-regulation and market principles, within the overall framework of the Energy Conservation Act, 2001 with the primary objective of reducing energy intensity of the Indian economy.
- Alliance for an Energy Efficient Economy (AEEE) was formed in 2008 as a not-for-profit society with the vision and financial support provided by various proactive energy efficiency equipment and services companies- Conzerv Systems Pvt Ltd (CSPL – since acquired by Schneider-Electric India in 2009), Schneider Electric, Thermax Limited, Grundfos Pumps India and Development Enviroenergy Services Ltd, and other practitioners and energy service companies to create a platform for the private sector to engage with government and help create an energy services market in India.
- **Hence option (a) is the correct answer**

Q 81.D

- Microalgae are usually made from one single cell, or a small number of cells put together in a very simple structure that can quickly grow and multiply into a large biomass rich in nutrients.
- Microalgae can grow in freshwater, marine, and highly saline environments. Microalgae are small-sized organisms found in fresh and saline waters, in both benthic and littoral habitats, and also throughout the ocean. **Hence statement 1 is not correct.**
- Seaweed is one of the most common types of microalgae formed in water bodies as a result of eutrophication. However Seagrasses are marine plants. Seaweed lacks the organized structure of true roots, stems, and leaves, while seagrass exhibits a more complex plant structure with distinct roots, stems, and leaves. **Hence statement 2 is not correct.**
- However Microalgae are not only safe to eat, but they're also associated with numerous health benefits. Microalgae are also a promising source of antioxidant vitamins such as provitamin A (β -carotene), vitamin B, vitamin C, and E, as well as folic acid, inositol, and biotin. **Hence statement 3 is not correct.**

Q 82.B

- **United States Agency for International Development (USAID), an independent agency of the United States government primarily responsible for administering civilian foreign aid and development assistance has organized a virtual 'Global Summit on Extreme Heat' along with the International Federation of Red Cross and Red Crescent Societies (IFRC).**
- It brought together political and civil society leaders, and representatives of the private sector and the world's most affected communities.
- The year 2023 was the hottest on record. It coincided with deadly heat waves on nearly every continent, from Pakistan to Tunisia to Texas.

- The IFRC and USAID have jointly announced a “sprint of action” on extreme heat which will run up to the global heat action day on 2 June.
- Hence option (b) is the correct answer.

Q 83.A

- The Northern river terrapin is found in India and Bangladesh (Sundarbans), Myanmar, Malaysia (peninsular), Indonesia (Sumatra), Thailand, and Cambodia. It is regionally extinct in Myanmar, Singapore, Thailand and Vietnam. **Hence statement 1 is not correct.**
- It lives in coastal mangrove estuaries and creeks, but ventures far upstream during the breeding season. An omnivore, it feeds on waterside plants and small animals such as clams. **Hence statement 2 is correct.**
- Females nest in sandbars and banks far upstream from the normal estuarine habitat during the breeding season, December to March.
- The northern river terrapin is one of Asia's largest freshwater and brackwater turtles, reaching a carapace length of up to 60 cm and a maximum weight of 18 kg. **Hence statement 3 is not correct.**
 - **Olive ridley turtles** are among the smallest of the world's sea turtles and is found primarily in the tropical regions of the Pacific, Indian, and Atlantic oceans.



Q 84.C

- **Brumation** is a physiological response observed primarily in reptiles, particularly those living in temperate or colder regions. It is similar to hibernation in mammals but differs in some key aspects.
- **During brumation, reptiles and amphibians undergo a period of reduced activity, metabolism, and energy expenditure.** They may seek out sheltered locations such as underground burrows, rock crevices, or other protected areas where temperatures are more stable and warmer than the external environment.
- Unlike hibernation, which is typically triggered by low temperatures and reduced food availability during winter, brumation can occur in response to various environmental cues such as changes in photoperiod (day length) and temperature. It is a way for these ectothermic animals to conserve energy and survive through periods of environmental stress.
- During brumation, the animal's metabolic rate decreases, its heart rate slows down, and it may go for extended periods without eating. However, it is not a deep sleep like hibernation, and the animal may still be somewhat responsive to external stimuli.
- Overall, brumation is a vital adaptive strategy that enables reptiles and amphibians to cope with seasonal changes in their environment and enhance their chances of survival.
- **Hence option (c) is the correct answer.**

Q 85.B

- **Recently, Goa has set up a Blended Finance Facility in the State in partnership with the World Bank.**
- This will be first-of-its-kind, climate-focused, multi-sectoral facility at a subnational level that will enable Goa to access and mobilize concessional finance to implement low-carbon and climate-resilient investments.

- Blended finance- inclusive of public, private, international, and domestic climate finance- has the potential to help Goa and other coastal states in India build greater resilience to the impacts of a rapidly warming climate.
- Blended Finance facility will play key roles in assisting projects with existing funding and/or established commercial viability to access financing, and to channelize financing to projects/sectors with no existing funding, below market rate returns, or unattractive risk/return profile.
- Blended Finance facility will serve as a crucial bridge between green financing mechanisms and domestic capital markets, attracting private capital investments for climate action. By integrating carbon markets and harnessing green credits, the programme will ensure its sustainable operation and continuous re-capitalization.
- **Hence option (b) is the correct answer.**

Q 86.D

- Before the 1860s, three-fourths of raw cotton imports into Britain came from America. In 1857 the Cotton Supply Association was founded in Britain, and in 1859 the Manchester Cotton Company was formed. Their objective was “to encourage cotton production in every part of the world suited for its growth”.
- **When the American Civil War broke out in 1861**, a wave of panic spread through cotton circles in Britain. Raw cotton imports from America fell to less than three per cent of the normal. Frantic messages were sent to India and elsewhere to **increase cotton exports to Britain. As cotton prices soared, export merchants in Bombay were keen to secure as much cotton as possible to meet the British demand.** So they gave advances to urban sahuakars who in turn extended credit to those rural moneylenders who promised to secure the produce. When there is a boom in the market credit flows easily, for those who give out loans feel secure about recovering their money. **Hence Statement-I is incorrect but Statement-II is correct.**
- The ryots in the Deccan villages suddenly found access to seemingly limitless credit. **By 1865, as the Civil War ended, cotton production in America revived and Indian cotton exports to Britain steadily declined.** Export merchants and sahuakars in Maharashtra were no longer keen on extending long-term credit. The refusal of moneylenders to extend loans enraged the ryots. Ryots had got deeper and deeper into debt, or that they were utterly dependent on the moneylender for survival, but that moneylenders were being insensitive to their plight. The moneylenders were violating the customary norms of the countryside. The ryots came to see the moneylender as devious and deceitful. They complained of moneylenders manipulating laws and forging accounts. **In 1859 the British passed a Limitation Law that stated that the loan bonds signed between moneylenders and ryots would have validity for only three years.** The moneylender, however, turned the law around, forcing the ryot to sign a new bond every three years.
- **Deeds and bonds appeared as symbols of the new oppressive system.** Over time, peasants came to associate the misery of their lives with the new regime of bonds and deeds. It led to Deccan Riots of 1875 which targeted conditions of debt peonage (kamiuti) to moneylenders. The rioters' specific purpose was to obtain and destroy the bonds, decrees, and other documents in the possession of the moneylenders. Hence statement -I is correct.
- **Hence option (d) is the correct answer.**

Q 87.A

- **Pitt's India Act 1784, named after British Prime Minister William Pitt, was a significant piece of legislation.** It established a system of dual control with creation of the Board of Control under the British Government to supervise the East India Company's civil, military and revenue affairs. **For the first time, the East India Company's territory in India were referred to as British possession in India. It also provided a prohibition on the aggressive war and treaties.**
- **With a parliamentary prohibition on the imperial expansion, the major policy of the Board of Control during that time was to protect British possessions and promote trading interests through a careful balance of power between the Indian states, thus reducing the imperial military liabilities. However, that cautious policy was abandoned when Lord Wellesley arrived as Governor-General in 1798 with a dream of building empire and used 'subsidiary alliance' to expand British influence and control over the subcontinent. Hence statement I is correct.**
 - Under subsidiary alliance, a treaty was signed between the British East India Company and Indian princely state, where Indian rulers accepted a permanent contingent of the British troops stationed within their territory.

- They also paid for the maintenance of these British forces and disband their own armies.
- A British resident was also placed in their court acting as a political advisor.
- The Indian rulers need to sought British permission before entering into any foreign alliance or engaging in warfare.
- **Napoleon Bonaparte invaded Egypt in the summer of 1798.** Although Lord Wellesley personally never believed for a moment that there was any danger of French invasion of British India either overland from Egypt or a naval attack around the Cape of Good Hope. But Lord Wellesley saw this an opportunity to soften the Parliament's resistance to expansionist policy. Hence, to assuage the London's concerns, he evolved the policy of Subsidiary Alliance which would only established control over the internal affairs of an Indian state without incurring any direct imperial liability. The policy of Subsidiary Alliance was more of Lord Wellesley's personal agenda for expansion rather than a genuine threat from French invasion. Hence statement II is correct.

Q 88.B

- **Sir Herbert Hope Risley was an Indian civil servant and British ethnographer.** He was also part of the Statistical Survey of India in 1869 conducted by another civil servant William Wilson Hunter. He was appointed as Under Secretary in the Home Department of the Government of India, where he was involved in ethnographic study of India.
- **After the 1857 rebellion the British Government wanted to study and understand the Indian diversities to avoid further discontent.** Risley in 1891 published four volumes of the book titled "The Tribes and Castes of Bengal" with the first two volumes comprising "Ethnographical Glossary" and another two containing "Anthropometric Data". Risley also viewed the continued practice of endogamy had ensured strict delineations of the various communities by caste and that consequently caste could be viewed as identical to race.
- He also saw a linkage between the nasal index and the definition of a community as either a tribe or a Hindu caste. He believed that the caste system had its basis in race rather than in occupation.
- **In 1899 he was appointed Census Commissioner,** tasked with reporting on the decennial census of 1901. He formally applied the caste system to the entire Hindu population of British India in the 1901 census. The census of 1901 revealed that the Bengal population reached 78.5 million.
- **The Viceroy Lord Curzon drew up a scheme for the partition of Bengal in his Minute on Territorial Redistribution in India,** it was later published as the Risley Papers on 3rd December 1903. It proposed the transfer of Chittagong, Dacca and Mymensingh districts to Assam and Chota Nagpur to the Central Province. The partition of Bengal was formally announced on 19th July 1905 and implemented on 16th October 1905. **Hence option (b) is the correct answer.**
- **The official reason for the partition of Bengal was an administrative one.** However, Sumit Sarkar highlighted that the administrative reason was not hold ground because Lord Curzon rejected another best alternative plan on linguistic division rather than religious one on political ground that linguistic unity would further consolidate Bengali people. It became clearer when Herber Risley in 1904 wrote "Bengal united is a power".

Q 89.D

- **After the abolition of slavery in the British Empire, there was a need for a new source of cheap labour in European colonies.** British India, with its large population and widespread poverty, became a prime target for the recruitment of cheap labour. They adopted a new instrument called indentured labour. **Under this system, a labour signed a contract to work for an employer for a specific amount of time generally of five years. Hence statement 1 is not correct.**
- In 19th century, indenture has been described as a 'new system of slavery'. On arrival at plantation, labourers found conditions to be different from what they had imagined. Living and working conditions were harsh and, there were few legal rights. Initially, the agreement was for a fixed period, however, most of indentured workers stay on after their contract ended or return to their new homes after a short spell in India.
- **Most of the indentured labourers were employed particularly for sugar plantation in Caribbean, Mauritius, Fiji and other part of Africa.** Consequently, there are large communities of people of Indian descent in these countries. Nobel Prize winning writer V.S. Naipaul captured the sense of loss and alienation of indentured workers in these countries. **Indentured labourers were not just employed outside India, but they were also employed in the plantations of Assam and other regions as well. Hence statement 2 is not correct.**

- In 1998, UNESCO designated August 23 as the International Day for Remembrance of the Slave Trade and Abolition to commemorate the tragedy of slave trade in the memory of all peoples. In Mauritius, the Immigration Depot or the Aapravasi Ghat in Port Louis was declared a UNESCO World Heritage Site in 2006. **Mauritius was the first British colony to receive indentured labour.** On the bank of Hooghly, near the port of Kolkata, the Suriname Ghat is named after one of the colonies. At the Suriname Ghat, the Mai-Baap statue was erected. This statue is a replica of the Baba and Mai monument in Suriname that marked the first Indian migrant in Suriname. **Hence statement 3 is not correct.**

Q 90.C

- The **Indian languages can be divided into 6 groups** based on the development of different ethnic groups. These are Indo-Aryan, Dravidian, Sino-Tibetan, Negroid, Austric and others.
- Recently the Odisha Cabinet has proposed including the Kui language in the 8th Schedule of the Indian Constitution. **Kui language belongs to the Dravidian family of languages and is spoken mainly by the people of the Kondh/Kandha tribe, the largest tribe of Odisha.** Kui language uses Odia script and has two dialects namely Kondi and Gumsai. **Hence option (c) is the correct answer.**
- Other languages in the Dravidian family of languages are Kudu, Kolami, Konda, Koya, Gundi, Oraon, Parji etc. Kaikadi is a major Dravidian tribal language spoken in several parts of Karnataka and Maharashtra.
- **Tibeto-Burman tribal language** is mainly spoken in the northern and northeast regions of India. Some of the prominent Tibetan-Burman family languages are Bodo, Mikir, Lushai, Mizo, Rabha, etc. Mizo language is spoken by Mizo people
- **Indo-Aryan tribal language** is one of the most prominent speaking languages in India mainly in northern, eastern, western, and central parts. Bhili, Khandesi, Halabi, Koch, Kishtwari, Konkani, etc. are the prominent languages of this family. Katkari tribes of Maharashtra mainly speak the Katkari language.
- **Austro-Asiatic tribal language** is mainly spoken in the state of West Bengal, Jharkhand and some other parts of north eastern states of India. Kharia, Santhali, and Khasi are the prominent languages of this family. Khasi language is spoken by the Khasi tribes of Meghalaya.
- **8th Schedule of Indian Constitution:** This schedule deals with the officially recognized languages of India. It lists the 22 languages. Originally it had 14 languages. However, with subsequent constitutional amendments, more languages are added to it, the latest one is Bodo through the 92nd Amendment Act in 2003. There are no fixed criteria for a language to be considered for inclusion in the 8th Schedule. Though two committees have been constituted namely Pothota in 1996 and Mohapatra in 2003 to evolve a fixed criteria, however, this objective is yet achieved.

Q 91.B

- **Khalimpur copper plate inscription was showcased by India during its G20 presidency because of its significance in depicting the democratic culture in India.** This historical artifact is inscribed with details about the Pala Empire. This plate inscription mentions about excavations of a lake at Mahabodhi, the establishment of an image of Chaturmukha Mahadev, and the military victories of the Pala King.
- This copper inscription belongs to the Pala ruler Dharmapala found at Khalimpur of Maldah district of West Bengal. This inscription mentions the state of affairs that prevailed in Bengal before the rise of Gopala as 'matsyanyayam'. Apart from this, it also talks of **how King Gopala was elected by the people to replace an unfit ruler.** Hence option (b) is the correct answer.
- **Doddahundi inscription mentions the ritual death through Sallekhana and Samadhi of the western Ganga dynasty king Ereganga Nitimarga I in the 19th century AD.** The spot where an ascetic, king or queen has died voluntarily is known as Nishidhi in Jainism.
- **Sohgaura Copper plate inscription is a pre-Mauryan record that mentions famine relief effort. It records the setting up of two-grain depots to fight against the famine.** It was written in the Prakrit language using Brahmi script. It is the oldest known copper plate inscription.
- **Halmidi inscription is the oldest known Kannada language inscriptions in the Kadamba script dated back around 450-500 AD.** It is the earliest evidence of the use of Kannada as an administrative language.

Q 92.C

- The city of Fatehpur Sikri was founded by Akbar (ruled from 1556 to 1605) as a token of gratitude to Sheikh Salim Chisti. Hence statement 1 is correct.
- The construction of the city began in 1569 and was completed in 1574. The city is a modest and compact township, consisting of halls, palaces, offices, gardens, pleasure resorts, baths, mosques, and tombs. Almost all the structures are based on trabeate construction. Hence statement 3 is correct.
 - The Muslim style of construction was based on arches, vaults, and domes, on columns and pyramidal towers or slender spires, called the trabeate style of architecture.
 - Only one building at Fatehpur Sikri is built of white marble, the tomb of Sheikh Salim Chisti, Akbar's spiritual preceptor. It is a square chamber, having a screened verandah, containing latticed panels of exquisite design, and is an elegant structure. Hence statement 2 is correct.
 - Fatehpur Sikri was a town planned as an administrative unit consisting of public buildings as well as private residences.
- Fatehpur Sikri has a long cultural history prior to it.
 - Sikri has been mentioned in the Mahabharata as 'Saik' meaning 'a region surrounded by water'.
 - An inscription found on the stone sculpture of Jaina Saraswati (dated 1010 A.D.) discovered from excavation at Bir Chhabili Tila mentions this place as 'Sekrikyā', which seems to be a similar derivative.
- Babur visited here on the eve of the Khanwah battle in A.D. 1527 and mentioned it as 'Sikri' in his Memoirs. He founded here a garden, a Jal-Mahal (water palace) and a baoli (step-well) to commemorate his victory in the Khanwah battle.
- Akbar gave it the name of Fatehabad after victory over Gujarat and which in later days "came to be known as Fatehpur Sikri".
 - After Gujarat victory he constructed Buland Darwaza of Dargah Complex which now regarded as the tallest gateway of Asia.
 - English traveler Ralph Fitch who visited Fatehpur Sikri in 1585 A.D. mentions "Agra and Fatepore are two very great cities, either of them much greater than London and very populous". However, in 1585 A.D., Akbar and his court left Fatehpur Sikri for Lahore never to return again.
 - Political exigencies made him move his capital to Lahore.
 - Jahangir stayed here for three months and the 28th Solar birthday of the prince Khurram was celebrated here in 1619 A.D.
 - Babur visited here on the eve of the Khanwah battle in A.D. 1527 and mentioned it as 'Sikri' in his Memoirs. He founded here a garden, a Jal-Mahal (water palace) and a baoli (step-well) to commemorate his victory in the Khanwah battle.

Q 93.A

- Thailand has recently objected to India's customs regulations Customs (Administration of Rules of Origin under Trade Agreements) Rules, 2020. The regulation, CAROTAR, is aimed at strictly verifying the rules of origin under the free trade agreements (FTAs) to guard against misuse of benefits. Hence option (a) is the correct answer.
- It is a set of regulations that went into effect in 2020 that require importers to follow certain rules before importing goods. The rules are intended to allow importers to receive benefits under Free Trade Agreements (FTAs) by providing preferential rates.
- The rules also include procedures for: Denial of preferential tariff, Temporary suspension of preferential tariff, and Prospective restoration of preferential tariff.
- The intention of the government is to only allow genuine importers to receive preferential benefits. The Union Finance Ministry has advised customs field officers to ask the importer for further information if they have reasons to believe that the country-of-origin criteria have not been met.

Q 94.B

- In its 63rd council meeting, the International Sugar Organization (ISO), headquartered in London, announced India to be the chair of the organization for 2024. This is a huge achievement for India to lead the global sugar sector which reflects the growing stature of the country in this domain. While attending the ISO Council Meeting, Shri Sanjeev Chopra, Secretary (Food), Government of India remarked that during its period of chairmanship of ISO in 2024, India seeks support and cooperation from

all member countries and would like to focus on bringing together all member countries to adopt more sustainable practices in sugarcane cultivation, sugar and ethanol production and better utilization of by-products. **Hence, statement 1 is correct.**

- The International Sugar Organization is a unique intergovernmental body devoted to improving conditions of the world's sugar market through debate, analysis, special Studies, transparent statistics, seminars, conferences, and workshops. **The proactive attitude to include ethanol from sugar crops in our activities already in the mid-90s helped to promote the increasing role of biofuels in the future energy mix worldwide.**
- **The ISO is the only worldwide forum for discussions on an inter-governmental level: sugar policy, sugar trade policy, and related matters and gives access to members to the know-how and experience of other countries in restructuring their sugar economies. Hence, statement 2 is correct.**
- India has been the largest consumer and second-largest producer of sugar in the world. The average annual production of sugarcane in India is around 35.5 crore tonnes which is used to produce around 3 crore tonnes of sugar. **With about 15% share in global sugar consumption and about 20% production of sugar, Indian sugar trends affect the global markets profusely.** This leading position makes India the most suitable nation to lead the International Sugar Organisation (ISO) which is the apex international body on sugar and related products having about 90 countries as members. **Hence, statement 3 is not correct.**
- With Brazil in the Western Hemisphere, India is the market leader in the Eastern Hemisphere for the sugar market. Now, being the 3rd largest country in the world in ethanol production after the USA and Brazil, India has shown commitment towards green energy and its capability to twist the challenges of surplus sugar in the domestic market to the solution of fossil fuels imports and a tool to meet COP 26 targets for India.
- Quite remarkably, the ethanol blending percentage in India has increased from 5% in 2019-20 to 12% in 2022-23 while the production has increased from 173 crore litres to more than 500 crore litres during the same period.

Q 95.A

- **The World Bank Group consists of several institutions working toward the goal of reducing poverty and promoting sustainable development worldwide. The five main institutions within the World Bank Group are:**
- **International Bank for Reconstruction and Development (IBRD): Established in 1944,** IBRD provides loans and development assistance to middle-income and creditworthy low-income countries for projects aimed at reducing poverty and promoting sustainable development. It primarily focuses on financing infrastructure, agriculture, education, healthcare, and other development projects. **India is one of the founding members of the IBRD. Hence, option 1 is correct.**
- **International Development Association (IDA): Founded in 1960,** IDA is the concessional arm of the World Bank Group. It provides low-interest loans and grants to the world's poorest countries, including those in sub-Saharan Africa and South Asia. IDA focuses on projects that address basic needs, such as healthcare, education, water supply, sanitation, and rural development. **India is one of the founding members of the IDA. Hence, option 2 is correct.**
- **International Finance Corporation (IFC): Established in 1956,** IFC is the private sector arm of the World Bank Group. It promotes private sector development in emerging markets by providing investment, advisory services, and mobilizing capital for private sector projects. IFC focuses on sectors such as infrastructure, manufacturing, agribusiness, and financial services. **India is one of the founding members of the IFC. Hence, option 3 is correct.**
- **Multilateral Investment Guarantee Agency (MIGA): Founded in 1988,** MIGA provides political risk insurance and credit enhancement to encourage foreign direct investment (FDI) in developing countries. It helps investors mitigate risks associated with projects in emerging markets, thereby promoting economic growth and development. **India became a member of MIGA in January 1994. Hence, option 4 is not correct.**
- **International Centre for Settlement of Investment Disputes (ICSID): Established in 1966,** ICSID provides a forum for the arbitration and conciliation of international investment disputes between investors and host countries. It aims to encourage foreign investment by providing a transparent and impartial mechanism for resolving disputes. **India is not a member of ICSID. Hence, option 5 is not correct.**

Q 96.A

- **Greenium or green-premium** refers to the savings an issuer of a green bond realizes on the associated coupon payment because the bond is green. Green bonds are debt instruments that fund specific projects/activities categorized as 'green' under national or international green taxonomies. Such activities may include renewable energy projects, electric buses, and energy efficiency initiatives. Each bond has an issuer, i.e., those who want to raise money with a promise of a regular fixed payment and investors, i.e. those who provide that money with a promise of a regular fixed income. The logic behind a greenium is that because of **the appeal of sustainability, green bonds attract investors who are ready to forego some of their returns and accept lower payments (called a lower yield)**. Further, long-term green projects are associated with a reduction in both physical and financial risk. Thus, investors settle for lower returns, making green projects financially attractive for the issuers of these debt instruments. **Hence option (a) is the correct answer.**

Q 97.C

- **Foreign capital** plays a significant role in the economic development and growth of countries, especially in emerging economies where domestic savings and investment may be insufficient to meet the capital requirements for infrastructure development, industrial expansion, and technological advancement. Indian companies can raise foreign capital through various means such as:
- **External Commercial Borrowings (ECBs):** Indian companies can raise funds from foreign sources through external commercial borrowings. ECBs include loans, buyer's credit, supplier's credit, and foreign currency convertible bonds (FCCBs). This method allows Indian companies to borrow funds in foreign currency to finance their operations or investments. **Hence, option 1 is correct.**
- **Depository Receipts (DRs):** Indian companies can issue Depository Receipts, such as American Depository Receipts (ADRs) or Global Depository Receipts (GDRs), to raise funds from international investors. A Depository Receipt (DR) is a negotiable economic instrument issued through an agency in a foreign jurisdiction traded on the local stock exchange. A company can issue ADRs / GDRs if it is eligible to issue shares to persons resident outside India under the FDI Scheme.
- However, **an Indian-listed company, that is not eligible to raise funds from the Indian Capital Market including a company that has been restrained from accessing the securities market by the Securities and Exchange Board of India (SEBI) will not be eligible to issue ADRs/GDRs. Hence, option 2 is correct.**
- **Participatory Notes:** Participatory Notes (P-notes) are issued by foreign institutional investors (FIIs) to overseas investors who wish to invest in Indian securities markets without directly registering with the Securities and Exchange Board of India (SEBI). **P-notes indirectly enable Indian companies to raise funds from international investors by attracting foreign investment into Indian securities markets. Hence, option 3 is correct.**
- **Direct listing of shares on foreign stock exchanges:** The Indian government has permitted certain Indian companies to directly list on foreign stock exchanges to access global capital. This provision, effective since 30th October 2023, was introduced through the Companies (Amendment) Act, 2020. **It allows certain classes of domestic public companies to list their securities on foreign stock exchanges, including the GIFT International Financial Services Centre (IFSC) in Ahmedabad, Gujarat.**
- **Direct listing** is a process by which a company can list its shares on a foreign stock exchange without issuing new shares or raising capital from investors. **It allows a company to access a larger and more diverse pool of investors, enhance its visibility and brand value, and improve its corporate governance and compliance standards.**

Q 98.D

- The government's borrowings may be categorised into market and non-market borrowings. Non-market borrowings include receipts under the public account such as small savings and provident fund, external assistance, investment in 14-day intermediate treasury bills by state governments and other items. Market borrowings for cash management purposes can be defined to include dated securities, treasury bills and cash management bills. Market borrowings, particularly treasury bills and cash management bills, are planned in view of the cash flow pattern of the government.
- **Cash Management Bills (CMBs):** In 2010, Government of India, in consultation with RBI introduced a new short-term instrument, known as **Cash Management Bills (CMBs)**, to meet the temporary

mismatches in the cash flow of the Government of India. The CMBs have the generic character of T-bills but are issued for maturities less than 91 days. **Hence option (d) is the correct answer.**

- **Ways and Means Advances:** Apart from this, the government also receives ways and means advances from the Reserve Bank, which are basically provided to manage temporary mismatches in cash flows. **The Ways and Means advances was introduced in 1997.** Prior to 1997, issuance of ad hoc treasury bills was the primary instrument for making up the shortfall in minimum cash requirements which was initially fixed at ₹4 crore
- **Treasury-bills:** Treasury bills or T-bills, which are money market instruments, are short term debt instruments issued by the Government of India and are presently issued in three tenors, namely, 91 day, 182 day and 364 day. Treasury bills are zero coupon securities and pay no interest. Instead, they are issued at a discount and redeemed at the face value at maturity. **Treasury bills were first issued in India in 1917.**
- **Special Zero Coupon Recapitalisation Bonds:** These are special types of bonds issued by the Central government specifically to a particular institution. These bonds are not G-secs or ways of borrowing for the government. Their primary aim is to recapitalise banks.

Q 99.D

- Neerakshi is an **autonomous underwater vehicle (AUV) that has been specifically designed to detect mines.** The name "Neerakshi" means "Eyes in the Water" and it is the first of its kind in India. The AUV is expected to undergo user trials by the Indian Navy, Coast Guard, and Army before it is commercially launched. It is a result of a collaboration between Kolkata-based warship maker Garden Reach Shipbuilders and Engineers (GRSE) Ltd, and MSME entity Aerospace Engineering Private Ltd (AEPL).
- The AUV can detect mines, and it can also be used for other functions like mine disposal and underwater surveys. It is a cylindrical unmanned vehicle that is 2.1 meters long, approximately a foot in diameter, and weighs around 45 kg. It has an endurance of nearly 4 hours and can operate up to a depth of 300 meters. The development of Neerakshi is part of GRSE's broader ambitions to create autonomous sea surface vehicles, and sea-based drones, and explore green propulsion technologies.
- **Hence option (d) is the correct answer.**

Q 100.A

- **Recent Context:** Core-loading process in the indigenous Prototype Fast Breeder Reactor (PFBR) was initiated at Madras Atomic Power Station in Kalpakkam, Tamil Nadu.
 - PFBR has been designed and constructed indigenously by Bharatiya Nabhikiya Vidyut Nigam Ltd or BHAVINI with significant contribution from more than 200 Indian industries.
 - Core loading is the process of placing nuclear fuel assemblies inside the core of a nuclear reactor.
 - The completion of core loading will effectively mark the first approach to 'criticality'.
 - ✓ Criticality means the initiation of a selfsustaining nuclear fission reaction that will eventually lead to the generation of power by the 500-megawatt electric (MWe) FBR.
- **Fast Breeder Reactor (FBR)** is a nuclear reactor that uses fast neutron to generate more nuclear fuel than they consume while generating power.
 - FBR will use Uranium-Plutonium Mixed Oxide (MOX) fuel.
 - The Uranium-238 "blanket" surrounding the fuel core will undergo nuclear transmutation to produce more fuel, which is why they are termed "breeders."
- **The operationalisation of PFBR will mark the start of stage II of India's three-stage nuclear power programme.**
- **India's 3 stage Nuclear Power Programme:**
 - India holds only about 2-3% of the world's uranium reserves, but it possesses one of the largest shares of global thorium reserves.
 - Dr Homi J Bhabha, father of India's nuclear programme, therefore, devised a three-stage nuclear power programme in 1950s to make the most of India's limited uranium reserves and abundant thorium reserves. **Hence option (a) is the correct answer.**
 - **Stage 1:** Pressurised heavy water reactors
 - **Stage 2:** FBRs using plutonium as fuel
 - **Stage 3:** Advanced heavy water reactor (AHWR)

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